Derandomizing Unique-SVP
Lattice Reductions Without Gap Restrictions

Abstract. The unique shortest vector problem uSVP_γ and bounded distance decoding BDD_α are linked by Kannan's embedding and by sparsification, and they underlie lattice cryptanalysis. However, two gaps remain. Bai et al. (ICALP'16) reduce $\mathsf{BDD}_{1/(\sqrt{2}\gamma)}$ to uSVP_γ, improving the factor $1/(2\gamma)$ of Lyubashevsky-Micciancio (CRYPTO 2009), but their reduction is randomized and runs in polynomial time only for polynomially bounded γ. For rotations of $\mathbb{Z}^n$, the only deterministic dimension-preserving reduction from the problem ZSVP of finding a unit vector to approximate SVP stops at the factor $\sqrt{2}$, and extending it is an open problem.
In this work, we remove both restrictions. We prove a tube lemma, which bounds by $O(n/\varepsilon)$ the lattice points near the segment of all multiples of the target offset for a slack ε in the radius, so the number of unwanted vectors does not depend on γ. With an explicit family of p^3 sparsification vectors built from pencils of the moment curve, we obtain a deterministic polynomial-time reduction from $\mathsf{BDD}_{1/(\sqrt{2}\gamma)}$ to uSVP_γ for every $\gamma \geq 1$, including super-polynomial γ, and a deterministic dimension-preserving reduction from ZSVP to uSVP_γ and to approximate SVP for every constant factor. Then, we build Tube Calibrated Sparsified Embedding (TSE), a randomized reduction whose sparsification prime is $\Theta(n/\varepsilon)$ for every γ, against more than $4\gamma n^2$ before. With one precise promise oracle, TSE and the reduction of Bai et al. are the only ones that solve all 700 instances, TSE solves every instance that an embedding with one call solves with the same call, and it uses $\times 16$ fewer oracle calls and $\times 21$ less time than Bai et al., with shorter oracle inputs.

Keywords: Unique Shortest Vector Problem · Bounded Distance Decoding · Lattice Sparsification · Derandomization · Rotations of $\mathbb{Z}^n$

1 Introduction

A lattice $L \subset \mathbb{R}^n$ is the set of integer combinations of n linearly independent vectors, and its minimum distance $\lambda_1(L)$ is the length of a shortest nonzero lattice vector. The unique shortest vector problem uSVP_γ asks for a shortest nonzero vector of a lattice in which every vector outside the line of the shortest one is at least $\gamma \geq 1$ times longer, and it is the problem behind the first public key cryptosystems with worst-case guarantees [1, 2]. Bounded distance decoding BDD_α asks for the lattice point nearest to a target that lies within distance $\alpha\lambda_1(L)$ of the lattice, and learning with errors is its average-case form [3]. Kannan's embedding [4] turns a decoding instance into a unique shortest vector instance by appending the target, with an extra coordinate M, as a new basis vector; it is both a reduction in complexity theory [5, 6] and the core of the primal attack on learning with errors [7, 8]. Rotations of the integer lattice $\mathbb{Z}^n$ are the simplest lattices whose shortest vectors are unusually short, and finding a unit vector in such a rotation, the problem ZSVP, is the first step of recovering the rotation, on which cryptography based on lattice isomorphism depends [9–11].

The known relations between these problems are proven by sparsification [12–14]. A sublattice of prime index p is selected so that one chosen vector remains in it while each of N unwanted short vectors is removed with probability about $1-1/p$, so the union bound forces p to exceed about $2N$ and the reduction repeats about p times. The cost and the randomness of these reductions are therefore governed by the number of unwanted vectors, and this number is the subject of this work.

Each of the two known restrictions has been named as an open problem. Bennett et al. reduce ZSVP to γ-uSVP for every constant γ with sparsification, which is randomized, and their only deterministic dimension-preserving reduction reaches the factor $\sqrt{2}$ of SVP; they state the hope that it generalizes to larger, perhaps superconstant, approximation factors [11]. Ducas and van Woerden ask for a formal reduction from unusual-SVP, in which the shortest vector is much shorter than the Gaussian heuristic predicts, to uSVP [9], and the reduction of Jiang et al. from ZSVP to its approximate version for constant factors is randomized as well [15]. Problem 1 is the question of [11].

Problem 1. *Find a deterministic dimension-preserving reduction from* ZSVP*, and more generally from* SVP *on lattices with few short vectors, to* γ-uSVP *and to* γ-SVP *for factors* $\gamma > \sqrt{2}$ *[11, 9, 15].*

As we identified, the best known relation between BDD and uSVP is only proven for polynomially bounded gaps and with randomness. Lyubashevsky and Micciancio proved $\mathsf{BDD}_{1/(2\gamma)} \leq \mathsf{uSVP}_\gamma$ for every γ and $\mathsf{uSVP}_\gamma \leq \mathsf{BDD}_{1/\gamma}$ for polynomial γ, and they asked both to remove the factor 2 and to extend their reductions beyond polynomial approximation factors [5]. Bai, Stehlé and Wen proved $\mathsf{BDD}_{1/(\sqrt{2}\gamma)} \leq \mathsf{uSVP}_\gamma$ with sparsification [6], but their reduction selects a prime $p > 4\gamma n^2$ and succeeds with probability about $1/p$ per trial, so it is randomized and runs in polynomial time only when $\gamma \leq \mathrm{poly}(n)$. Bai et al. list both restrictions as open [6], and the later analysis of Zhao and Zheng keeps them [16]. Problem 2 is the question of [5, 6].

Problem 2. *Reduce* $\mathsf{BDD}_{1/(\sqrt{2}\gamma)}$ *to* uSVP_γ *deterministically and in time polynomial in* n *for every gap* $\gamma \geq 1$*, including super-polynomial* γ *[5, 6, 16].*

Since sparsification repeats an oracle call about p times, the unwanted vectors determine the cost of every reduction based on sparsification. In the analysis of [6] their number grows linearly with γ, and a reduction that reaches the decoding radius $1/(\sqrt{2}\gamma)$ with a number of oracle calls independent of γ has not been available. Problem 3 is the efficiency of the reduction itself.

Problem 3. *Construct a reduction from* $\mathsf{BDD}_{(1-\varepsilon)/(\sqrt{2}\gamma)}$ *to* uSVP_γ *whose number of oracle calls, time and memory are below those of all published reductions for every gap* γ*, and confirm the ordering both by a complexity analysis and by measuring all competing reductions with the same oracle [5, 6].*

Problem 1 and Problem 2 share a tool, since both need an explicit family of sublattices that keeps one vector and removes a few others. Problem 2 and Problem 3 share a quantity, the number of unwanted vectors of the embedding, which fixes both the size of such a family and the prime of a randomized reduction.

Table 1: Comparison of the reductions in three parts, the proven guarantees of the reductions from BDD_α to uSVP_γ, their measurements on the 700 instances of Section 9 with the same promise oracle, and the dimension-preserving reductions from ZSVP with measurements on 75 rotations of $\mathbb{Z}^n$ with $4 \le n \le 8$ and $\gamma^2 \in \{2,3,4\}$. The primes $p_B > 4\gamma n^2$ and $p = \Theta(n/\varepsilon)$ are those of [6] and of TSE.

Proven guarantees of the reductions from BDD_α *to* uSVP_γ

Reduction	Radius α	Gap γ	Coins	Expected calls
$M = d/\sqrt{n}$ [4, 8]	$\Theta(1/(\gamma\sqrt{n}))$	all	no	1
$M = d$ [5]	$1/(2\gamma)$	all	no	1
Balanced [6]	$1/\gamma_1$	all	no	1
Sparsified [6]	$(1-\frac{1}{n})/(\sqrt{2}\gamma)$	poly(n)	yes	$\approx 2p_B$
TSE, $\varepsilon = 0.1$ (Thm. 7.2)	$(1-\varepsilon)/(\sqrt{2}\gamma)$	all	yes	$\le 1+2p$
Pencils (Thm. 6.4)	$(1-\varepsilon)/(\sqrt{2}\gamma)$	all	no	$\le (N(q-1)+2)q^3$
Pencils (Cor. 6.5)	$1/(\sqrt{2}\gamma)$	all	no	poly(n)

Measured on the 700 instances with $12 \le n \le 28$ *and* $\gamma \in \{1.25, 1.5, 2, 3\}$

Reduction	Solved	Hard	Calls	Time (ms)	KiB	Ratios
$M = d/\sqrt{n}$ [4, 8]	350	0	1	3.5	48.2	1.00, 1.08, 1.02
$M = d$ [5]	475	0	1	3.4	46.7	1.00, 1.05, 1.00
Balanced [6]	500	0	1	3.3	46.7	1.00, 1.03, 1.00
Sparsified [6]	700	200	3199.2	37401	80.7	16.18, 21.51, 1.44
TSE, $\varepsilon = 0.1$ (Thm. 7.2)	**700**	**200**	197.7	1739	55.8	1, 1, 1
Pencils (Thm. 6.4)	120 of 120 hard instances with $n \le 20$ solved, 8096.3 calls, 58.5 s, 53.3 KiB					
Pencils (Cor. 6.5)	the self-reduction of [5] followed by the row above					

Dimension-preserving reductions from ZSVP

Reduction	Oracle	Gap γ	Coins	Oracle calls	Oracle inputs	Solved, calls
Projection [11, Thm. 5.1]	γ-SVP	$\sqrt{2}$	no	≤ 2	projections of L	–
Sparsification [11, Cor. 5.4]	uSVP_γ	$O(1)$	yes	(A/G)poly(n)	sublattices of L	–
Automorphisms [15, Thm. 3.2]	γ-SVP on $\mathbf{R}\mathbb{Z}^n$	$O(1)$	yes	poly(n)	rotations of $\mathbb{Z}^n$	–
Pencils (Cor. 5.2, 5.3)	uSVP_γ, γ'-SVP	$O(1)$	no	$\le p^3 = n^{O(\gamma^2)}$	sublattices of L	**75**, 265.3

1.1 Related work

Kannan's embedding appeared in [4], following the reduction of low-density subset sum to shortest vectors of [17]. Lyubashevsky and Micciancio proved that BDD, uSVP and GapSVP are equivalent up to polynomial factors [5], Liu et al. refined the analysis of their embedding for $\gamma < 1.9318$ [18], and the analysis in the appendix of [6], attributed there to Bai, Galbraith and Micciancio, gives the radius $1/\gamma_1$ with $\gamma_1 = (2\gamma^2 + 2\lfloor\gamma\rfloor\lfloor\gamma + 1\rfloor)/(2\lfloor\gamma\rfloor + 1)$ with one oracle call, for an embedding that we call balanced in Section 7. Bai, Stehlé and Wen introduced sparsification into the embedding [6], and Zhao and Zheng improved the gap of their instances for specific parameters [16]. Sparsification comes from the hardness proof of [12], and its randomized versions with a shift are analyzed in [14, 19, 20]; a deterministic sparsification in exponential time is given in [13]. The derandomization of hardness proofs for SVP is a long line of work [21, 22], in which the sets to be sparsified are exponentially large, while in our setting they are polynomial and small bias constructions in the sense of [23, 24] over extension fields apply; our Pencil[1] family is a direct construction for the specific isolation property that

[1] In projective geometry, a *Pencil* is a one-parameter family of geometric objects. For linearly independent $\mathbf{u}, \mathbf{v} \in V$, where V is a vector space over $\mathbb{F}$, the pencil they span is $\{\mathbb{F}(a\mathbf{u} + b\mathbf{v}) : (a,b) \in \mathbb{F}^2 \setminus \{(0,0)\}\}$, the projective line through $\mathbb{F}\mathbf{u}$ and $\mathbb{F}\mathbf{v}$ in $\mathbb{P}(V)$, containing $|\mathbb{F}| + 1$ points if $\mathbb{F}$ is finite. Dually, it comprises the hyperplanes containing $\mathbf{u}^\perp \cap \mathbf{v}^\perp$, recovering the classical pencil of lines through a point when $\dim V = 3$.

sparsification needs. For rotations of $\mathbb{Z}^n$, ZSVP and the isomorphism problem ZLIP were studied in [25–27, 11, 15], provable reduction with block size $n/2$ is in [28, 29], and the isomorphism problem for general lattices is in [30, 9]; uSVP_γ in the Euclidean norm is known to be NP-hard only for gaps γ very near 1 [31], and deterministic hardness results for uSVP in ℓ_p norms with $p>2$ are recent [22]. In cryptanalysis, the success of the primal attack depends on the gap of the embedding [7, 8].

2 Contributions

In this work, our results solve Problems 1 to 3 in turn as we go through below.

2.1 Deterministic reductions for unusual lattices

We address Problem 1 in Section 5. We construct an explicit family of at most p^3 vectors of $\mathbb{F}_p^n$, the lines of the planes spanned by two points of the moment curve, which for every nonzero $\bar{\mathbf{x}} \in \mathbb{F}_p^n$ and every A vectors linearly independent from $\bar{\mathbf{x}}$ contains a vector perpendicular to $\bar{\mathbf{x}}$ and not perpendicular to any of the A vectors, as soon as $p \geq 2A(n-1)+2n$ (Lemma 4.3). Replacing random sublattices by the sublattices defined by this family gives a deterministic dimension-preserving reduction from SVP on any lattice with at most A pairs of primitive vectors of length at most $\gamma\lambda_1$ to uSVP_γ, with at most p^3 oracle calls (Theorem 5.1). For rotations of $\mathbb{Z}^n$, this is a deterministic reduction from ZSVP to uSVP_γ and to γ'-SVP for every $\gamma' < \gamma$, with $n^{O(\gamma^2)}$ oracle calls (Corollaries 5.2 and 5.3), which answers the question of [11] for all constant factors. A thinner family with about np^2 members and a prime half as large (Theorem A.2) is within a factor of about n^2 of the lower bound $(A+1)p$ that we prove for every family with the isolation property (Theorem A.3).

2.2 Decoding to unique vectors for all gaps

We address Problem 2 in Section 6. We prove a tube lemma, which states that a segment of length ℓ has at most $n(\lfloor(\ell+2r)/(\lambda_1\sqrt{\eta})\rfloor+1)$ lattice points within distance $r=\lambda_1\sqrt{(1-\eta)/2}$ (Lemma 4.2). In Kannan's embedding with the extra coordinate $M=\sqrt{\varepsilon}d$, where d is the decoding distance and $\varepsilon \in (0,1/2]$ is a slack in the radius, all unwanted vectors lie in the tube around one segment through the origin whose length does not depend on γ, so at most $N(n,\varepsilon)=O(n/\varepsilon)$ vectors must be removed (Lemma 6.2). Consequently, $\mathsf{BDD}_{(1-\varepsilon)/(\sqrt{2}\gamma)}$ reduces to uSVP_γ with a prime $p=\Theta(n/\varepsilon)$ for every $\gamma \geq 1$ (Theorem 6.3), the pencil family makes the reduction deterministic (Theorem 6.4), and with the self-reduction of [5] we obtain $\mathsf{BDD}_{1/(\sqrt{2}\gamma)} \leq \mathsf{uSVP}_\gamma$ deterministically for every $\gamma \geq 1$, including super-polynomial γ (Corollary 6.5). This removes both restrictions of [6]. Shifts of the target make the deterministic reduction much smaller, with $O(n^3)$ oracle calls for a constant slack (Theorem A.4).

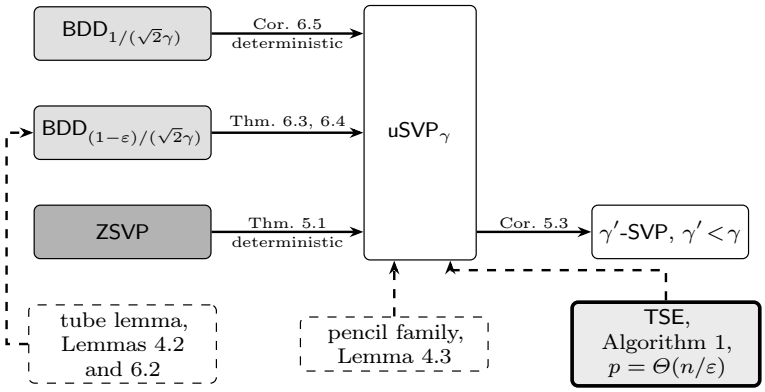


Fig. 1: Map of the results. Solid arrows are reductions, and dashed arrows show where the tools enter.

2.3 Our TSE reduction and its evaluation

We address Problem 3 in Sections 7 to 9. TSE, for tube-calibrated sparsified embedding, first makes one oracle call with the embedding analyzed in the appendix of [6], which we call balanced because its scaling equalizes the bounds for the two critical multiples of the offset, whose proven radius is the largest that any embedding with one call reaches (Theorem 8.3), and otherwise repeats the sparsified embedding of Section 2.2 with the prime $p=\Theta(n/\varepsilon)$ (Algorithm 1 and Theorem 7.2). Whenever an embedding with one call is guaranteed to succeed, TSE succeeds with the same single call, and for $n \geq 13$ its prime never exceeds the prime $p_B > 4\gamma n^2$ of [6] and is smaller as soon as $\gamma \geq 1.2$ or $n \geq 93$, which makes its guaranteed number of calls smaller (Corollary 7.3). We implemented TSE and the reductions of [4–6] with one oracle that checks the uSVP$_\gamma$ promise precisely and fails whenever the promise does not hold (Section 9). Table 1 shows that TSE and the reduction of [6] are the only ones that solve all 700 instances, that TSE solves every instance solved by an embedding with one call with the same single call, and that it uses 16 times fewer oracle calls and 21 times less time than the reduction of [6], with oracle inputs whose entries are shorter by 8 bits on average. An adaptive variant that tries small primes first keeps this guarantee up to a factor of two and needs $O(\ell+1/\sqrt{\varepsilon})$ calls in expectation when the unwanted vectors lie on ℓ lines (Theorem 10.3), and it solves all 700 instances with 3.2 calls on average, 61 times fewer than TSE (Section 10).

Figure 1 organizes the contributions, where a solid arrow from one problem to another means that the first reduces to the second, so that hardness of the first implies hardness of the second. The light gray boxes are the decoding problems with the radius $1/(\sqrt{2}\gamma)$ and with a slack ε, the dark gray box is ZSVP, and the white boxes are the unique and approximate shortest vector problems; the label above each arrow names the result, and the label below an arrow marks a deterministic reduction. The dashed boxes are the two counting tools of Section 4; the tube lemma bounds the unwanted vectors of the decoding reductions, and the pencil family makes the reductions of Theorems 5.1 and 6.4 deterministic. The box with the thick frame is TSE, the efficient randomized instance of the decoding reduction.

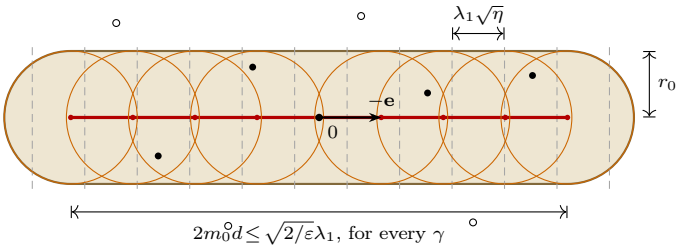


Fig. 2: The tube of unwanted vectors of the embedding.

Table 1 shows all reductions side by side. Its first part gives, for each reduction from BDD_α to uSVP_γ, the guaranteed radius, the range of gaps with polynomial running time, the use of coins and the expected number of oracle calls. Its second part gives the instances solved among all 700 and among the 200 hard ones, which have an offset along a shortest vector and a radius beyond the radius $1/\gamma_1$ of the balanced embedding, and the mean calls, time and memory; the column Ratios divides them by those of TSE on the instances that the reduction solves. Its third part lists the dimension-preserving reductions from ZSVP with their oracle, factors, coins, calls and oracle inputs. Bold entries mark the reductions that solve all instances of their part.

2.4 Technical overview

Where the polynomial gap comes from. Let L' be a sublattice of L and let $\mathbf{c}' \in L'$ be the lattice point that should be recovered for the target $\mathbf{t}'$. The embedding lattice generated by a basis of L', padded with a zero coordinate, and by the column $(\mathbf{t}', M)$ contains precisely the vectors $(\mathbf{y} + m\mathbf{e}, mM)$ with $\mathbf{y} \in L'$ and $m \in \mathbb{Z}$, where $\mathbf{e} = \mathbf{t}' - \mathbf{c}'$ is the offset of the target and $d = \|\mathbf{e}\|$ (Lemma 6.1). The wanted vector is $(\mathbf{e}, M)$, obtained for $\mathbf{y} = 0$ and $m = 1$, and a vector with $\mathbf{y} \neq 0$ is short only if $\mathbf{y}$ is near $-m\mathbf{e}$ and $|m|M$ is small. Reaching the radius $1/(\sqrt{2}\gamma)$ requires M to be a small fraction of d, so all multiples m up to about $\gamma d/M$ matter, and the analysis of [6] bounds the lattice points in one ball per multiple by $2n$, which gives $N \le 2\gamma n^2$ unwanted vectors and a prime $p > 4\gamma n^2$.

Tubes instead of balls. The points $-m\mathbf{e}$ with $|m| \le m_0$ lie on the segment of length $2m_0 d$ through the origin in the direction of $\mathbf{e}$, and $m_0 = r_0/M$ with $r_0 = \gamma\sqrt{d^2 + M^2}$ the radius that the promise requires. Because $d \le (1-\varepsilon)\lambda_1/(\sqrt{2}\gamma)$, the length $m_0 d = r_0 d/M$ is at most $\lambda_1/\sqrt{2\varepsilon}$ for $M = \sqrt{\varepsilon} d$, whatever γ is, so the union of all balls is a tube of bounded length (Figure 2). We cut the tube into slices of length $\lambda_1\sqrt{\eta}$ with $\eta = 1 - 2r_0^2/\lambda_1^2$. Two lattice points in one slice are at distance at least λ_1, so their projections onto the hyperplane perpendicular to $\mathbf{e}$ are at distance more than $\sqrt{2}r_0$ while both have norm at most r_0, which forces a negative inner product; at most n vectors of $\mathbb{R}^{n-1}$ have pairwise negative inner products (Lemma 4.1), so each slice has at most n points and the tube has $O(n/\varepsilon)$.

Figure 2 shows the plane that contains the offset $\mathbf{e}$ of the target. The red points are the multiples $-m\mathbf{e}$ for $1 \le |m| \le m_0$, and the orange circles are the balls of radius r_0 around them, one per multiple, that [6] counts separately. Their union lies in the shaded tube around the red segment, whose half-length $m_0 d$ is at most $\lambda_1/\sqrt{2\varepsilon}$

because d shrinks when γ grows. The dashed lines cut the tube into slices of length $\lambda_1\sqrt{\eta}$, each of which contains at most n lattice points by Lemma 4.2. The filled black points are lattice points inside the tube, which sparsification must remove, and the hollow ones are outside it and never shorten the second minimum of the embedding.

Pencils instead of random vectors. Sparsification keeps a vector $\mathbf{x}$ and removes $\mathbf{y}_1,\ldots,\mathbf{y}_A$ when the vector $\mathbf{z}$ that defines the sublattice satisfies $\langle \mathbf{z},\bar{\mathbf{x}}\rangle = 0$ and $\langle \mathbf{z},\bar{\mathbf{y}}_i\rangle \neq 0$, where the bar denotes coordinates modulo p. For two points $\mu(\alpha)$ and $\mu(\alpha')$ of the moment curve, the vector $f_{\bar{\mathbf{x}}}(\alpha')\mu(\alpha) - f_{\bar{\mathbf{x}}}(\alpha)\mu(\alpha')$ is always perpendicular to $\bar{\mathbf{x}}$, where $f_{\bar{\mathbf{x}}}$ is the polynomial with coefficient vector $\bar{\mathbf{x}}$, and its inner product with $\bar{\mathbf{y}}_i$ is a nonzero polynomial of degree at most $2(n-1)$ in (α,α'). The Schwartz–Zippel lemma [32, 33] then gives a pair (α,α') that works for all $\bar{\mathbf{y}}_i$, and all lines of all planes spanned by two moment points form a family of size at most p^3 that works for every $\bar{\mathbf{x}}$ and every choice of the $\bar{\mathbf{y}}_i$ at once.

3 Preliminaries

Lattices. For a nonsingular matrix $\mathbf{B} \in \mathbb{Q}^{n\times n}$ with columns $\mathbf{b}_1,\ldots,\mathbf{b}_n$, the lattice with basis $\mathbf{B}$ is $L = L(\mathbf{B}) = \{\mathbf{Bx} : \mathbf{x} \in \mathbb{Z}^n\}$, of rank n. We write $\|\cdot\|$ for the Euclidean norm, $\langle\cdot,\cdot\rangle$ for the inner product, $\operatorname{dist}(\mathbf{t},S) = \inf_{\mathbf{s}\in S}\|\mathbf{t}-\mathbf{s}\|$ for the distance from a point to a set, and $\lambda_1(L)$, respectively $\lambda_2(L)$, for the least r such that the ball of radius r around 0 contains one, respectively two, linearly independent lattice vectors. A nonzero $\mathbf{v} \in L$ is primitive if $\mathbf{v}/k \notin L$ for every integer $k \geq 2$, and every nonzero lattice vector is a positive integer multiple of a primitive one. For a prime p, $\mathbb{F}_p$ is the field with p elements, and for $\mathbf{v} \in L$ the vector $\bar{\mathbf{v}} = \mathbf{B}^{-1}\mathbf{v} \bmod p \in \mathbb{F}_p^n$ is its reduced coordinate vector; $\bar{\mathbf{v}} = 0$ precisely when $\mathbf{v} \in pL$. A rotation of $\mathbb{Z}^n$ is a lattice $\mathbf{R}\mathbb{Z}^n$ with $\mathbf{R}^\top\mathbf{R} = \mathbf{I}$, and its unit vectors are the images $\pm\mathbf{Re}_i$ of the standard basis vectors $\mathbf{e}_i$.

The following definitions fix the promise problems, where the input lattice is given by a basis.

Definition 3.1 (Unique shortest vector problem). *For $\gamma \geq 1$, uSVP_γ is the problem that, given a basis of a lattice L with $\lambda_2(L) \geq \gamma\lambda_1(L)$, asks for a shortest nonzero vector of L.*

Definition 3.2 (Bounded distance decoding). *For $\alpha > 0$, BDD_α is the problem that, given a basis of a lattice L and a target $\mathbf{t} \in \mathbb{Q}^n$ with $\operatorname{dist}(\mathbf{t},L) \leq \alpha\lambda_1(L)$, asks for a lattice vector $\mathbf{c} \in L$ with $\|\mathbf{t}-\mathbf{c}\| = \operatorname{dist}(\mathbf{t},L)$.*

Definition 3.3 (Approximate and integer shortest vectors). *For $\gamma \geq 1$, γ-SVP asks, given a basis of L, for a vector $\mathbf{v} \in L$ with $0 < \|\mathbf{v}\| \leq \gamma\lambda_1(L)$. The problem ZSVP asks, given a basis of a rotation of $\mathbb{Z}^n$, for a lattice vector of norm 1.*

We follow the conventions of [6], which allow equality in the promise of Definition 3.1; the conventions with strict inequality are equivalent by [5]. A reduction from a problem to a promise problem is a polynomial-time algorithm with access to an oracle that answers correctly on every input that satisfies the promise and may answer arbitrarily otherwise. A reduction is dimension-preserving if the oracle inputs have dimension at most n, and for BDD we also allow dimension $n+1$, as in [5, 6].

Sparsified sublattices. For a prime p and a nonzero $\mathbf{z} \in \mathbb{F}_p^n$, we write $L_{\mathbf{z}} = \{\mathbf{v} \in L : \langle \mathbf{z}, \bar{\mathbf{v}} \rangle = 0\}$, which is a sublattice of index p. The Lemma 3.4 addresses its basis.

Lemma 3.4. *Let $\mathbf{z} \in \mathbb{F}_p^n$ with $z_j \neq 0$, and let $c_i \in \{0,...,p-1\}$ with $c_i \equiv z_i z_j^{-1} \pmod{p}$. Then the n vectors $\mathbf{b}_i - c_i \mathbf{b}_j$ for $i \neq j$ and $p\mathbf{b}_j$ form a basis of $L_{\mathbf{z}}$, which is computed with $O(n^2)$ arithmetic operations on integers of bit length at most $\log_2 p$ plus the bit length of the entries of $\mathbf{B}$, plus one.*

Proof. Each listed vector lies in $L_{\mathbf{z}}$, because the coordinate vector $\mathbf{e}_i - c_i \mathbf{e}_j$ satisfies $z_i - c_i z_j \equiv 0$ and $p\mathbf{e}_j$ vanishes modulo p. Conversely, let $\mathbf{Bx} \in L_{\mathbf{z}}$ and write $\mathbf{x} = \sum_{i \neq j} x_i (\mathbf{e}_i - c_i \mathbf{e}_j) + s\mathbf{e}_j$ with $s = x_j + \sum_{i \neq j} c_i x_i$. Then $z_j s \equiv z_j x_j + \sum_{i \neq j} z_i x_i = \langle \mathbf{z}, \mathbf{x} \rangle \equiv 0$, so p divides s and $\mathbf{Bx}$ is an integer combination of the listed vectors. The n listed vectors are linearly independent, since their coordinate matrix is triangular after moving index j last, with determinant $\pm p$, so they form a basis and $L_{\mathbf{z}}$ has index p. Each vector costs $O(n)$ operations. □

The following two statements are used as black boxes, which are results from the literature. The Proposition 3.5 addresses slack in the decoding radius, and Proposition 3.6 goes through the short vectors of $\mathbb{Z}^n$.

Proposition 3.5. *For every $\alpha > 0$ and every constant c, there is a deterministic polynomial-time reduction from BDD_α to $\mathsf{BDD}_{\alpha(1-1/n)^c}$ whose oracle inputs have dimension n if $\alpha \leq 1/2$ and $n+1$ otherwise [5, Corollary 3.3], see also [6, Lemma 5].*

Proposition 3.6. *For every $n \geq 1$ and every $r \geq 1$ with $r^2 \in \mathbb{Z}$ and $r \leq \sqrt{n}$, the number of vectors of $\mathbb{Z}^n$ of norm at most r is at most $(2e^3 n / r^2)^{r^2}$ [34, Claim 8.2], as used in [11, Proposition 2.12].*

4 Two formal counting tools

Here, we prove the two combinatorial statements behind all reductions. The first is a bound on vectors with pairwise negative inner products. The Lemma 4.1 handles pairwise negative inner products.

Lemma 4.1. *If $\mathbf{v}_1,...,\mathbf{v}_k \in \mathbb{R}^m$ satisfy $\langle \mathbf{v}_i, \mathbf{v}_j \rangle < 0$ for all $i \neq j$, then $k \leq m+1$.*

Proof. Assume $k \geq m+2$. The vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{m+1}$ are linearly dependent, so $\sum_{i \leq m+1} a_i \mathbf{v}_i = 0$ for real a_i that are not all zero, and after negating all a_i we may assume that $P = \{i : a_i > 0\}$ is not empty. Let $N = \{i : a_i < 0\}$ and $\mathbf{w} = \sum_{i \in P} a_i \mathbf{v}_i = \sum_{j \in N} (-a_j) \mathbf{v}_j$. Then $\|\mathbf{w}\|^2 = \sum_{i \in P} \sum_{j \in N} a_i (-a_j) \langle \mathbf{v}_i, \mathbf{v}_j \rangle \leq 0$, so $\mathbf{w} = 0$. Hence $0 = \langle \mathbf{w}, \mathbf{v}_{m+2} \rangle = \sum_{i \in P} a_i \langle \mathbf{v}_i, \mathbf{v}_{m+2} \rangle < 0$, which is a contradiction. □

4.1 The tube lemma

For a unit vector $\mathbf{u} \in \mathbb{R}^n$ and reals $a \leq b$, let $S = \{s\mathbf{u} : a \leq s \leq b\}$ be the segment from $a\mathbf{u}$ to $b\mathbf{u}$, and for $r > 0$ let $\mathcal{T}(S,r) = \{\mathbf{x} \in \mathbb{R}^n : \mathrm{dist}(\mathbf{x}, S) \leq r\}$ be the tube of radius r around it. The Lemma 4.2 addresses lattice points in a tube.

Lemma 4.2. *Let $L \subset \mathbb{R}^n$ be a lattice with $\lambda = \lambda_1(L)$, let S be a segment of length $\ell = b-a$ in the direction of a unit vector $\mathbf{u}$, let $0 < r < \lambda/\sqrt{2}$ and let $\eta = 1 - 2r^2/\lambda^2$. Then $|L \cap \mathcal{T}(S,r)| \le n\big(\lfloor(\ell+2r)/(\lambda\sqrt{\eta})\rfloor + 1\big)$.*

Proof. For $\mathbf{x} \in \mathbb{R}^n$ let $\sigma(\mathbf{x}) = \langle \mathbf{x}, \mathbf{u} \rangle$ and $\pi(\mathbf{x}) = \mathbf{x} - \sigma(\mathbf{x})\mathbf{u}$, which lies in the hyperplane H perpendicular to $\mathbf{u}$. For every s, $\|\mathbf{x} - s\mathbf{u}\|^2 = (\sigma(\mathbf{x}) - s)^2 + \|\pi(\mathbf{x})\|^2$, so every $\mathbf{x} \in \mathcal{T}(S,r)$ satisfies $\|\pi(\mathbf{x})\| \le r$ and $\sigma(\mathbf{x}) \in [a-r, b+r]$. Let $\delta = \lambda\sqrt{\eta}$ and cover $[a-r, b+r]$ by the $K = \lfloor(\ell+2r)/\delta\rfloor + 1$ half-open intervals $[a-r+(k-1)\delta, a-r+k\delta)$ for $k = 1, \ldots, K$. Fix one interval and two distinct lattice points $\mathbf{x}, \mathbf{x}'$ of the tube whose values σ lie in it. Since $\mathbf{x} - \mathbf{x}'$ is a nonzero lattice vector and $|\sigma(\mathbf{x}) - \sigma(\mathbf{x}')| < \delta$,

$$\begin{aligned} \|\pi(\mathbf{x}) - \pi(\mathbf{x}')\|^2 &= \|\mathbf{x} - \mathbf{x}'\|^2 - (\sigma(\mathbf{x}) - \sigma(\mathbf{x}'))^2 \\ &> \lambda^2 - \delta^2 = 2r^2 \ge \|\pi(\mathbf{x})\|^2 + \|\pi(\mathbf{x}')\|^2, \end{aligned}$$

so $\langle \pi(\mathbf{x}), \pi(\mathbf{x}') \rangle < 0$. The projections of the lattice points of the tube in one interval are therefore vectors of the $(n-1)$-dimensional space H with pairwise negative inner products, and Lemma 4.1 bounds their number by n. Summing over the K intervals gives the claim. □

The bound depends on ℓ only through the number of slices and on n only linearly, while a covering of the tube by balls of radius r with the bound of [35, Theorem 5.2] for each ball would count every lattice point once for each ball that contains it. The condition $r < \lambda/\sqrt{2}$ cannot be dropped, since for every $r' > \lambda/\sqrt{2}$ there are lattices with $2^{n^{\Omega(1)}}$ points in a single ball of radius r' [6, 35].

4.2 Isolation with pencils of the moment curve

For $\alpha \in \mathbb{F}_p$ let $\mu(\alpha) = (1, \alpha, \ldots, \alpha^{n-1}) \in \mathbb{F}_p^n$ be the point of the moment curve, and let

$$\mathcal{Z}_{n,p} = \{\mu(\alpha) : \alpha \in \mathbb{F}_p\} \cup \{\mu(\alpha) + \beta\mu(\alpha') : \alpha, \alpha', \beta \in \mathbb{F}_p,\ \alpha \ne \alpha'\}.$$

For $n \ge 2$ and $\alpha \ne \alpha'$, the vectors $\mu(\alpha)$ and $\mu(\alpha')$ are linearly independent, since their first two coordinates are, so the set $\mathcal{Z}_{n,p}$ contains one nonzero vector on each line of each plane $\mathrm{span}(\mu(\alpha), \mu(\alpha'))$, and $|\mathcal{Z}_{n,p}| \le p + p^2(p-1) \le p^3$. The Lemma 4.3 addresses isolation.

Lemma 4.3. *Let $n \ge 2$, $A \ge 0$ and let $p \ge 2A(n-1) + 2n$ be a prime. For every nonzero $\mathbf{x} \in \mathbb{F}_p^n$ and every set $Y \subseteq \mathbb{F}_p^n$ of at most A vectors that are each linearly independent from $\mathbf{x}$, there is $\mathbf{z} \in \mathcal{Z}_{n,p}$ with $\langle \mathbf{z}, \mathbf{x} \rangle = 0$ and $\langle \mathbf{z}, \mathbf{y} \rangle \ne 0$ for all $\mathbf{y} \in Y$.*

Proof. For $\mathbf{v} \in \mathbb{F}_p^n$ let $f_{\mathbf{v}}(X) = \sum_{i=1}^{n} v_i X^{i-1} \in \mathbb{F}_p[X]$, so that $\langle \mu(\alpha), \mathbf{v} \rangle = f_{\mathbf{v}}(\alpha)$ and $\mathbf{v} \mapsto f_{\mathbf{v}}$ is linear and injective. For $\mathbf{y} \in Y$ let $D_{\mathbf{y}}(X, X') = f_{\mathbf{x}}(X') f_{\mathbf{y}}(X) - f_{\mathbf{x}}(X) f_{\mathbf{y}}(X')$, of total degree at most $2(n-1)$. The polynomial $D_{\mathbf{y}}$ is not zero, because otherwise $f_{\mathbf{y}}(X) f_{\mathbf{x}}(X') = f_{\mathbf{x}}(X) f_{\mathbf{y}}(X')$, and comparing the coefficients of X^{i-1} for an index i with $x_i \ne 0$ gives $f_{\mathbf{y}}(X') = (y_i/x_i) f_{\mathbf{x}}(X')$, so $\mathbf{y} = (y_i/x_i)\mathbf{x}$, which contradicts independence. A pair $(\alpha, \alpha') \in \mathbb{F}_p^2$ is bad if $\alpha = \alpha'$, or if $f_{\mathbf{x}}(\alpha) = f_{\mathbf{x}}(\alpha') = 0$, or if $D_{\mathbf{y}}(\alpha, \alpha') = 0$ for some $\mathbf{y} \in Y$. There are p pairs of the first kind, at most $(n-1)^2$

of the second since $f_{\mathbf{x}} \neq 0$ has at most $n-1$ roots, and at most $2(n-1)p$ for each $\mathbf{y}$ by the Schwartz–Zippel lemma [32, 33]. Since $p \geq 2A(n-1)+2n$, we have $p(p-1-2A(n-1)) \geq p(2n-1) > (n-1)^2$, so there are fewer than p^2 bad pairs and a good pair (α,α') exists. Let $\mathbf{z}^* = f_{\mathbf{x}}(\alpha')\mu(\alpha) - f_{\mathbf{x}}(\alpha)\mu(\alpha')$, which is nonzero because the two coefficients are not both zero and the two moment points are independent. Then $\langle \mathbf{z}^*, \mathbf{x} \rangle = f_{\mathbf{x}}(\alpha')f_{\mathbf{x}}(\alpha) - f_{\mathbf{x}}(\alpha)f_{\mathbf{x}}(\alpha') = 0$ and $\langle \mathbf{z}^*, \mathbf{y} \rangle = D_{\mathbf{y}}(\alpha,\alpha') \neq 0$ for every $\mathbf{y} \in Y$. If $f_{\mathbf{x}}(\alpha') \neq 0$, then $\mathbf{z} = \mathbf{z}^*/f_{\mathbf{x}}(\alpha') = \mu(\alpha) + \beta\mu(\alpha')$ with $\beta = -f_{\mathbf{x}}(\alpha)/f_{\mathbf{x}}(\alpha')$ lies in $\mathcal{Z}_{n,p}$; otherwise $f_{\mathbf{x}}(\alpha) \neq 0$ and $\mathbf{z} = \mu(\alpha') = -\mathbf{z}^*/f_{\mathbf{x}}(\alpha)$ lies in $\mathcal{Z}_{n,p}$. Multiplication by a nonzero scalar keeps the conditions, which proves the claim. □

5 Deterministic reductions for unusual lattices

For a lattice L and $\gamma \geq 1$, let $A_\gamma(L)$ be the number of pairs $\{\mathbf{y}, -\mathbf{y}\}$ of primitive vectors of L with $\|\mathbf{y}\| \leq \gamma\lambda_1(L)$, which counts the shortest vectors themselves. Lattices with an unusually short vector, such as rotations of $\mathbb{Z}^n$, have few such pairs for constant γ, and this is the quantity that governs sparsification [11]. The Theorem 5.1 tackles deterministic isolation of a shortest vector.

Theorem 5.1. *Let $\gamma \geq 1$ and $n \geq 2$. There is a deterministic algorithm that receives a basis $\mathbf{B}$ of a lattice $L \subset \mathbb{R}^n$ and an integer $A \geq A_\gamma(L)$, has access to a uSVP_γ oracle, and outputs a shortest nonzero vector of L. It makes at most p^3 oracle calls, where p is the least prime with $p \geq \max(2A(n-1)+2n, 2\gamma+1)$, each on a sublattice of L of index p, and it runs in time polynomial in the bit length of $\mathbf{B}$ and in p.*

Proof (sketch). The algorithm calls the oracle on a basis of $L_{\mathbf{z}}$, computed by Lemma 3.4, for every $\mathbf{z} \in \mathcal{Z}_{n,p}$, discards the answers that are not nonzero vectors of L, and outputs the shortest remaining answer. Let $\mathbf{x}$ be a shortest nonzero vector of L, which is primitive, so $\bar{\mathbf{x}} \neq 0$. Let Y contain $\bar{\mathbf{y}}$ for one vector $\mathbf{y}$ of each pair $\{\mathbf{y}, -\mathbf{y}\}$ of primitive vectors with $\|\mathbf{y}\| \leq \gamma\lambda_1(L)$ and $\mathbf{y} \neq \pm\mathbf{x}$. If $\bar{\mathbf{y}} = c\bar{\mathbf{x}}$ with an integer $|c| \leq (p-1)/2$, then $\mathbf{y} - c\mathbf{x} \in pL$ and $\|\mathbf{y} - c\mathbf{x}\| \leq (\gamma + (p-1)/2)\lambda_1(L) < p\lambda_1(L)$ because $p > 2\gamma - 1$, so $\mathbf{y} = c\mathbf{x}$, which is impossible for a primitive $\mathbf{y} \neq \pm\mathbf{x}$; hence each element of Y is independent from $\bar{\mathbf{x}}$. Lemma 4.3 gives $\mathbf{z} \in \mathcal{Z}_{n,p}$ with $\mathbf{x} \in L_{\mathbf{z}}$ and $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle \neq 0$ for all $\bar{\mathbf{y}} \in Y$. A vector $\mathbf{v} = k\mathbf{y}$ of $L_{\mathbf{z}}$ with $\mathbf{y}$ primitive and not parallel to $\mathbf{x}$ has either $\|\mathbf{y}\| > \gamma\lambda_1(L)$ or $p \mid k$, so $\|\mathbf{v}\| > \gamma\lambda_1(L)$ in both cases, and $L_{\mathbf{z}}$ satisfies the promise of uSVP_γ with $\lambda_1(L_{\mathbf{z}}) = \lambda_1(L)$. The full proof is in the Appendix. □

The algorithm does not need to know $\lambda_1(L)$ or the vector $\mathbf{x}$, since the pencil family works for every nonzero $\bar{\mathbf{x}}$ at once, and the only input that depends on the lattice is the bound A. For rotations of $\mathbb{Z}^n$ this bound is known in advance. The Corollary 5.2 handles rotations of $\mathbb{Z}^n$.

Corollary 5.2. *Let $\gamma \geq 1$ with $s = \lceil \gamma^2 \rceil \leq n$. There is a deterministic dimension-preserving reduction from ZSVP in dimension n to uSVP_γ that makes at most p^3 oracle calls, where $p = O(n(2e^3 n/s)^s)$. For every constant γ the reduction runs in time polynomial in n and in the bit length of the input.*

Proof. A rotation $L=\mathbf{R}\mathbb{Z}^n$ has $\lambda_1(L)=1$, and its primitive vectors of norm at most γ are the images of the primitive vectors of $\mathbb{Z}^n$ of norm at most γ, which have norm at most $\sqrt{s}$. By Proposition 3.6 with $r=\sqrt{s}$, $A_\gamma(L)\le A:=\lfloor(2e^3n/s)^s/2\rfloor$, which depends only on n and γ. Theorem 5.1 with this A outputs a shortest vector of L, which has norm 1, with at most p^3 calls on sublattices of rank n, where $p\le 2\max(2A(n-1)+2n,2\gamma+1)$ by Bertrand's postulate. For constant γ, s is constant, so A, p and the running time are polynomial in n. □

The Corollary 5.3 addresses the approximate shortest vector problem.

Corollary 5.3. *For every $1\le\gamma'<\gamma$ with $\lceil\gamma^2\rceil\le n$, there is a deterministic dimension-preserving reduction from* ZSVP *to* γ'-SVP *with the same number of oracle calls as in Corollary 5.2. In particular,* ZSVP *reduces deterministically to* γ'-SVP *for every constant $\gamma'\ge 1$ in polynomial time.*

Proof. It suffices to answer each uSVP_γ query of Corollary 5.2 with one γ'-SVP query on the same lattice L'. On a valid query, $\lambda_2(L')\ge\gamma\lambda_1(L')>\gamma'\lambda_1(L')$, so the answer $\mathbf{v}$ satisfies $0<\|\mathbf{v}\|\le\gamma'\lambda_1(L')<\lambda_2(L')$, and all lattice vectors shorter than $\lambda_2(L')$ lie on the line of a shortest vector $\mathbf{s}$. Hence $\mathbf{v}=k\mathbf{s}$ for a nonzero integer k, and $\mathbf{s}=\pm\mathbf{v}/g$, where g is the greatest common divisor of the coordinates of $\mathbf{v}$ in the basis of L', which is computed in polynomial time. On invalid queries the answer is arbitrary, and the reduction of Theorem 5.1 only uses the answers to valid queries. □

Corollary 5.3 with $\gamma'=\sqrt{2}$ recovers the factor of the projection reduction of [11], and for every constant $\gamma'>\sqrt{2}$ it gives the deterministic generalization that [11] asks for. The same argument applies to any family with a known bound on A_γ. For example, a semi-stable lattice, in which every sublattice has determinant at least 1, has at most $3e^{\pi t\rho^2/2}$ vectors of norm at most $\rho\ge 1$ with $t=10(\log n+2)$ [34], as used in [11, Proposition 2.13]. For semi-stable lattices with $\lambda_1(L)\le c$ for a known constant c, the choice $A=\lfloor\frac{3}{2}e^{\pi t\gamma^2c^2/2}\rfloor$ is polynomial in n for every constant γ, so Theorem 5.1 gives a deterministic polynomial-time reduction for this family as well, which is the deterministic counterpart of the randomized reduction for the same family in [11].

6 Decoding to unique vectors for all gaps

For a sublattice $L'\subseteq L$ with basis $\mathbf{B}'$, a point $\mathbf{t}'\in\mathbb{Q}^n$ and $M>0$, the embedding lattice $\mathcal{E}(L',\mathbf{t}',M)\subset\mathbb{R}^{n+1}$ is generated by the columns of the matrix $\left(\begin{smallmatrix}\mathbf{B}' & \mathbf{t}'\\ 0 & M\end{smallmatrix}\right)$. Throughout this section $\lambda=\lambda_1(L)$, $\gamma\ge 1$ is the gap of the oracle and $\varepsilon\in(0,1/2]$ is the slack in the decoding radius. The Lemma 6.1 handles the shape of the embedding.

Lemma 6.1. *Let $L'\subseteq L$, $\mathbf{c}'\in L'$ and $\mathbf{t}'\in\mathbb{Q}^n$ with $\mathbf{e}=\mathbf{t}'-\mathbf{c}'\ne 0$ and $d=\|\mathbf{e}\|$, and let $M>0$, $r_0=\gamma\sqrt{d^2+M^2}$ and $m_0=r_0/M$. Assume $r_0<\lambda$ and that no $\mathbf{y}\in L'\setminus\{0\}$ and integer m with $1\le|m|\le m_0$ satisfy $\|\mathbf{y}+m\mathbf{e}\|\le r_0$. Then $\pm(\mathbf{e},M)$ are the only shortest nonzero vectors of $\mathcal{E}(L',\mathbf{t}',M)$, and every vector of $\mathcal{E}(L',\mathbf{t}',M)$ that is not parallel to $(\mathbf{e},M)$ has norm larger than r_0. In particular, $\mathcal{E}(L',\mathbf{t}',M)$ is an instance of* uSVP_γ.

Proof. The vectors of $\mathcal{E}(L',\mathbf{t}',M)$ are $(\mathbf{b}+m\mathbf{t}',mM)$ with $\mathbf{b}\in L'$ and $m\in\mathbb{Z}$. With $\mathbf{y}=\mathbf{b}+m\mathbf{c}'\in L'$ such a vector equals $\mathbf{v}=(\mathbf{y}+m\mathbf{e},mM)$, and $(\mathbf{e},M)$ is obtained for $\mathbf{b}=-\mathbf{c}'$ and $m=1$. If $\mathbf{y}=0$, then $\mathbf{v}=m(\mathbf{e},M)$. If $\mathbf{v}=s(\mathbf{e},M)$ for a real s, then $s=m$ and $\mathbf{y}=0$, so $\mathbf{y}\neq 0$ precisely when $\mathbf{v}$ is not parallel to $(\mathbf{e},M)$. Let $\mathbf{y}\neq 0$. If $m=0$, then $\|\mathbf{v}\|=\|\mathbf{y}\|\geq\lambda>r_0$. If $|m|>m_0$, then $\|\mathbf{v}\|\geq|m|M>m_0M=r_0$. If $1\leq|m|\leq m_0$, then $\|\mathbf{v}\|\geq\|\mathbf{y}+m\mathbf{e}\|>r_0$ by assumption. Hence all vectors that are not parallel to $(\mathbf{e},M)$ are longer than $r_0=\gamma\|(\mathbf{e},M)\|\geq\|(\mathbf{e},M)\|$, while the nonzero parallel vectors $m(\mathbf{e},M)$ are shortest precisely for $m=\pm1$. Consequently $\lambda_1(\mathcal{E})=\|(\mathbf{e},M)\|$ and $\lambda_2(\mathcal{E})>r_0=\gamma\lambda_1(\mathcal{E})$. □

We now fix the parameters of the reduction. Let $d\leq(1-\varepsilon)\lambda/(\sqrt{2}\gamma)$ be the decoding distance, let M be a rational number with $\sqrt{\varepsilon}d\leq M\leq 1.04\sqrt{\varepsilon}d$, and set

$$r_0=\gamma\sqrt{d^2+M^2},\quad m_0=\frac{r_0}{M},\quad h(\varepsilon)=\frac{1+1/\sqrt{\varepsilon}}{\sqrt{2}},$$

$$N(n,\varepsilon)=n\left(\left\lfloor 1.5\left(\frac{1}{\varepsilon}+\frac{1}{\sqrt{\varepsilon}}\right)\right\rfloor+1\right).$$

For the offset $\mathbf{e}$ of the target, the unwanted set $\mathcal{Y}$ contains the nonzero $\mathbf{y}\in L$ for which some integer m with $1\leq|m|\leq m_0$ satisfies $\|\mathbf{y}+m\mathbf{e}\|\leq r_0$; these are the vectors that Lemma 6.1 requires to be absent from the sparsified lattice. The Lemma 6.2 goes through the number of unwanted vectors.

Lemma 6.2. *With the parameters above,* $r_0^2\leq(1-0.918\varepsilon)\lambda^2/2$, $m_0d+r_0<h(\varepsilon)\lambda$, *every* $\mathbf{y}\in\mathcal{Y}$ *satisfies* $\|\mathbf{y}\|<h(\varepsilon)\lambda$, *and* $|\mathcal{Y}|\leq N(n,\varepsilon)$.

Proof. Since $M^2\leq 1.0816\,\varepsilon d^2$, we have $r_0^2\leq\gamma^2d^2(1+1.0816\varepsilon)\leq(1-\varepsilon)^2(1+1.0816\varepsilon)\lambda^2/2$. Expanding, $(1-\varepsilon)^2(1+1.0816\varepsilon)=1-0.9184\varepsilon-\varepsilon^2(1.1632-1.0816\varepsilon)\leq 1-0.918\varepsilon$, which is the first claim. Next, $m_0d=r_0d/M\leq r_0/\sqrt{\varepsilon}$, so $m_0d+r_0\leq r_0(1+1/\sqrt{\varepsilon})<h(\varepsilon)\lambda$ because $r_0<\lambda/\sqrt{2}$. For $\mathbf{y}\in\mathcal{Y}$ with multiple m, $\|\mathbf{y}\|\leq\|\mathbf{y}+m\mathbf{e}\|+|m|d\leq r_0+m_0d<h(\varepsilon)\lambda$. Finally, $-m\mathbf{e}$ lies on the segment S from $-m_0\mathbf{e}$ to $m_0\mathbf{e}$, of length $\ell=2m_0d$, so $\mathcal{Y}\subseteq L\cap\mathcal{T}(S,r_0)$. Lemma 4.2 applies with $r=r_0<\lambda/\sqrt{2}$ and $\eta=1-2r_0^2/\lambda^2\geq 0.918\varepsilon$, and

$$\frac{\ell+2r_0}{\lambda\sqrt{\eta}}\leq\frac{2r_0(1+1/\sqrt{\varepsilon})}{\lambda\sqrt{0.918\varepsilon}}<\frac{\sqrt{2}}{\sqrt{0.918}}\left(\frac{1}{\varepsilon}+\frac{1}{\sqrt{\varepsilon}}\right)<1.5\left(\frac{1}{\varepsilon}+\frac{1}{\sqrt{\varepsilon}}\right),$$

so $|\mathcal{Y}|\leq|L\cap\mathcal{T}(S,r_0)|\leq N(n,\varepsilon)$. □

The count does not depend on γ, because a larger gap shrinks the admissible distance d and lengthens the range of multiples m_0 by the same factor, which leaves the tube length $2m_0d$ unchanged. For $\varepsilon=1/n$ it is $N(n,1/n)=1.5n^2+O(n^{3/2})$, against the count $2\gamma n^2$ of [6], and for constant ε it is linear in n.

One trial. Let p be a prime with $p\geq\max(2N(n,\varepsilon),h(\varepsilon)+1)$. A trial with vectors $\mathbf{z},\mathbf{u}\in\mathbb{F}_p^n$, $\mathbf{z}\neq 0$, lifts $\mathbf{u}$ to $\tilde{\mathbf{u}}\in\{0,\ldots,p-1\}^n$, sets the shift $\mathbf{w}=\mathbf{B}\tilde{\mathbf{u}}\in L$, and calls the oracle on $\mathcal{E}(L_{\mathbf{z}},\mathbf{t}+\mathbf{w},M)$. If the answer is a vector of this lattice whose last coordinate is $\pm M$, the trial returns the candidate $\mathbf{t}\mp\mathbf{v}'$, where $\mathbf{v}'$ is the answer without its last coordinate. By the proof of Lemma 6.1, such an answer is $(\mathbf{b}\pm(\mathbf{t}+\mathbf{w}),\pm M)$ with

$\mathbf{b} \in L_{\mathbf{z}}$, so the candidate equals $-\mathbf{b}-\mathbf{w}$ and lies in L. The unwanted set $\mathcal{Y}$ depends only on L and $\mathbf{e}$, and not on the shift, which only randomizes the residue $\overline{\mathbf{c}+\mathbf{w}} = \bar{\mathbf{c}}+\mathbf{u}$ of the point that must remain. The Theorem 6.3 tackles all gaps with randomness.

Theorem 6.3. *For every $\gamma \geq 1$, which may be any function of n, and every $\varepsilon \in (0,1/2]$, there is a randomized reduction from $\mathsf{BDD}_{(1-\varepsilon)/(\sqrt{2}\gamma)}$ to uSVP_γ whose oracle inputs have dimension $n+1$. With p the least prime at least $\max(2N(n,\varepsilon), h(\varepsilon)+1)$, which is $\Theta(n/\varepsilon)$, each trial with uniform $\mathbf{z} \neq 0$ and $\mathbf{u}$ and the right guess of d succeeds with probability at least $1/(2p)$, and $\lceil 2pn \rceil$ trials for each of $G = 1+\lceil 35.01n \rceil$ guesses succeed with probability at least $1-e^{-n}$. The running time is polynomial in n, $1/\varepsilon$ and the input length, and it does not depend on γ.*

Proof (sketch). If $\mathbf{t} \in L$, the reduction outputs $\mathbf{t}$. Otherwise Babai's algorithm with an LLL-reduced basis [36, 37] finds $\mathbf{c}_0 \in L$ with $d \leq D = \|\mathbf{t}-\mathbf{c}_0\| \leq 2^n d$ [5, Lemma 2.7], and for $0 \leq i < G$ the guess M_i is a rational number in $[\sqrt{\varepsilon}D1.02^{-i}, 1.01\sqrt{\varepsilon}D1.02^{-i}]$; the largest i with $D1.02^{-i} \geq d$ gives $\sqrt{\varepsilon}d \leq M_i \leq 1.0302\sqrt{\varepsilon}d$. For that guess, let $\mathbf{c} \in L$ be a nearest vector and $\mathbf{e} = \mathbf{t}-\mathbf{c}$. A trial with uniform $\mathbf{z} \neq 0$ and $\mathbf{u}$ succeeds if $\langle \mathbf{z}, \bar{\mathbf{c}}+\mathbf{u} \rangle = 0$, which puts $\mathbf{c}+\mathbf{w}$ in $L_{\mathbf{z}}$, and $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle \neq 0$ for all $\mathbf{y} \in \mathcal{Y}$, which removes $\mathcal{Y}$ from $L_{\mathbf{z}}$. For fixed $\mathbf{z} \neq 0$ the first event has probability $1/p$ over $\mathbf{u}$, independently of $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle$, and each $\bar{\mathbf{y}}$ is nonzero because $\|\mathbf{y}\| < h(\varepsilon)\lambda < p\lambda$ by Lemma 6.2, so $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle = 0$ has probability at most $1/p$, and the union bound over $|\mathcal{Y}| \leq N(n,\varepsilon) \leq p/2$ vectors gives probability at least $1/p - N(n,\varepsilon)/p^2 \geq 1/(2p)$. On success, every nonzero $\mathbf{y} \in L_{\mathbf{z}}$ with $\|\mathbf{y}+m\mathbf{e}\| \leq r_0$ for some $1 \leq |m| \leq m_0$ would lie in $\mathcal{Y}$, which is disjoint from $L_{\mathbf{z}}$, and $r_0 < \lambda$ by Lemma 6.2, so Lemma 6.1 with $L' = L_{\mathbf{z}}$, $\mathbf{c}' = \mathbf{c}+\mathbf{w}$ and $\mathbf{t}' = \mathbf{t}+\mathbf{w}$ shows that the oracle input is a valid instance whose shortest vectors are $\pm(\mathbf{e}, M)$, which yield the candidate $\mathbf{c}$. The reduction outputs the candidate nearest to $\mathbf{t}$. The full proof is in the Appendix. □

The pencil family of Lemma 4.3 replaces the random choice of $(\mathbf{z}, \mathbf{u})$ by an explicit list. The Theorem 6.4 tackles all gaps without randomness, and Corollary 6.5 handles the precise radius.

Theorem 6.4. *For every $\gamma \geq 1$ and every $\varepsilon \in (0,1/2]$ with $1/\varepsilon \leq \mathrm{poly}(n)$, there is a deterministic polynomial-time reduction from $\mathsf{BDD}_{(1-\varepsilon)/(\sqrt{2}\gamma)}$ to uSVP_γ whose oracle inputs have dimension $n+1$. Let $N = N(n,\varepsilon)$ and let q be the least prime at least $\max(2N(n-1)+2n, h(\varepsilon)+1)$. The reduction makes at most $G(N(q-1)+2)q^3$ oracle calls, with G as in Theorem 6.3, and this number does not depend on γ.*

Proof (sketch). For each guess, the reduction runs one trial of the modulus q for each pair $(\mathbf{z}, \mathbf{u})$ with $\mathbf{z} \in \mathcal{Z}_{n,q}$ and $\mathbf{u}$ in a fixed set $U \subset \mathbb{F}_q^n$ of $N(q-1)+2$ distinct vectors. For the right guess, the set $\{0\} \cup \bigcup_{\mathbf{y} \in \mathcal{Y}} \mathbb{F}_q \bar{\mathbf{y}}$ has at most $1+N(q-1) < |U|$ elements, and $\mathbf{u} \mapsto \bar{\mathbf{c}}+\mathbf{u}$ is injective, so some $\mathbf{u} \in U$ makes $\mathbf{x} = \bar{\mathbf{c}}+\mathbf{u}$ nonzero and independent from every $\bar{\mathbf{y}}$ with $\mathbf{y} \in \mathcal{Y}$. Lemma 4.3 with $A = N$ gives $\mathbf{z} \in \mathcal{Z}_{n,q}$ with $\langle \mathbf{z}, \mathbf{x} \rangle = 0$ and $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle \neq 0$ for all $\mathbf{y} \in \mathcal{Y}$, and the argument of Theorem 6.3 shows that this trial returns a nearest vector. The full proof is in the Appendix. □

Corollary 6.5. *For every $\gamma \geq 1$, including super-polynomial γ, there is a deterministic reduction from $\mathsf{BDD}_{1/(\sqrt{2}\gamma)}$ to uSVP_γ that runs in time polynomial in n and*

the input length, independently of γ, and whose oracle inputs have dimension $n+1$ if $\gamma \geq \sqrt{2}$ and $n+2$ otherwise.

Proof. Proposition 3.5 with $c=1$ reduces $\mathsf{BDD}_{1/(\sqrt{2}\gamma)}$ in dimension n deterministically to polynomially many instances of $\mathsf{BDD}_{(1-1/n)/(\sqrt{2}\gamma)}$ in dimension $n'=n$ if $1/(\sqrt{2}\gamma) \leq 1/2$, that is $\gamma \geq \sqrt{2}$, and $n' \leq n+1$ otherwise. Each such instance is an instance of $\mathsf{BDD}_{(1-\varepsilon)/(\sqrt{2}\gamma)}$ in dimension n' with $\varepsilon = 1/n \leq 1/2$, which Theorem 6.4 solves deterministically in time polynomial in n, with oracle inputs of dimension $n'+1$. Neither step depends on γ. □

For $\gamma \geq 2^{n/2}$ both problems are solved by the LLL algorithm and Babai's algorithm [37, 36], so Corollary 6.5 is new for the range $\mathrm{poly}(n) \ll \gamma < 2^{n/2}$, and it answers the question of [6] on this range and on determinism. Together with $\mathsf{uSVP}_\gamma \leq \mathsf{BDD}_{1/\gamma}$ for polynomial γ [5], it shows that the two problems are equivalent up to the factor $\sqrt{2}$ in the gap under deterministic reductions for all polynomial γ.

7 Our TSE reduction

TSE combines the best known embedding with one oracle call and the sparsified embedding of Section 6. For $\gamma \geq 1$ let $a = \lfloor \gamma \rfloor$,

$$\gamma_1 = \frac{2\gamma^2 + 2a(a+1)}{2a+1} \quad \text{and} \quad k_\gamma = \frac{\sqrt{4\gamma^2 - 1}}{2a+1},$$

which are the radius parameter and the scaling of the embedding analyzed in the appendix of [6]. We call it the balanced embedding, because with $M = k_\gamma d$ the lower bounds in the proof of Lemma 7.1 for the two critical multiples a and $a+1$ of the offset coincide. Since $(\gamma - a)(\gamma - a - 1) \leq 0$, we have $\gamma_1 \leq 2\gamma$, with equality precisely for integer γ, so its radius $1/\gamma_1$ is at least the radius $1/(2\gamma)$ of [5]. As in the analyses of [5, 6], this section assumes that the distance $d = \mathrm{dist}(\mathbf{t}, L)$ is known; Theorem 6.3 shows how to remove this assumption with $O(n)$ guesses that are the same for all compared reductions, and our experiments give the same value of d to all of them. The Lemma 7.1 goes through the embedding with one call.

Lemma 7.1. *Let $\gamma > 1$, let $\mathbf{c} \in L$ be a nearest vector to $\mathbf{t}$ with $0 < d = \|\mathbf{t} - \mathbf{c}\| \leq \lambda/\gamma_1$, and let $M = k_\gamma d$. Then $\pm(\mathbf{t} - \mathbf{c}, M)$ are the only shortest nonzero vectors of $\mathcal{E}(L, \mathbf{t}, M)$ and $\lambda_2 \geq \gamma \lambda_1$ holds for this lattice.*

Proof. As in the proof of Lemma 6.1, every vector that is not parallel to $(\mathbf{e}, M)$ with $\mathbf{e} = \mathbf{t} - \mathbf{c}$ equals $(\mathbf{y} + m\mathbf{e}, mM)$ with $\mathbf{y} \in L \setminus \{0\}$, and by symmetry we may take $m \geq 0$. Let $s = \gamma^2 + a^2 + a$, so that $\gamma_1 = 2s/(2a+1)$, $1 + k_\gamma^2 = 4s/(2a+1)^2$ and the target value is $\gamma^2(1 + k_\gamma^2)d^2 = 4\gamma^2 s d^2/(2a+1)^2$. For $m = 0$, $\|\mathbf{y}\|^2 \geq \lambda^2 \geq \gamma_1^2 d^2 = 4s^2 d^2/(2a+1)^2$, which is at least the target value because $s \geq \gamma^2$. For $1 \leq m < \gamma_1$, $\|\mathbf{y} + m\mathbf{e}\| \geq \lambda - md \geq (\gamma_1 - m)d$, and the quadratic $f(m) = (\gamma_1 - m)^2 + m^2 k_\gamma^2$ has its minimum at $m = \gamma_1/(1 + k_\gamma^2) = a + 1/2$, so over the integers it is minimal at $m = a$ and $m = a + 1$, where $f = \gamma_1^2 k_\gamma^2/(1 + k_\gamma^2) + (1 + k_\gamma^2)/4 = 4\gamma^2 s/(2a+1)^2$. For

Algorithm 1 TSE with a uSVP_γ oracle $\mathcal{O}$, slack ε and budget T

function TSE($\mathbf{B}$,$\mathbf{t}$,d)
  $\hat{\mathbf{c}} \leftarrow$ CAND($\mathcal{O}(\mathcal{E}(L,\mathbf{t},k_\gamma d))$,$\mathbf{t}$,$k_\gamma d$)
  if $\hat{\mathbf{c}} \neq \perp$ **and** $\|\mathbf{t}-\hat{\mathbf{c}}\| = d$ **then return** $\hat{\mathbf{c}}$
  end if
  $p \leftarrow$ least prime $\geq 2N(n,\varepsilon)$; $M \leftarrow \sqrt{\varepsilon} d$
  for $j = 1,\ldots,T$ **do**
    $\mathbf{z} \leftarrow \mathbb{F}_p^n \setminus \{0\}$, $\mathbf{u} \leftarrow \mathbb{F}_p^n$ uniform; $\mathbf{w} \leftarrow \mathbf{B}\tilde{\mathbf{u}}$
    $\mathbf{B}_\mathbf{z} \leftarrow$ basis of $L_\mathbf{z}$ from Lemma 3.4
    $\hat{\mathbf{c}} \leftarrow$ CAND($\mathcal{O}(\mathcal{E}(L_\mathbf{z},\mathbf{t}+\mathbf{w},M))$,$\mathbf{t}$,$M$)
    if $\hat{\mathbf{c}} \neq \perp$ **and** $\|\mathbf{t}-\hat{\mathbf{c}}\| = d$ **then return** $\hat{\mathbf{c}}$
    end if
  end for
  return $\perp$
end function

$m \geq \gamma_1$, the norm is at least $m^2 k_\gamma^2 d^2 \geq \gamma_1^2 k_\gamma^2 d^2$, and $\gamma_1^2 k_\gamma^2 \geq \gamma^2(1+k_\gamma^2)$ is equivalent to $4\gamma^4 - 2\gamma^2 - a^2 - a \geq 0$, which holds because $\gamma \geq a \geq 1$. Hence every vector that is not parallel to $(\mathbf{e},M)$ has norm at least $\gamma\|(\mathbf{e},M)\| > \|(\mathbf{e},M)\|$. □

Algorithm 1 states TSE for a basis $\mathbf{B}$ of L, a target $\mathbf{t}$ and its distance d to L. The helper CAND receives an oracle answer, checks that it is a vector of the queried embedding lattice with last coordinate $\pm M$, and in that case returns the candidate $\mathbf{t} \mp \mathbf{v}'$ described in Section 6, where $\mathbf{v}'$ is the answer without its last coordinate; otherwise it returns the failure symbol $\perp$. TSE first asks the oracle for the shortest vector of the balanced embedding with scaling $k_\gamma d$ and stops if the candidate is at distance d. Otherwise it fixes the least prime p above $2N(n,\varepsilon)$ and the tube-calibrated scaling $M = \sqrt{\varepsilon} d$, and it runs up to T trials. Each trial draws a nonzero sparsification vector $\mathbf{z}$ and a residue $\mathbf{u}$, forms the shift $\mathbf{w}$ from the lift $\tilde{\mathbf{u}} \in \{0,\ldots,p-1\}^n$, builds the index-$p$ sublattice $L_\mathbf{z}$, and queries the embedding of $L_\mathbf{z}$ with the shifted target. The loop stops at the first candidate at distance d. In the implementation, M is rounded to an integer after scaling the lattice by 2^{12}, which changes it by a relative error below $2 \cdot 10^{-4}$; the proof of Lemma 6.2 absorbs this error, since its constant $\sqrt{2}/\sqrt{0.918} < 1.477$ is below 1.5, and the promise oracle of Section 9 checks the resulting instances precisely. The Theorem 7.2 goes through the guarantee of our TSE.

Theorem 7.2. *Let $\gamma > 1$, $\varepsilon \in (0, 1/2]$, and let $(\mathbf{B}, \mathbf{t})$ be an instance with $d = \mathrm{dist}(\mathbf{t},L) > 0$. If $d \leq \lambda/\gamma_1$, then TSE returns a nearest vector after one oracle call. If $d \leq (1-\varepsilon)\lambda/(\sqrt{2}\gamma)$, then each trial succeeds with probability at least $1/(2p)$ with p the least prime at least $2N(n,\varepsilon)$, so the expected number of oracle calls is at most $1+2p$, and with $T = \lceil 2pn \rceil$ the failure probability is at most e^{-n}. All oracle inputs have dimension $n+1$ and entries of bit length at most $\beta + \lceil \log_2 p \rceil + \lceil \log_2(n+1) \rceil + 1$, where β bounds the bit length of the entries of $\mathbf{B}$, $\mathbf{t}$ and M.*

Proof (sketch). The first claim is Lemma 7.1, since the promise holds and the only shortest vectors are $\pm(\mathbf{t}-\mathbf{c},M)$, whose candidate is $\mathbf{c}$. The second is the trial analysis of Theorem 6.3 with the precise value $M = \sqrt{\varepsilon} d$, which satisfies $\sqrt{\varepsilon} d \leq M \leq 1.04\sqrt{\varepsilon} d$. The expected number of trials of a geometric variable with success probability at

least $1/(2p)$ is at most $2p$, and $(1-1/(2p))^{2pn} \le e^{-n}$. The bit lengths follow from Lemma 3.4 and from $\|\mathbf{w}\|_\infty \le n(p-1)\max|B_{ij}|$. The full proof is in the Appendix. □

The Corollary 7.3 handles the comparison of guarantees.

Corollary 7.3. *Let $n \ge 13$, $\gamma > 1$ and $\varepsilon = 1/n$. On each instance on which the embedding of [5] or the balanced embedding of [6] is guaranteed to succeed, TSE succeeds with the same single oracle call. On each instance with $d \le (1-1/n)\lambda/(\sqrt{2}\gamma)$, on which the reduction of [6] is guaranteed to succeed when $\gamma \le \mathrm{poly}(n)$, TSE uses a prime $p \le p_B$, where p_B is the least prime larger than $4\gamma n^2$ used by [6]. If moreover $\gamma \ge 1.2$ or $n \ge 93$, then $p < p_B$, and the bound $1+2p$ on the expected number of calls of TSE is smaller than $2p_B$, which is below the bound implied by the success probability $1/p_B - N_B/p_B^2 - N_B/p_B^{n-1}$ with $N_B \le p_B/2$ of [6, Lemma 13].*

Proof. The guarantee of [5] holds for $d \le \lambda/(2\gamma)$ and that of the balanced embedding for $d \le \lambda/\gamma_1$, and $\lambda/(2\gamma) \le \lambda/\gamma_1$ because $\gamma_1 \le 2\gamma$. In both cases Theorem 7.2 shows that TSE stops after its first call. For the second claim, $N = N(n,1/n) = n(\lfloor 1.5(n+\sqrt{n})\rfloor + 1)$ gives $2N \le 3n^2 + 3n^{3/2} + 2n < 4n^2$ for $n \ge 13$. Since p_B is a prime larger than $4\gamma n^2 > 2N$ and p is the least prime at least $2N$, we get $p \le p_B$. By the theorem of Nagura [38], every interval $[x,1.2x]$ with $x \ge 25$ contains a prime, so there is a prime in $[2N,2.4N]$. If $\gamma \ge 1.2$, then $2.4N \le 1.2 \cdot 4n^2 \le 4\gamma n^2 < p_B$, and if $n \ge 93$, then $2.4N \le 3.6n^2 + 3.6n^{3/2} + 2.4n < 4n^2 < p_B$. In both cases $p < p_B$, hence $p \le p_B - 1$ and $1 + 2p < 2p_B$. □

The comparison with the reduction of [6] is only possible when $\gamma \le \mathrm{poly}(n)$, since for larger γ the prime p_B is super-polynomial, while TSE keeps the prime $p = \Theta(n^2)$ for $\varepsilon = 1/n$ and $p = \Theta(n)$ for constant ε.

8 Computational costs analysis

We compare the reductions from BDD to uSVP_γ in the oracle model, in which one call is charged one unit and the oracle input is an $(n+1)$-dimensional basis. Let β be the bit length of the entries of the input basis and target, and let $T_{\mathcal{O}}(n+1,b)$ denote the time of one oracle call on entries of bit length b. The time of a reduction is its number of calls times $T_{\mathcal{O}}$ plus its own arithmetic, and its space is the size of the largest oracle input together with its own working memory. We derived all entries of Table 2 from the algorithms as stated in the cited works and in Section 7, as follows.

Radius and range. The embedding of [5] uses $M = d$ and is guaranteed for $\alpha = d/\lambda \le 1/(2\gamma)$, and the balanced embedding uses $M = k_\gamma d$ and is guaranteed for $\alpha \le 1/\gamma_1$ by Lemma 7.1. The embedding with $M = d/\sqrt{n}$, which is the usual choice $M \approx \|\mathbf{e}\|/\sqrt{n}$ of the primal attack [8], satisfies the condition of Lemma 6.1 without sparsification only if the multiples $m \le \gamma\sqrt{n+1}$ avoid the lattice, which the argument of Lemma 7.1 guarantees for $\alpha \le 1/(\gamma(\sqrt{n+1}+\sqrt{1+1/n})) = \Theta(1/(\gamma\sqrt{n}))$. The reduction of [6] is guaranteed for $\alpha \le (1-1/n)/(\sqrt{2}\gamma)$ and $\gamma \le \mathrm{poly}(n)$, and TSE for $\alpha \le (1-\varepsilon)/(\sqrt{2}\gamma)$ and every γ.

Oracle calls. Table 2 counts calls when d is known, which is the setting of the analyses of [5, 6] and of our experiments. The three embeddings without

Table 2: Complexity of the reductions from BDD_α to uSVP_γ in the oracle model, for a lattice of dimension n with entries of bit length β. The coins are counted per call, the arithmetic is that of the reduction per call, the last three rows are Theorem 7.2, Corollary 7.3 and Theorem 6.4, and $q=\Theta(n^2/\varepsilon)$ is the prime of the last one.

Reduction	Guaranteed radius α	Range of γ	Coins	Expected calls	Arithmetic	Bits per oracle entry
$M=d/\sqrt{n}$ [4, 8]	$\Theta(1/(\gamma\sqrt{n}))$	all	none	1	$O(n^2)$	$\beta+O(1)$
$M=d$ [5]	$1/(2\gamma)$	all	none	1	$O(n^2)$	$\beta+O(1)$
Balanced [6]	$1/\gamma_1$	all	none	1	$O(n^2)$	$\beta+O(1)$
Sparsified [6]	$(1-1/n)/(\sqrt{2}\gamma)$	poly(n)	$2n\log_2 p_B$	$\approx 2p_B > 8\gamma n^2$	$\Theta(n^3)$	$\beta+\log_2 p_B+\log_2(n^2\lambda)$
TSE, ε constant	$(1-\varepsilon)/(\sqrt{2}\gamma)$	all	$2n\log_2 p$	$\le 1+2p$, $p=\Theta(n/\varepsilon)$	$O(n^2)$	$\beta+\log_2 p+\log_2 n+1$
TSE, $\varepsilon=1/n$	$(1-1/n)/(\sqrt{2}\gamma)$	all	$2n\log_2 p$	$\le 1+2p$, $p<4n^2$	$O(n^2)$	$\beta+\log_2 p+\log_2 n+1$
Deterministic	$(1-\varepsilon)/(\sqrt{2}\gamma)$	all	none	$\le (N(q-1)+2)q^3$	$O(n^2)$	$\beta+\log_2 q+\log_2 n+1$

sparsification make one call. The reduction of [6] succeeds in each trial with probability $1/p_B - N_B/p_B^2 - N_B/p_B^{n-1} \ge 1/(2p_B) - N_B/p_B^{n-1}$ with $N_B \le 2\gamma n^2$, so its expected number of trials is about $2p_B > 8\gamma n^2$. TSE makes one call and then at most $2p$ expected trials with $p \le 4N(n,\varepsilon)$ by Bertrand's postulate. Removing the knowledge of d multiplies these counts by the number of guesses, which is $\Theta(n^2)$ for the embeddings of [5, 6] that guess d within a factor $1\pm 1/n$, $\Theta(n^4)$ for the sparsified reduction of [6], which guesses both d and λ with this precision, and $O(n)$ for the sparsified phase of TSE, which only guesses M within the factor 1.02 (Theorem 6.3).

Arithmetic and space. The sparsified basis of Lemma 3.4 costs $O(n^2)$ operations, and the shift $\mathbf{w}=\mathbf{B}\tilde{\mathbf{u}}$ costs $O(n^2)$ operations. The reduction of [6] computes the basis of $L_{\mathbf{z}}$ through the kernel of the linear form and a Hermite normal form of an $n\times 2n$ integer matrix [6, Lemma 9], which costs $\Theta(n^3)$ operations with algorithms based on elimination, and it chooses a lift $\mathbf{w}$ with $\|\mathbf{t}+\mathbf{w}\| \ge (n+1)\lambda/\sqrt{2}$, which adds $\log_2(n\lambda)$ bits to the last column. The entries of the sparsified basis grow by at most $\log_2 p$ bits in both reductions by Lemma 3.4, so TSE saves $\log_2(p_B/p)=\log_2(\gamma n\varepsilon)+O(1)$ bits per entry. All reductions store one $(n+1)\times(n+1)$ basis at a time besides the input.

Table 2 lists, for each reduction, the radius α for which it is proven, the range of the gap γ, the number of random bits per call, the expected number of oracle calls when d is known, the arithmetic of the reduction per call and the bit length of the entries of the oracle input. The first three rows are embeddings with one call, which differ only in the scaling M of the last coordinate, and the fourth row is the sparsified reduction of [6] with its prime $p_B > 4\gamma n^2$. The last three rows are our reductions, where $N(n,\varepsilon)=O(n/\varepsilon)$ is the tube count of Lemma 6.2 and q is the prime of Theorem 6.4. For every $\gamma \le \text{poly}(n)$, the reduction of [6] and TSE with $\varepsilon=1/n$ share the radius, while TSE has a smaller prime by Corollary 7.3, fewer guesses of the unknown distances, less arithmetic per call and fewer bits per oracle entry; for super-polynomial γ, only TSE and the deterministic reduction reach the radius $1/(\sqrt{2}\gamma)$ up to the slack.

Figure 3 shows the proven normalized radius $\gamma\alpha$ for $1 \le \gamma \le 6$, beyond which all curves are flat. For one call with $M=kd$, the argument of Lemma 7.1 proves precisely the radius γ/β_k with $\beta_k = \max_m(m+(\gamma^2(1+k^2)-m^2k^2)^{1/2})$ over the integers $0 \le m < \gamma\sqrt{1+k^2}/k$, which offsets along a shortest vector attain. The balanced embedding reaches $\gamma/\gamma_1 \le 3\sqrt{2}/8 \approx 0.530$, and the maximum over 599 scalings coincides with it, as Theorem 8.3 predicts. The scaling $M=d$, for which [5]

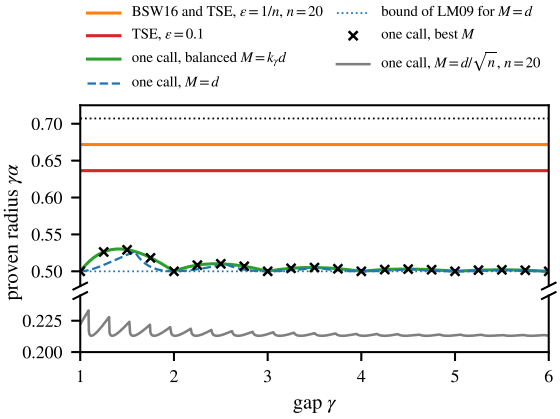


Fig. 3: Proven normalized radius $\gamma\alpha$ of the reductions from BDD_α to uSVP_γ as a function of the gap. The curves of one call are computed precisely for each scaling M with the argument of Lemma 7.1, the crosses are the maximum over all scalings, and the lower panel shows the scaling $M = d/\sqrt{n}$ on its own axis. The dotted line is the limit $1/\sqrt{2}$.

prove $1/(2\gamma)$, reaches up to 0.527 between integers, and $M = d/\sqrt{n}$ stays between $\sqrt{n}/(n+1) \approx 0.213$ and 0.234 for $n = 20$. The reduction of [6] and TSE with $\varepsilon = 1/n$ share the value $(1-1/n)/\sqrt{2}$, but the line of [6] is only valid while $p_B > 4\gamma n^2$ is polynomial, and TSE with $\varepsilon = 0.1$ gives up about 5% of it with a prime linear in n.

Refinements and lower bounds. The count of Lemma 6.2 is linear in n because Lemma 4.1 allows n points per slice, and a Gram argument on shorter slices removes this dependence for a constant slack. For $\theta \in (0,1)$ and $\eta \in (0,1)$ let $\kappa_\theta(\eta) = \lceil (1-\eta)/((1-\theta^2)\eta) \rceil$. The Lemma 8.1 handles slices of any length.

Lemma 8.1. *With the notation of Lemma 4.2 and $\theta \in (0,1)$, $|L \cap \mathcal{T}(S,r)| \leq \left(\lfloor (\ell+2r)/(\theta\lambda\sqrt{\eta}) \rfloor + 1\right) \min(n, \kappa_\theta(\eta))$.*

Proof. Cover $[a-r, b+r]$ by half-open intervals of length $\delta = \theta\lambda\sqrt{\eta}$ as in the proof of Lemma 4.2, and let $\mathbf{x}_1, \ldots, \mathbf{x}_k$ be the lattice points of the tube whose values σ lie in one interval, with projections $\pi_i = \pi(\mathbf{x}_i)$. Lemma 4.1 gives $k \leq n$ as in that proof, since the interval is shorter than $\lambda\sqrt{\eta}$. For $i \neq j$, $\|\pi_i - \pi_j\|^2 > \lambda^2 - \delta^2 = \lambda^2(1-\theta^2\eta)$ and $\|\pi_i\|^2, \|\pi_j\|^2 \leq r^2 = \lambda^2(1-\eta)/2$, so $\langle \pi_i, \pi_j \rangle < -\lambda^2\eta(1-\theta^2)/2$. If $k \geq 2$, then $0 \leq \|\sum_i \pi_i\|^2 < kr^2 - k(k-1)\lambda^2\eta(1-\theta^2)/2$, hence $k - 1 < (1-\eta)/((1-\theta^2)\eta)$ and $k \leq \kappa_\theta(\eta)$. □

With $\theta = 1/\sqrt{3}$ and the parameters of Lemma 6.2, which give $\eta \geq 0.918\varepsilon$ and $(\ell + 2r_0)/(\lambda\sqrt{\eta}) < 1.5(1/\varepsilon + 1/\sqrt{\varepsilon})$, Lemma 8.1 bounds the unwanted set by $N'(\varepsilon) = (\lfloor 2.6(1/\varepsilon + 1/\sqrt{\varepsilon}) \rfloor + 1)\lceil 1.64/\varepsilon \rceil$, because $1.5\sqrt{3} < 2.6$ and $1.5/0.918 < 1.64$, and this bound does not depend on n. The Corollary 8.2 tackles a prime independent of n.

Corollary 8.2. *Theorems 6.3, 6.4 and 7.2 hold with $N(n, \varepsilon)$ replaced by $N_0 = \min(N(n, \varepsilon), N'(\varepsilon))$. For a constant slack ε, TSE with the least prime*

$p \geq \max(2N_0, h(\varepsilon)+1)$ therefore makes $O(1/\varepsilon^2)$ oracle calls in expectation when d is known, for every n and every γ.

Proof. The proofs of the three theorems use the count only through $|\mathcal{Y}| \leq N(n,\varepsilon)$, and $|\mathcal{Y}| \leq N'(\varepsilon)$ by Lemma 8.1 with $\theta = 1/\sqrt{3}$. For constant ε, Bertrand's postulate gives $p \leq 2\max(2N'(\varepsilon), h(\varepsilon)+1) = O(1/\varepsilon^2)$, and Theorem 7.2 bounds the expected number of calls by $1+2p$. □

The next two results are lower bounds on the offsets along a shortest vector that Section 9 calls hard. The Theorem 8.3 tackles the limit of one call.

Theorem 8.3. *Let $\gamma > 1$, $1/\gamma_1 < \alpha < 1/2$, $\lambda > 0$, $L = \lambda\mathbb{Z}^n$ and $\mathbf{t} = \alpha\lambda\mathbf{e}_1$, whose nearest lattice vector is 0. For every $M > 0$, $\mathcal{E}(L,\mathbf{t},M)$ is not an instance of uSVP_γ whose shortest vectors are $\pm(\mathbf{t},M)$.*

Proof (sketch). In units of $\alpha\lambda$, with $\beta = 1/\alpha$, $k = M/(\alpha\lambda)$ and $a = \lfloor\gamma\rfloor$, the embedding contains $(\beta,0)$, $(a-\beta,ak)$ and $(a+1-\beta,(a+1)k)$ in the plane of $\mathbf{e}_1$ and the last coordinate, none parallel to $(1,k)$. If all three had squared norm at least $\gamma^2(1+k^2)$, the first would give $\beta > \gamma$, and the other two would give an upper and a lower bound on k^2 that are compatible only if $\beta \geq \gamma_1$, because the only positive root of the resulting quadratic in $\beta - a$ is $\gamma_1 - a$. The full proof is in the Appendix. □

Since every vector $\mathbf{w} \in L$ gives $\mathcal{E}(L,\mathbf{t}+\mathbf{w},M) = \mathcal{E}(L,\mathbf{t},M)$, Theorem 8.3 and Lemma 7.1 show that $1/\gamma_1$ is the largest radius for which Kannan's embedding with one call and any scaling yields an instance whose shortest vectors reveal the nearest point, so sparsification or more calls are necessary beyond it.

Figure 5 tests Theorem 8.3 with the promise oracle of Section 9. For each gap $\gamma \in \{1.5,2,3\}$, three random q-ary lattices of dimension 20 and each of 25 normalized radii $\gamma\alpha \in [0.40,0.64]$, we build the target along a shortest vector and call the oracle once for each of 45 scalings k between 0.03 and 5 on a logarithmic grid. The largest normalized radius at which some scaling succeeds is 0.52 for $\gamma = 1.5$, where the balanced radius is 0.529, and 0.50 for $\gamma = 2$ and $\gamma = 3$, where it is 0.5; no scaling succeeds at any larger radius. For $\gamma = 3$ the column $\gamma\alpha = 0.50$ lies on the balanced radius itself, and the rounding of the offset to integers puts it on either side of the limit, which explains the gray cell. The set of successful scalings is an interval that shrinks toward k_γ as the radius grows, as the analysis of the limit $\beta = \gamma_1$ predicts, and scalings below 0.5, which include the scaling $M = d/\sqrt{n}$ of the primal attack and the scaling $M = d/(n-1)$ of [6] for $n = 20$, fail already at the smallest radius $\gamma\alpha = 0.40$ of the grid, where the successful scalings form the interval [0.50,2.0] for all three gaps. The Theorem 8.4 tackles the smallest usable prime.

Theorem 8.4. *Let $\mathbf{s}$ be a shortest vector of L, $\lambda = \|\mathbf{s}\|$, $\mathbf{c} \in L$, $0 < \alpha < 1/2$, $\mathbf{t} = \mathbf{c} + \alpha\mathbf{s}$, $d = \alpha\lambda$ and $M = kd$ with $k > 0$. If $p < \alpha(\sqrt{\gamma^2 + (\gamma^2 - 1/4)/k^2} - 1/2)$, then for every nonzero $\mathbf{z} \in \mathbb{F}_p^n$ and $\mathbf{w} \in L$, $\mathcal{E}(L_\mathbf{z},\mathbf{t}+\mathbf{w},M)$ is not an instance of uSVP_γ whose shortest vectors are $\pm(\mathbf{t}-\mathbf{c},M)$.*

Proof (sketch). If $\mathbf{c}+\mathbf{w} \in L_\mathbf{z}$, then with m the integer nearest to p/α the embedding contains $(m\alpha - p)\mathbf{s}$ padded with mM, which comes from $-p\mathbf{s} \in pL \subseteq L_\mathbf{z}$, is not parallel to $(\mathbf{t}-\mathbf{c},M)$ and has squared norm at most $d^2/4 + (p/\alpha + 1/2)^2 M^2 < \gamma^2(d^2 + M^2)$. The full proof is in the Appendix. □

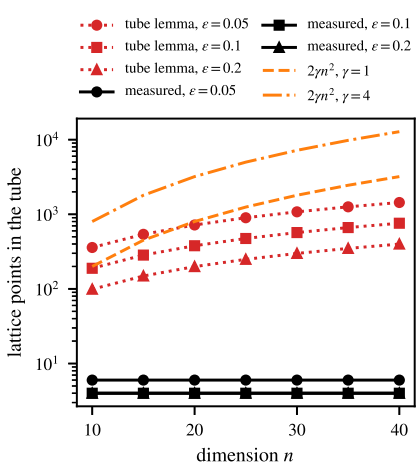


Fig. 4: Lattice points in the tube of Lemma 6.2 for random q-ary lattices and for $\mathbb{Z}^n$, with the largest count over offsets along a shortest vector and random offsets, against the tube lemma and the count $2\gamma n^2$ of [6].

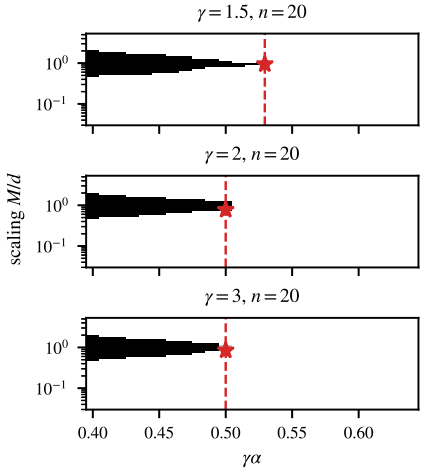


Fig. 5: Fraction of three lattices of dimension $n = 20$ on which one call with the embedding $\mathcal{E}(L, \mathbf{t}, kd)$ returns the nearest vector, for targets along a shortest vector, as a function of the normalized radius $\gamma\alpha$ and the scaling $k = M/d$; black means success on all three lattices. The dashed line is the balanced radius γ/γ_1 and the star is the balanced scaling k_γ.

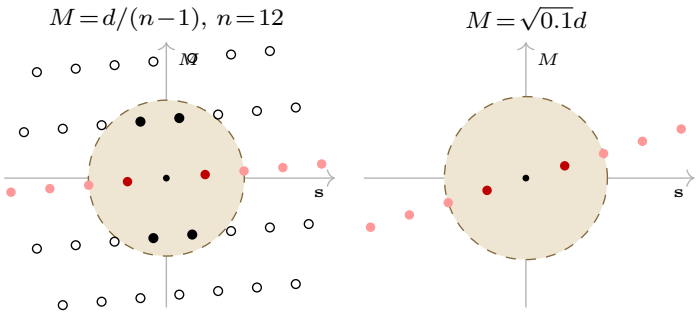


Fig. 6: The plane of $\mathbf{s}$ and the last coordinate in the sparsified embedding for $\alpha = 0.3$, $\gamma = 2$ and $p = 5$, in units of λ_1, when the sublattice keeps the nearest point and removes $\mathbf{s}$. Red points are multiples of $(\mathbf{e}, M)$, the dashed circle has radius $\gamma\|(\mathbf{e}, M)\|$, and black points come from $p\mathbf{s}$; filled black points lie inside the circle.

For the scaling $M = d/(n-1)$ of [6] and $\alpha \geq 0.9/(\sqrt{2}\gamma)$, Theorem 8.4 excludes every prime below $0.55(n-1) - 0.32$ on these instances, even though their unwanted vectors lie on one line, while for the scaling $M = \sqrt{\varepsilon}d$ of TSE the threshold is below $h(\varepsilon)$, above which Theorem 6.3 applies; Section 10 turns this gap into an adaptive variant of TSE.

Figure 6 illustrates Theorem 8.4. The embedding restricted to this plane contains the points $((pj+\alpha m)\lambda, mM)$ for integers j and m. With the scaling $M=d/11$ of [6] for $n=12$, the points with $j=\pm 1$ and $m=\mp 16, \mp 17$, whose first coordinates ± 0.1 and ± 0.2 are small because $m\alpha$ is near p, have heights of only about 0.45 and lie inside the circle of radius 0.60. The input therefore violates the promise however the other unwanted vectors are removed, which is why all primes up to 5 fail in Figure 10 for $n=12$. With the scaling $\sqrt{0.1}d$ of TSE, the same points have height 1.6, far outside the circle of radius 0.63, and the window contains only multiples of $(\mathbf{e}, M)$.

9 Experimental evaluation

Setup. All of our practical findings are available at the repository of the https://github.com/ethanwaker9/work-45. We conducted our experiments on an Apple MacBook Air (2022) with the M2 chip, A15 Bionic (64-bit ARM-based), an 8-core CPU featuring a base clock speed of 3.49 GHz, and 16GB of memory, Tahoe OS 26.2 build 25C56, where we implemented all reductions in Python 3.12 on top of fplll 5.5.0 and its Python interface fpylll 0.6.4 [39], with seven worker processes that each run one reduction at a time. All reductions call the same promise oracle for uSVP_γ. The oracle LLL-reduces its input, enumerates the $2\lceil\gamma\rceil+3$ shortest vectors within radius $\gamma\|\mathbf{b}_1\|$, and decides with precise integer arithmetic whether a vector that is not parallel to the shortest one is shorter than γ times the minimum. If the promise holds it returns a shortest vector, and otherwise it returns a failure symbol, which is the worst behavior that the definition of a promise problem allows; a reduction therefore only succeeds through valid queries. We implemented the embedding with $M = d/\sqrt{n}$ used by the primal attack [8], the embedding of [5] with $M = d$, the balanced embedding of [6] with $M = k_\gamma d$, the sparsified reduction of [6] with p_B the least prime above $4\gamma n^2$, $M=d/(n-1)$ and a shift with $\|\mathbf{t}+\mathbf{w}\| \geq (n+1)\lambda/\sqrt{2}$, and TSE with $\varepsilon=0.1$. The sparsified bases of both sparsified reductions are computed with Lemma 3.4, which is cheaper than the Hermite normal form of [6] and therefore only helps that reduction. All reductions receive the distance d, stop at the first candidate at distance d, and may use up to $10p$ trials.

Instances. For each dimension $n \in \{12,16,20,24,28\}$ and each of five seeds, we generate a random q-ary lattice with fpylll and LLL-reduce it, and we scale it by 2^{12} so that the offsets have the prescribed norm up to a relative error below 10^{-4}. A target is a random lattice point plus an offset of norm $\alpha\lambda_1$, either in a uniformly random direction or along a shortest vector $\mathbf{s}$; the second family realizes the worst case of the embeddings with one call, since multiples of the offset then approach the lattice points $j\mathbf{s}$. For each gap $\gamma \in \{1.25,1.5,2,3\}$, the radius α takes one value in each of four regions, where R1 is $0.9/(2\gamma)$, below the radius of [5]; R2 lies between $1/(2\gamma)$ and $1/\gamma_1$, which is empty for integer γ; and R3a and R3b lie at one quarter and three quarters of the way from $1/\gamma_1$ to $0.9/(\sqrt{2}\gamma)$, where only the sparsified reductions have guarantees. This gives 700 instances, each solved by each of the five reductions.

Success by region. All five reductions return a nearest vector on every instance in which they succeed, since each candidate is checked against the distance d. Table 1 shows that TSE and the reduction of [6] solve all 700 instances, the balanced embed-

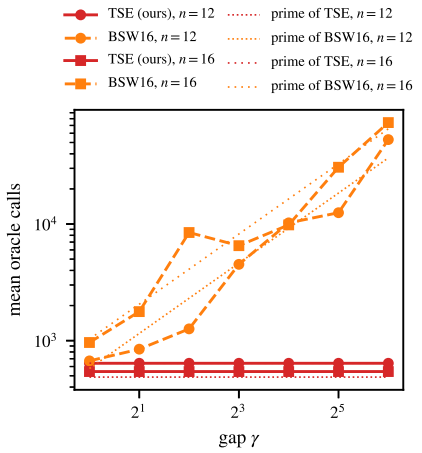


Fig. 7: Mean oracle calls over ten offsets along a shortest vector beyond the balanced radius as a function of the gap γ, for $n=12$ and $n=16$, with the primes of the two reductions as dotted lines.

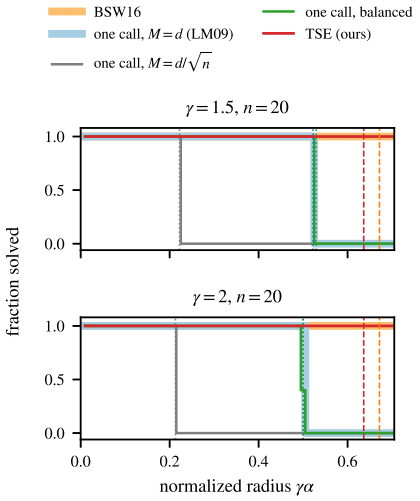


Fig. 8: Fraction of ten lattices of dimension $n=20$ solved at each normalized radius $\gamma\alpha$ on a grid of step 0.01, with offsets along a shortest vector. The dotted lines are the precise radii of one call from Figure 3, and the dashed lines are the proven radii of TSE with $\varepsilon=0.1$ and of [6]. Thick light lines lie below thin lines that coincide with them.

ding solves 500, the embedding of [5] solves 475 and the embedding with $M=d/\sqrt{n}$ solves 350. On each instance that one of the embeddings with one call solves, TSE also succeeds with one call, since its first call is the balanced call, and the paired ratios in the last column of Table 1 are 1.00 for the calls and at most 1.08 for the time and the memory, which is the variation between two runs of the same computation. On the instances that the reduction of [6] solves, which are all of them, TSE uses 16.2 times fewer oracle calls, 21.5 times less time and 1.44 times less memory on average.

Table 3 splits the results by region and by the direction of the offset, where a short offset points along a shortest vector; the normalized radius $\gamma\alpha$ in the caption is the mean over the gaps, and its proven limit is $1/\sqrt{2}\approx0.707$. With a random direction, all embeddings succeed in each region, because the multiples of a random offset stay far from the lattice. With an offset along a shortest vector, the embeddings fail once the radius is clearly beyond their proven radius, since the embedding with $M=d/\sqrt{n}$ fails in each region, the embedding of [5] still solves half of R2, which lies beyond its proven radius $1/(2\gamma)$, and the balanced embedding fails on all of R3a and R3b. Only the two sparsified reductions solve these 200 hard instances, where TSE needs 689.6 calls on average against 3460.5. The calls of both sparsified reductions coincide across regions, because the lattice, the nearest point and the random choices of each instance are the same in all regions, and in each region the first trial whose sublattice keeps the shifted

Table 3: Instances solved by each reduction per region of the radius and per direction of the offset, out of 100, or out of 50 for R2, and mean oracle calls of the two sparsified reductions. The mean normalized radius $\gamma\alpha$ is 0.450 in R1, 0.514 in R2, 0.545 in R3a and 0.606 in R3b.

		Solved					Mean calls	
Region	Offset	$d/\sqrt{n}$	[5]	Bal.	[6]	TSE	[6]	TSE
R1	short	0	100	100	100	100	3460.5	1.0
	random	100	100	100	100	100	3460.5	1.0
R2	short	0	25	50	50	50	1631.3	1.0
	random	50	50	50	50	50	1631.3	1.0
R3a	short	0	0	0	100	100	3460.5	689.6
	random	100	100	100	100	100	3460.5	1.0
R3b	short	0	0	0	100	100	3460.5	689.6
	random	100	100	100	100	100	3460.5	1.0

nearest point also removes all unwanted vectors; the regions differ in the number of calls only through the gaps that they contain, since R2 is empty for integer gaps.

Success along the radius. Figure 8 refines the regions with the 70 radii $\gamma\alpha \in \{0.01,...,0.70\}$, for $n = 20$, $\gamma \in \{1.5,2\}$ and ten lattices with an offset along a shortest vector per radius. Each embedding with one call solves all instances up to its precise radius from Figure 3 and none beyond it, to the resolution of the grid. For $\gamma = 1.5$ the transitions lie between 0.22 and 0.23 around the radius 0.222 of $M = d/\sqrt{n}$, and between 0.52 and 0.53 around the radii 0.523 of $M = d$ and 0.529 of the balanced embedding. For $\gamma = 2$ they lie between 0.21 and 0.22 around 0.213, and between 0.50 and 0.51 around the common radius 0.5, at which the balanced embedding solves 4 of 10 instances. The sparsified reductions solve all instances, also beyond their proven radii 0.636 and 0.672. TSE makes one call where the balanced embedding succeeds and 836.3 and 819.1 calls on average elsewhere, close to its prime 809, and 215.8 and 241.7 over all radii for $\gamma = 1.5$ and $\gamma = 2$, against 2457.5 and 3312.4 for [6].

Oracle calls and bit lengths. Figure 9 shows the cost of the two sparsified reductions on the 200 hard instances as a function of the dimension. The prime of TSE is 487,641,809,967 and 1123 for $n = 12,...,28$, the same for all gaps, while the prime of [6] grows from 727 for $n = 12$ and $\gamma = 1.25$ to 9413 for $n = 28$ and $\gamma = 3$. The mean number of calls follows the prime, as Theorem 7.2 predicts, and lies between 0.2 and 2.1 times the prime for both reductions, which agrees with a success probability of about $1/p$ per trial; the curves of TSE for the four gaps coincide because its prime and its random choices do not depend on γ. The entries of the oracle inputs of TSE are 2 to 4 bits shorter than those of [6] in each dimension, which matches the saving $\log_2(p_B/p)$ of Section 8, and the time per oracle call is correspondingly smaller, 13.0 ms against 18.7 ms for $n = 28$.

Dependence on the gap. Figure 7 varies the gap from $\gamma = 1$ to $\gamma = 64$ for $n = 12$ and $n = 16$, with ten instances per gap whose offset lies along a shortest vector, halfway between the balanced radius and $0.9/(\sqrt{2}\gamma)$. The balanced embedding fails on all of them, and both sparsified reductions solve all of them. The prime of TSE is 487 for $n = 12$ and 641 for $n = 16$ for every gap, and its mean number of calls is 639.8 and 543.4 for every gap. The prime of [6] grows linearly with γ, from 577 to 36871 for $n = 12$ and from 1031 to 65537 for $n = 16$, and its mean number of calls grows

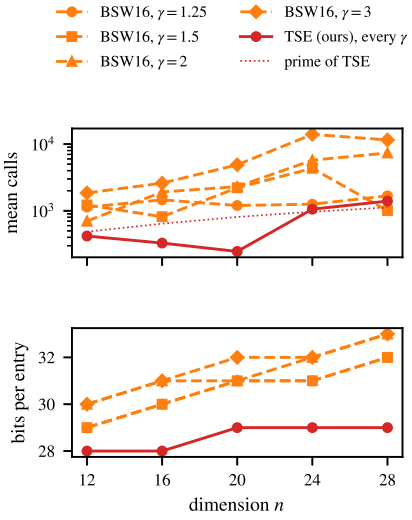


Fig. 9: Mean oracle calls (top) and largest bit length of an entry of an oracle input (bottom) of TSE and of the reduction of [6] on the hard instances, as a function of the dimension. The markers distinguish the gaps.

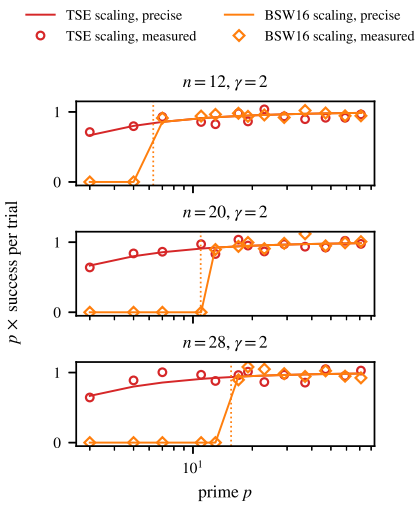


Fig. 10: Success probability of one sparsified trial times the prime p, for the scalings $M = \sqrt{\varepsilon}d$ of TSE with $\varepsilon = 0.1$ and $M = d/(n-1)$ of [6], on the five hard lattices with $\gamma = 2$ and the radius R3b. The lines are the precise values of Proposition 10.1, the markers are measured over $50p$ trials per lattice, and the dotted vertical line is the threshold of Theorem 8.4 for the scaling of [6].

with it to 52937.7 and 74077.4 for $\gamma = 64$, which is 83 and 136 times more than TSE. The entries of its oracle inputs grow from 29 to 35 bits for $n = 12$ and from 30 to 36 bits for $n = 16$, while those of TSE stay at 28 bits. The time of both reductions grows with γ, because our oracle enumerates $2\lceil\gamma\rceil + 3$ vectors within a radius proportional to γ, and for $\gamma = 64$ TSE takes 12.1 s and 14.6 s against 1333.6 s and 1996.8 s.

Time and memory. The time of a sparsified reduction is its number of calls times the time per call, and TSE is smaller in both factors, which gives the time ratio 21.5 in Table 1, above the call ratio 16.2. For $n = 28$, TSE takes 18.4 s on average on the hard instances against 101.0 s. The peak memory, measured with `tracemalloc` for the Python process, agrees within one percent between the two sparsified reductions on the hard instances, for example 134.2 KiB and 134.9 KiB for $n = 28$, because both hold one oracle input with $(n+1)^2$ entries and the working memory of one oracle call at a time, and the memory of a Python integer grows in steps of 30 bits, so the few bits saved per entry do not change the measured memory. Over all 700 instances, TSE uses 55.8 KiB against 80.7 KiB because it stops after the balanced call on 500 of them, and its oracle inputs are smaller in bits on every instance.

The tube lemma. Figure 4 counts, for $10 \le n \le 40$, three random q-ary lattices and $\mathbb{Z}^n$, the lattice points in the tube of radius $r = \lambda_1 \sqrt{(1-0.895\varepsilon)/2}$ around the segment of half-length $r/\sqrt{\varepsilon}$ in the direction of the offset, which contains the tube of Lemma 6.2, found with fplll's enumeration around points of the segment. The largest count is 6 for $\varepsilon = 0.05$ and 4 for $\varepsilon = 0.1$ and $\varepsilon = 0.2$, whose markers overlap, and it is attained by offsets along a shortest vector $\mathbf{s}$, where the tube contains the multiples $\pm j\mathbf{s}$; random offsets give at most 2 points. The bound of Lemma 4.2 grows linearly in n and lies below the count $2\gamma n^2$ of [6] already for $\gamma = 1$ when $n \ge 20$ and $\varepsilon \ge 0.05$, while the count of [6] grows with γ. The gap between the measured counts and the bound shows that the prime of TSE is conservative on these lattices, while the bound holds for every lattice.

Deterministic reductions. We ran the deterministic reduction of Theorem 6.4 with $\varepsilon = 0.1$ on the 120 hard instances with $n \le 20$. It tries the vectors $\mathbf{u} \in U$ in the order of their integer encodings, starting with $\mathbf{u} = 0$, and for each of them the pencil family in the order $\mu(0),\ldots,\mu(q-1)$ followed by the pencils. It solves all 120 instances with the primes $q = 5309$, 9643 and 15241 and with 2814.8, 7163.2 and 14310.8 calls on average for $n = 12$, 16 and 20, and it never needs more than $1.9q$ calls, far below the bound $(N(q-1)+2)q^3 > 10^{17}$ of Theorem 6.4. On these instances it takes 58.5 s and 53.3 KiB on average, against 1.07 s for TSE and 9.64 s for [6]. Its number of calls depends only on the lattice and the nearest point, and not on the gap or the region, which confirms that its cost does not depend on γ. The count is small when the coefficient vector of the nearest point is perpendicular to an early member of the family, for example 2 calls when its coordinates sum to zero modulo q.

Table 7 in Appendix A.10 shows the deterministic reduction from ZSVP of Corollary 5.2 on five random bases of $\mathbb{Z}^n$ per row, each generated by $40n$ random elementary operations with coefficients at most 3 and a random permutation, which is equivalent to a random rotation with the same Gram matrix because the reduction only uses inner products. It finds a unit vector on all 75 bases, with at most 4642 calls against family sizes up to $1.7 \cdot 10^{12}$. On every basis it stops within the first $0.94p$ members of the family, which are the moment points $\mu(\alpha)$, and its mean over all rows is 265.3 calls against 356.8 for uniform random $\mathbf{z}$ with the same prime, so on these instances the explicit family is not more expensive than random sparsification.

10 An adaptive variant of our TSE

The proofs of Theorems 6.3 and 7.2 bound the success probability of a trial from below by the union bound, and the prime of TSE protects against the worst case of Lemma 6.2. This section computes the success probability precisely and uses it to build a variant of TSE that tries small primes first, keeps the guarantee of TSE up to a factor of two, and needs about ten oracle calls on all our instances.

10.1 Precise success probability of a trial

Fix a target $\mathbf{t}$ with nearest vector $\mathbf{c}$, offset $\mathbf{e} = \mathbf{t} - \mathbf{c}$, distance $d = \|\mathbf{e}\|$ and a scaling $M > 0$, and let

$$\mathcal{Y}^* = \{\mathbf{y} \in L \setminus \{0\} \colon \|\mathbf{y} + m\mathbf{e}\|^2 + m^2M^2 < \gamma^2(d^2 + M^2) \text{ for some } m \in \mathbb{Z}\},$$

which is symmetric, and which is contained in the unwanted set $\mathcal{Y}$ of Lemma 6.2 under the parameters of that lemma, since a vector of $\mathcal{Y}^*$ satisfies $\|\mathbf{y}+m\mathbf{e}\|<r_0$ and $|m|M<r_0$, and $m\neq 0$ because $\|\mathbf{y}\|\geq\lambda_1(L)>r_0$. For a prime p and a nonzero $\mathbf{z}$, a trial succeeds when the oracle answer yields the candidate $\mathbf{c}$.

The Proposition 10.1 addresses the precise success probability.

Proposition 10.1. *Let $\gamma>1$ and assume $r_0=\gamma\sqrt{d^2+M^2}<\lambda_1(L)$. For uniform nonzero $\mathbf{z}\in\mathbb{F}_p^n$ and uniform $\mathbf{u}\in\mathbb{F}_p^n$, the input $\mathcal{E}(L_{\mathbf{z}},\mathbf{t}+\mathbf{w},M)$ of a trial is an instance of uSVP_γ with shortest vectors $\pm(\mathbf{e},M)$ precisely when $\langle\mathbf{z},\bar{\mathbf{c}}+\mathbf{u}\rangle=0$ and $\langle\mathbf{z},\bar{\mathbf{y}}\rangle\neq 0$ for all $\mathbf{y}\in\mathcal{Y}^*$. Every oracle therefore makes the trial succeed with probability at least π_p/p, and the oracle that answers invalid inputs with a failure symbol makes it succeed with probability precisely π_p/p, where $\pi_p=\Pr_{\mathbf{z}}[\langle\mathbf{z},\bar{\mathbf{y}}\rangle\neq 0\ \forall\mathbf{y}\in\mathcal{Y}^*]$. If some $\mathbf{y}\in\mathcal{Y}^*$ lies in pL, then $\pi_p=0$. Otherwise let Λ be the set of lines $\mathbb{F}_p\bar{\mathbf{y}}$ for $\mathbf{y}\in\mathcal{Y}^*$ and, for $S\subseteq\Lambda$, let $\mathrm{rk}S$ be the dimension of the span of the lines in S; then*

$$\pi_p=\frac{1}{p^n-1}\sum_{S\subseteq\Lambda}(-1)^{|S|}\left(p^{n-\mathrm{rk}S}-1\right).$$

Proof. Let $\mathbf{w}=\mathbf{B}\tilde{\mathbf{u}}$, $\mathbf{t}'=\mathbf{t}+\mathbf{w}$ and $\mathbf{c}'=\mathbf{c}+\mathbf{w}$, so that $\bar{\mathbf{c}}'=\bar{\mathbf{c}}+\mathbf{u}$ and $\mathbf{t}'-\mathbf{c}'=\mathbf{e}$. If $\mathbf{c}'\notin L_{\mathbf{z}}$, then $(\mathbf{e},M)$ is not in the embedding, as in the proof of Theorem 8.4. If $\mathbf{c}'\in L_{\mathbf{z}}$, the vectors of the embedding are $(\mathbf{y}+m\mathbf{e},mM)$ with $\mathbf{y}\in L_{\mathbf{z}}$ and $m\in\mathbb{Z}$, and those that are not parallel to $(\mathbf{e},M)$ are precisely those with $\mathbf{y}\neq 0$. If some $\mathbf{y}\in L_{\mathbf{z}}\cap\mathcal{Y}^*$ exists, the corresponding vector is not parallel to $(\mathbf{e},M)$ and is shorter than $\gamma\|(\mathbf{e},M)\|$, so the input is not an instance with shortest vectors $\pm(\mathbf{e},M)$. If $L_{\mathbf{z}}\cap\mathcal{Y}^*=\emptyset$, every vector that is not parallel to $(\mathbf{e},M)$ has norm at least $\gamma\|(\mathbf{e},M)\|>\|(\mathbf{e},M)\|$, and the parallel vectors $m(\mathbf{e},M)$ are shortest for $m=\pm 1$, so the input is such an instance. The event $\mathbf{c}'\in L_{\mathbf{z}}$ is $\langle\mathbf{z},\bar{\mathbf{c}}+\mathbf{u}\rangle=0$, and $L_{\mathbf{z}}\cap\mathcal{Y}^*=\emptyset$ is $\langle\mathbf{z},\bar{\mathbf{y}}\rangle\neq 0$ for all $\mathbf{y}\in\mathcal{Y}^*$, which proves the first claim. On a valid input with shortest vectors $\pm(\mathbf{e},M)$ every oracle returns one of them, which yields $\mathbf{c}$. On any other input, the oracle with a failure symbol returns either the symbol or a shortest vector $\mathbf{v}^*\neq\pm(\mathbf{e},M)$ of a valid input; a vector with last coordinate $\pm M$ yields the candidate $\mathbf{c}$ only if it equals $\pm(\mathbf{t}'-\mathbf{c}',M)=\pm(\mathbf{e},M)$, so $\mathbf{v}^*$ does not yield $\mathbf{c}$. Hence the success probability equals the probability of the two events for this oracle and is at least that for every oracle.

For fixed $\mathbf{z}\neq 0$, $\langle\mathbf{z},\bar{\mathbf{c}}+\mathbf{u}\rangle$ is uniform in $\mathbb{F}_p$ over the choice of $\mathbf{u}$, so the first event has probability $1/p$ independently of the second, which only depends on $\mathbf{z}$. If $\mathbf{y}\in\mathcal{Y}^*\cap pL$, then $\bar{\mathbf{y}}=0$ and the second event is impossible. Otherwise the second event only depends on the lines of Λ, and for $S\subseteq\Lambda$ the vectors $\mathbf{z}\in\mathbb{F}_p^n$ perpendicular to all lines of S form a subspace of dimension $n-\mathrm{rk}S$, of which $p^{n-\mathrm{rk}S}-1$ are nonzero. Inclusion and exclusion over S gives the number of nonzero $\mathbf{z}$ that are perpendicular to no line of Λ, and dividing by p^n-1 gives π_p. □

The Corollary 10.2 handles the expected number of trials.

Corollary 10.2. *Under the assumptions of Proposition 10.1, with the failure-symbol oracle, the number of trials with prime p until the first success is geometric with mean p/π_p. If the vectors of $\mathcal{Y}^*$ lie on ℓ lines through the origin of $\mathbb{R}^n$ and $\mathcal{Y}^*\cap pL=\emptyset$, then $\pi_p>1-\ell/p$. If Λ consists of k lines spanned by linearly independent vectors, then $\pi_p=(p-1)^k p^{n-k}/(p^n-1)$, and if $k=1$, then $p/\pi_p<p+1+1/(p-1)$.*

Proof. The trials are independent with success probability π_p/p by Proposition 10.1. Two vectors of $\mathcal{Y}^*$ on one line through the origin are integer multiples of one primitive lattice vector, so their residues lie on one line of $\mathbb{F}_p^n$, and $|\Lambda| \le \ell$. For each line of Λ the probability that $\mathbf{z}$ is perpendicular to it is $(p^{n-1}-1)/(p^n-1) < 1/p$, and the union bound gives $\pi_p > 1-\ell/p$. For k independent lines, the map $\mathbf{z} \mapsto (\langle \mathbf{z},\mathbf{v}_1\rangle,\ldots,\langle \mathbf{z},\mathbf{v}_k\rangle)$ is onto $\mathbb{F}_p^k$ with fibers of size p^{n-k}, so $(p-1)^k p^{n-k}$ vectors have all coordinates nonzero, none of them zero. For $k=1$, $\pi_p = (p-1)p^{n-1}/(p^n-1) > (p-1)/p$, so $p/\pi_p < p^2/(p-1) = p+1+1/(p-1)$. □

For k independent lines, p/π_p is about $p(1-1/p)^{-k} \ge pe^{k/p}$, which is minimized near $p=k$ with value about ek. The best prime therefore depends on the number of independent unwanted lines of the instance, and not on the worst-case count $N(n,\varepsilon)$ that fixes the prime of TSE; Section 10.2 exploits this.

Figure 10 measures the success probability of single trials for 13 primes between 3 and 71, far below the primes of both reductions, with 511500 trials in total. Where Proposition 10.1 predicts a positive probability, the 67 measured values deviate from it by 5.2% on average, and the mean absolute deviation in units of the binomial standard error is 0.82, with 3 of 67 values beyond two standard errors, which is what sampling alone produces. Where it predicts probability 0, no trial succeeded. For the scaling of TSE, p times the success probability equals $\pi_p = 1-(p^{n-1}-1)/(p^n-1)$, about $1-1/p$, for all primes from 3 on, as Corollary 10.2 states for one line. For the scaling of [6], all trials fail for $p \in \{3,5\}$ when $n=12$, for $p \le 11$ when $n=20$ and for $p \le 13$ when $n=28$, where Theorem 8.4 predicts failure for all primes below 6.29, 10.95 and 15.6; the precise failure sets are slightly larger because Theorem 8.4 only uses the multiple $p\mathbf{s}$, while $\mathcal{Y}^*$ may contain other multiples of $\mathbf{s}$ that are divisible by p. Above the threshold, both scalings behave identically, because both unwanted sets lie on one line. The scaling of [6] is therefore not worse than the scaling of TSE because of the number of unwanted vectors on these instances, but because it forces a prime linear in n.

Figure 11 compares whole distributions and not only means. Corollary 10.2 predicts that the number of calls of TSE minus one is geometric with success probability π_p/p, and π_p is the same for all five lattices because their unwanted vectors lie on one line. The Kolmogorov–Smirnov distance between the 200 runs of a dimension and this distribution is 0.067 for $n=12$ and 0.043 for $n=20$, below the threshold 0.096 of the test at level 5%. Per lattice, 8 of the 10 distances are below the threshold 0.215 for 40 runs, and the two others, for lattices 2 and 4 with $n=12$, are 0.23 and 0.25. The heavy tails of the distributions, with 7% and 4.5% of the runs above $3p$ calls for $n=12$ and $n=20$, near $e^{-3} \approx 5\%$, are the reason why a fixed budget of trials must be about pn to make the failure probability exponentially small, as in Theorem 7.2.

10.2 Interleaving small primes

Corollary 10.2 and Theorem 8.4 show that on the hard instances the scaling $M = \sqrt{\varepsilon}d$ of TSE works with every prime above $h(\varepsilon)$, and that the expected number of trials is near the prime. The prime p^* of TSE, the least prime at least $\max(2N_0, h(\varepsilon)+1)$ with $N_0 = \min(N(n,\varepsilon), N'(\varepsilon))$, protects against the worst case of Lemma 6.2, while the instance may need a much smaller prime. The variant below tries small primes

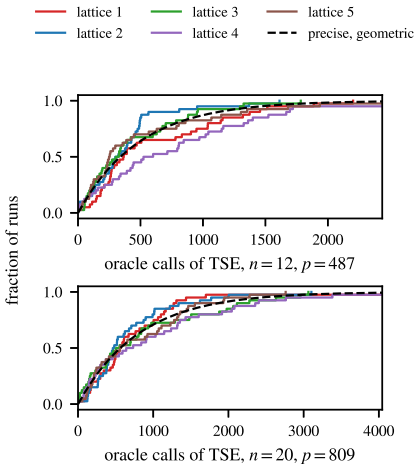


Fig. 11: Empirical distribution of the number of oracle calls of TSE over 40 runs with independent coins on each of the five hard lattices with $\gamma = 2$ and the radius R3b, against the geometric distribution of Corollary 10.2 shifted by the balanced call.

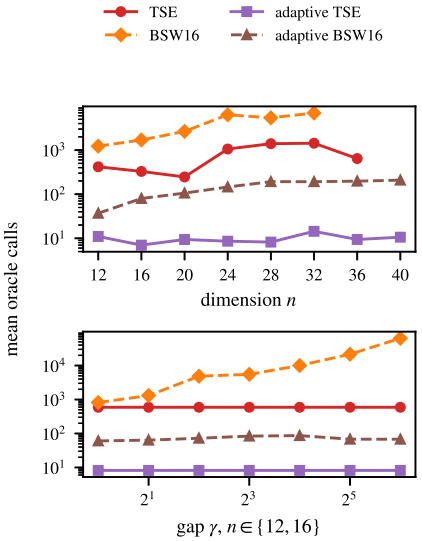


Fig. 12: Mean oracle calls on hard instances of TSE, of [6] and of their adaptive variants. Top, as a function of the dimension, with 40 runs per dimension up to $n = 28$ and 10 runs for $n \in \{32,36,40\}$, where only the adaptive variants run in all dimensions. Bottom, as a function of the gap for $n \in \{12,16\}$ and ten instances per gap and dimension.

first and runs the trials of TSE in parallel, so that it pays at most a factor of two over TSE in the worst case.

Algorithm 2 keeps the first call of TSE with the balanced embedding. It then alternates between two streams of trials of Section 6, each with fresh uniform $\mathbf{z} \neq 0$ and $\mathbf{u}$. The fixed stream uses the prime p^* of TSE forever. The adaptive stream runs through levels $j = 0,1,...$, where level j uses the least prime $p_j \geq p_0 2^j$ for $\lceil 2\ln 4 \cdot p_j \rceil$ trials, and it stops before the first level whose prime reaches p^*. Every prime of the adaptive stream exceeds $h(\varepsilon)$, so no unwanted vector lies in $p_j L$, and the number of trials per level is chosen so that a level whose prime is at least twice the number of unwanted lines fails with probability at most $1/4$. In the implementation the adaptive stream is generated lazily, so its memory does not depend on its length. The Theorem 10.3 goes through the cost of the adaptive variant.

Theorem 10.3. *Let $\gamma > 1$, $\varepsilon \in (0, 1/2]$ and let $(\mathbf{B}, \mathbf{t})$ be an instance with $0 < d = \mathrm{dist}(\mathbf{t}, L) \leq (1-\varepsilon)\lambda/(\sqrt{2}\gamma)$. Let ℓ be the number of lines through the origin that contain vectors of $\mathcal{Y}^*$ for $M = \sqrt{\varepsilon}d$, and let j^* be the least level of the adaptive stream with $p_{j^*} \geq 2\ell$, if there is one. Algorithm 2 returns a nearest vector with*

Algorithm 2 Interleaved adaptive TSE

function ATSE($\mathbf{B}$,$\mathbf{t}$,d)
  $\hat{\mathbf{c}} \leftarrow$ CAND($\mathcal{O}(\mathcal{E}(L,\mathbf{t},k_\gamma d))$,$\mathbf{t}$,$k_\gamma d$)
  if $\hat{\mathbf{c}} \neq \perp$ **and** $\|\mathbf{t}-\hat{\mathbf{c}}\| = d$ **then return** $\hat{\mathbf{c}}$
  end if
  $p^* \leftarrow$ least prime $\geq \max(2N_0, h(\varepsilon)+1)$; $p_0 \leftarrow$ least prime $> h(\varepsilon)$; $M \leftarrow \sqrt{\varepsilon} d$
  $\mathcal{S} \leftarrow$ for $j = 0,1,...$ while $p_j < p^*$, $\lceil 2\ln 4 \cdot p_j \rceil$ copies of p_j, where p_j is the least prime $\geq p_0 2^j$
  for $i = 1,2,...$ **do**
    $\hat{\mathbf{c}} \leftarrow$ one trial with prime p^* and scaling M; **if** $\|\mathbf{t}-\hat{\mathbf{c}}\| = d$ **return** $\hat{\mathbf{c}}$
    if $i \leq |\mathcal{S}|$ **then**
      $\hat{\mathbf{c}} \leftarrow$ one trial with prime $\mathcal{S}_i$ and scaling M; **if** $\|\mathbf{t}-\hat{\mathbf{c}}\| = d$ **return** $\hat{\mathbf{c}}$
    end if
  end for
end function

Table 4: The adaptive variant against all reductions on the 700 instances of Section 9, with the same oracle and a budget of 10 times the fixed prime. The last three columns divide the mean calls, time and memory of a reduction by those of adaptive TSE on the instances that the reduction solves.

			Mean calls		Mean time (ms)			Ratios to adaptive TSE		
Reduction	Solved	Hard	All	Hard	All	Hard	KiB	Calls	Time	Memory
$M = d/\sqrt{n}$ [4, 8]	350	0	1	1	3.5	3.7	48.2	1.00	1.08	1.02
$M = d$ [5]	475	0	1	1	3.4	3.5	46.7	1.00	1.04	1.00
Balanced [6]	500	0	1	1	3.3	3.3	46.7	1.00	1.02	1.00
Sparsified [6]	700	200	3199.2	3460.5	37400.9	41315.4	80.7	987.4	2524.1	1.46
Adaptive [6]	700	200	52.6	112.2	368.7	807.4	80.0	16.2	24.9	1.45
TSE (Alg. 1)	700	200	197.7	689.6	1738.7	6077.7	55.8	61.0	117.4	1.01
Adaptive TSE (Alg. 2)	**700**	**200**	**3.2**	**8.8**	**14.8**	**43.9**	55.4	1	1	1

probability 1*, and its expected number of oracle calls is at most* $1+4p^*$*, and at most* $1+2(18.7p_{j^*}+j^*+2)$ *when* j^* *exists, where* $p_{j^*} < \max(2p_0, 8\ell)$.

Proof (sketch). Let F and A be the indices of the first successful trials of the fixed and of the adaptive stream, which are independent, so the algorithm stops after at most $1+2\min(F,A)$ calls. Each trial of the fixed stream succeeds with probability at least $1/(2p^*)$ by the proof of Theorem 7.2, which gives termination with probability 1 and the bound $1+4p^*$. From level j^* on, Corollary 10.2 gives $\pi_{p_j} > 1-\ell/p_j \geq 1/2$, so all $\lceil 2\ln 4 \cdot p_j \rceil$ trials of a level fail with probability at most $1/4$. Summing the lengths of the levels weighted by the probability of reaching them, and bounding the case in which the adaptive stream ends without success by $E[F]$, gives $E[\min(F,A)] \leq 18.7p_{j^*}+j^*+2$, and Bertrand's postulate gives $p_{j^*} < \max(2p_0, 8\ell)$. The full proof is in the Appendix. □

Since $p_0 < 2(h(\varepsilon)+1) = O(1/\sqrt{\varepsilon})$, Theorem 10.3 bounds the expected number of calls by $O(\ell+1/\sqrt{\varepsilon})$ on every instance, independently of n and γ, and by $1+4p^*$ in the worst case, which is at most twice the bound $1+2p^*$ of TSE. On instances whose unwanted vectors lie on one line, such as all hard instances of Section 9, $\ell = 1$ and

the bound is $O(1/\sqrt{\varepsilon})$. The same adaptive schedule can be applied to the scaling of [6], which we call adaptive [6], with the prime p_B in the fixed stream and $p_0 = 3$; by Theorem 8.4, its adaptive stream cannot succeed before its primes exceed about $0.55(n-1)$ on the hard instances, which isolates the effect of the scaling from the effect of adaptivity.

10.3 Measurements

Table 4 adds the adaptive variants to the comparison of Table 1, with the oracle, the instances and the budgets of Section 9 and the parameters of Appendix A.25. Adaptive TSE solves all 700 instances with 3.2 calls on average and 8.8 calls on the hard instances, at most 27 on any instance, against 689.6 hard calls for TSE and 3460.5 for [6]. On the instances that each other reduction solves, it uses the same single call as the embeddings without sparsification, 61 times fewer calls and 117 times less time than TSE, and 987 times fewer calls and 2524 times less time than [6], whose calls are also more expensive because of their longer entries. Its memory equals that of TSE and the single-call embeddings within one percent, and it is 1.46 times smaller than that of [6]. Adaptive [6] also improves on [6] by a factor of 30 on the hard instances, but it needs 112.2 calls there. Of the factor 393 between [6] and adaptive TSE on the hard instances, a factor of 12.7 therefore remains between the two adaptive variants, which differ only in the scaling.

Figure 12 shows how the four reductions scale. As a function of the dimension, adaptive TSE stays between 7 and 14.4 calls for $12 \le n \le 40$, while adaptive [6] grows from 37 calls for $n = 12$ to 208 calls for $n = 40$, because the smallest usable prime of Theorem 8.4 grows linearly with n and moves its first successful level up. As a function of the gap, adaptive TSE uses 7.6 calls for $n = 12$ and 8.8 calls for $n = 16$ for every γ between 1 and 64, which confirms that neither its prime nor its cost depends on γ, while adaptive [6] needs between 31 and 138 calls, and [6] itself up to 74077 calls for $\gamma = 64$ and $n = 16$.

Future Directions

Extending our tube lemma beyond the radius $\lambda_1/\sqrt{2}$, where the number of lattice points near a single point is no longer polynomial, would decide whether the factor $\sqrt{2}$ between BDD and uSVP can be improved for any range of gaps. Moreover, a deterministic reduction from ZSVP to γ-SVP in time polynomial in n for superconstant γ, which the pencil family gives only in time $n^{O(\gamma^2)}$, remains our next aim for unusual lattices.

A Appendix

The Appendix gives the full proofs of the theorems of the main body in Appendices A.2 to A.8, derives the complexity comparison of Section 8 in Appendix A.9, and then extends the three contributions. For Contribution 1, Appendix A.10 replaces the pencil family by a family that is smaller by a factor of about $8A$ and proves a lower bound on the size of every family with the isolation property, and Appendix A.11 compares it with the sampler of Bennett et al. For Contribution 2, Appendix A.12 shows that shifts make the deterministic decoding reduction much smaller, Appendix A.13 quantifies the count of Corollary 8.2, and Appendix A.14 derives a deterministic reduction to GapSVP_γ. For Contribution 3, Appendix A.15 describes the precise unwanted sets of the hard instances and the distribution of the number of calls, Appendix A.16 measures the adaptive variant along the radius and up to dimension 40, Appendix A.17 studies the choice of the slack, and Appendix A.18 shows that the tube lemma is tight per slice and that sparsification with uniform sublattices needs a linear number of calls on some instances. Appendices A.19, A.20 and A.21 run the reductions beyond the proven radius, examine the role of the promise and test other lattice families, and Appendix A.23 document the oracle, the instances, the parameters and the implementation of all reductions, additional measurements and the scripts behind every table and figure. Appendix A.1 lists the notation.

A.1 Notation and parameters

Table 5 lists the symbols that the main body and the Appendix use across sections, with the place where each is defined.

A.2 Proof of Theorem 5.1

The algorithm computes the least prime $p \geq \max(2A(n-1)+2n, 2\gamma+1)$ by trial division, and for every $\mathbf{z} \in \mathcal{Z}_{n,p}$ it computes the basis of $L_{\mathbf{z}}$ of Lemma 3.4, calls the oracle on it, and keeps the answer if it is a nonzero vector $\mathbf{v}$ with $\mathbf{B}^{-1}\mathbf{v} \in \mathbb{Z}^n$. It outputs the shortest kept answer. Each $\mathbf{z} \in \mathcal{Z}_{n,p}$ is nonzero, and $|\mathcal{Z}_{n,p}| \leq p^3$.

Let $\mathbf{x}$ be a shortest nonzero vector of L. If $\mathbf{x} = k\mathbf{x}'$ with $\mathbf{x}' \in L$ and $k \geq 2$, then $\mathbf{x}'$ is shorter, so $\mathbf{x}$ is primitive, hence $\mathbf{x} \notin pL$ and $\bar{\mathbf{x}} \neq 0$. Let $Y \subset \mathbb{F}_p^n$ contain $\bar{\mathbf{y}}$ for precisely one vector $\mathbf{y}$ of each pair $\{\mathbf{y}, -\mathbf{y}\}$ of primitive vectors with $\|\mathbf{y}\| \leq \gamma\lambda_1(L)$ and $\mathbf{y} \notin \{\mathbf{x}, -\mathbf{x}\}$, so that $|Y| \leq A_\gamma(L) - 1 < A$. Each element of Y is linearly independent from $\bar{\mathbf{x}}$. Indeed, if $\bar{\mathbf{y}}$ were a multiple of $\bar{\mathbf{x}}$, there would be an integer c with $|c| \leq (p-1)/2$ and $\bar{\mathbf{y}} = c\bar{\mathbf{x}}$, so the coordinate vector of $\mathbf{y} - c\mathbf{x}$ would vanish modulo p and $\mathbf{y} - c\mathbf{x} \in pL$. If $\mathbf{y} \neq c\mathbf{x}$, this nonzero vector of pL would have norm at least $p\lambda_1(L)$, while $\|\mathbf{y} - c\mathbf{x}\| \leq \|\mathbf{y}\| + |c|\,\|\mathbf{x}\| \leq (\gamma + (p-1)/2)\lambda_1(L) < p\lambda_1(L)$ because $p > 2\gamma - 1$. Hence $\mathbf{y} = c\mathbf{x}$, and since $\mathbf{y}$ is primitive, $c = \pm 1$, which contradicts $\mathbf{y} \notin \{\mathbf{x}, -\mathbf{x}\}$.

Lemma 4.3 with $\bar{\mathbf{x}}$, the set Y and A gives $\mathbf{z} \in \mathcal{Z}_{n,p}$ with $\langle \mathbf{z}, \bar{\mathbf{x}} \rangle = 0$ and $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle \neq 0$ for all $\bar{\mathbf{y}} \in Y$, so $\mathbf{x} \in L_{\mathbf{z}}$. Let $\mathbf{v} \in L_{\mathbf{z}}$ be nonzero and not parallel to $\mathbf{x}$, and write $\mathbf{v} = k\mathbf{y}$ with $\mathbf{y}$ primitive and $k \geq 1$. If $\|\mathbf{y}\| > \gamma\lambda_1(L)$, then $\|\mathbf{v}\| > \gamma\lambda_1(L)$. Otherwise $\mathbf{y}$ or $-\mathbf{y}$ contributes an element of Y, so $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle \neq 0$, and $k\langle \mathbf{z}, \bar{\mathbf{y}} \rangle = \langle \mathbf{z}, \bar{\mathbf{v}} \rangle = 0$ in $\mathbb{F}_p$ forces $p \mid k$, so $\|\mathbf{v}\| \geq p\|\mathbf{y}\| \geq p\lambda_1(L) > \gamma\lambda_1(L)$. Since $\mathbf{x} \in L_{\mathbf{z}} \subseteq L$, $\lambda_1(L_{\mathbf{z}}) = \lambda_1(L)$, and every vector of

Table 5: Symbols used across sections, with their definitions and the place where they are introduced.

Symbol	Meaning	Introduced in
$\bar{\mathbf{v}}$, $L_{\mathbf{z}}$	reduced coordinates of $\mathbf{v}$ modulo p, sublattice $\{\mathbf{v}\in L:\langle\mathbf{z},\bar{\mathbf{v}}\rangle=0\}$ of index p	Section 3, Lemma 3.4
$\mathcal{E}(L',\mathbf{t}',M)$	embedding lattice generated by a basis of L' padded with 0 and by $(\mathbf{t}',M)$	Section 6
$\mathbf{e}$, d, M, k	offset $\mathbf{t}-\mathbf{c}$, distance $\|\mathbf{e}\|$, scaling of the last coordinate, $k=M/d$	Lemma 6.1, Appendix A.6
r_0, m_0	$r_0=\gamma\sqrt{d^2+M^2}$ and $m_0=r_0/M$, the radius and range of multiples of the promise	Lemma 6.1
ε, $h(\varepsilon)$	slack in the radius $(1-\varepsilon)/(\sqrt{2}\gamma)$, $h(\varepsilon)=(1+1/\sqrt{\varepsilon})/\sqrt{2}$	Section 6
η, θ, $\kappa_\theta(\eta)$	$\eta=1-2r^2/\lambda^2$, relative slice length, bound per slice $\lceil(1-\eta)/((1-\theta^2)\eta)\rceil$	Lemmas 4.2 and 8.1
$N(n,\varepsilon)$, $N'(\varepsilon)$, N_0	tube count, dimension-free count, $N_0=\min(N(n,\varepsilon),N'(\varepsilon))$	Lemma 6.2, Corollary 8.2
$\mathcal{Y}$, $\mathcal{Y}^*$	unwanted set of the tube, precise unwanted set of the promise	Lemma 6.2, Section 10.1
π_p, ℓ	probability that $\mathbf{z}$ avoids all residues of $\mathcal{Y}^*$, number of lines of $\mathcal{Y}^*$	Proposition 10.1, Corollary 10.2
γ_1, k_γ, a	balanced radius parameter, balanced scaling, $a=\lfloor\gamma\rfloor$	Section 7
p, p^*, p_B, q	prime of TSE, prime of TSE with N_0, prime of [6], prime of the deterministic reductions	Algorithms 1, 2, 3
p_0, p_j, T_j	first prime of the adaptive stream, prime and number of trials of level j	Algorithm 2
G, T	number of guesses of M, number of trials per guess	Theorem 6.3
$\mu(\alpha)$, $f_{\mathbf{x}}$	point $(1,\alpha,\ldots,\alpha^{n-1})$ of the moment curve, polynomial with coefficient vector $\mathbf{x}$	Section 4.2
$\mathcal{Z}_{n,p}$, $\mathcal{Z}^S_{n,p}$	pencil family, thin pencil family	Section 4.2, Appendix A.10
$A_\gamma(L)$, A	number of pairs of primitive vectors of norm at most $\gamma\lambda_1$, an upper bound on it	Section 5
$L_{n,b}$, $\mathbf{w}_i$	lattice generated by $(b-1)\mathbf{1}+n\mathbf{e}_i$	Appendix A.18

$L_{\mathbf{z}}$ that is not parallel to $\mathbf{x}$ is longer than $\gamma\lambda_1(L_{\mathbf{z}})$, so $\lambda_2(L_{\mathbf{z}})>\gamma\lambda_1(L_{\mathbf{z}})$. The oracle therefore answers the query for this $\mathbf{z}$ with a shortest nonzero vector of $L_{\mathbf{z}}$, which is a nonzero vector of L of norm $\lambda_1(L)$ and is kept. All kept answers are nonzero vectors of L, of norm at least $\lambda_1(L)$, so the output is a shortest nonzero vector of L.

The number of calls is $|\mathcal{Z}_{n,p}|\le p^3$, each call is on a basis of a sublattice of index p and rank n, and each iteration costs $O(n^2)$ operations for Lemma 3.4 and one rational linear system of dimension n for the membership test, on numbers whose bit length is polynomial in the bit length of $\mathbf{B}$ and in $\log p$. The prime p is at most $2\max(2A(n-1)+2n,2\gamma+1)$ by Bertrand's postulate and is found in time polynomial in p. The total running time is therefore polynomial in the bit length of $\mathbf{B}$ and in p.

A.3 Proof of Theorem 6.3

On input $(\mathbf{B},\mathbf{t})$, the reduction outputs $\mathbf{t}$ if $\mathbf{B}^{-1}\mathbf{t}\in\mathbb{Z}^n$. Otherwise it runs Babai's nearest plane algorithm on an LLL-reduced basis [37, 36], which returns $\mathbf{c}_0\in L$ with $d\le D=\|\mathbf{t}-\mathbf{c}_0\|\le 2^n d$ [5, Lemma 2.7], where $d=\mathrm{dist}(\mathbf{t},L)$. It computes the least prime $p\ge\max(2N(n,\varepsilon),h(\varepsilon)+1)$ by trial division. For each $i\in\{0,\ldots,G-1\}$ with $G=1+\lceil 35.01n\rceil$, it computes a rational number M_i in the interval $[\sqrt{\varepsilon}D1.02^{-i},1.01\sqrt{\varepsilon}D1.02^{-i}]$, which is possible in polynomial time because εD^2 is rational, and it runs $T=\lceil 2pn\rceil$ trials of Section 6 with $M=M_i$, each with fresh uniform

$\mathbf{z} \in \mathbb{F}_p^n \setminus \{0\}$ and $\mathbf{u} \in \mathbb{F}_p^n$. It outputs the vector nearest to $\mathbf{t}$ among $\mathbf{c}_0$ and all candidates returned by the trials. Each candidate is a lattice vector, since a returned candidate equals $-\mathbf{b}-\mathbf{w}$ with $\mathbf{b} \in L_{\mathbf{z}} \subseteq L$ and $\mathbf{w} \in L$, and the reduction checks that each answer belongs to the queried embedding lattice by solving one rational linear system.

For the analysis, let i^* be the largest index with $D1.02^{-i^*} \geq d$. It exists because $D \geq d$, and $i^* \leq G-1$ because $D1.02^{-(G-1)} \leq 2^n d\, 1.02^{-35.01n} < d$, since $35.01 \ln 1.02 > \ln 2$. As $D1.02^{-i^*-1} < d$, we have $d \leq D1.02^{-i^*} < 1.02d$, so $\sqrt{\varepsilon}d \leq M_{i^*} \leq 1.0302\sqrt{\varepsilon}d \leq 1.04\sqrt{\varepsilon}d$, and Lemma 6.2 applies with $M = M_{i^*}$ and its unwanted set $\mathcal{Y}$ for the offset $\mathbf{e} = \mathbf{t}-\mathbf{c}$ of a fixed nearest vector $\mathbf{c}$.

Consider a trial of the guess i^* with vectors $\mathbf{z} \neq 0$ and $\mathbf{u}$, and let $\mathbf{c}_s = \mathbf{c}+\mathbf{w}$ and $\mathbf{t}_s = \mathbf{t}+\mathbf{w}$, so $\mathbf{t}_s - \mathbf{c}_s = \mathbf{e}$ and $\bar{\mathbf{c}}_s = \bar{\mathbf{c}}+\mathbf{u}$ because $\mathbf{w} = \mathbf{B}\tilde{\mathbf{u}}$. Let E be the event that $\langle \mathbf{z}, \bar{\mathbf{c}}+\mathbf{u} \rangle = 0$ and $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle \neq 0$ for every $\mathbf{y} \in \mathcal{Y}$. Under E, the vector $\mathbf{c}_s$ lies in $L_{\mathbf{z}}$. If some nonzero $\mathbf{y}' \in L_{\mathbf{z}}$ and integer m with $1 \leq |m| \leq m_0$ satisfied $\|\mathbf{y}'+m\mathbf{e}\| \leq r_0$, then $\mathbf{y}' \in \mathcal{Y}$ by definition, and $\mathbf{y}' \in L_{\mathbf{z}}$ would give $\langle \mathbf{z}, \bar{\mathbf{y}}' \rangle = 0$, which contradicts E. Since also $r_0 < \lambda/\sqrt{2} < \lambda$ by Lemma 6.2, Lemma 6.1 applies to $L' = L_{\mathbf{z}}$, $\mathbf{c}' = \mathbf{c}_s$ and $\mathbf{t}' = \mathbf{t}_s$, so the oracle input satisfies the promise of uSVP_γ and its only shortest nonzero vectors are $\pm(\mathbf{e}, M)$. The oracle returns one of them. If it returns $(\mathbf{e}, M)$, the last coordinate is M and the candidate is $\mathbf{t}-\mathbf{e} = \mathbf{c}$; if it returns $-(\mathbf{e}, M)$, the last coordinate is $-M$ and the candidate is $\mathbf{t}+(-\mathbf{e}) = \mathbf{c}$.

It remains to bound the probability of E. Fix $\mathbf{z} \neq 0$. As $\mathbf{u}$ is uniform in $\mathbb{F}_p^n$ and independent of $\mathbf{z}$, the value $\langle \mathbf{z}, \bar{\mathbf{c}}+\mathbf{u} \rangle$ is uniform in $\mathbb{F}_p$, so it is zero with probability $1/p$, independently of the values $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle$ that do not involve $\mathbf{u}$. Each $\mathbf{y} \in \mathcal{Y}$ satisfies $0 < \|\mathbf{y}\| < h(\varepsilon)\lambda < p\lambda$ by Lemma 6.2, so $\mathbf{y} \notin pL$ and $\bar{\mathbf{y}} \neq 0$, and for uniform $\mathbf{z} \in \mathbb{F}_p^n \setminus \{0\}$ we have $\Pr[\langle \mathbf{z}, \bar{\mathbf{y}} \rangle = 0] = (p^{n-1}-1)/(p^n-1) < 1/p$. By the union bound,

$$\Pr[E] \geq \frac{1}{p} - \sum_{\mathbf{y} \in \mathcal{Y}} \frac{1}{p} \Pr[\langle \mathbf{z}, \bar{\mathbf{y}} \rangle = 0] \geq \frac{1}{p} - \frac{N(n,\varepsilon)}{p^2} \geq \frac{1}{2p},$$

because $|\mathcal{Y}| \leq N(n,\varepsilon) \leq p/2$. The T trials of the guess i^* are independent, so all of them fail with probability at most $(1-1/(2p))^T \leq e^{-T/(2p)} \leq e^{-n}$. When one of them succeeds, $\mathbf{c}$ is a candidate, and since all candidates lie in L, the output is at distance precisely d from $\mathbf{t}$, so it is a nearest vector.

Finally, the prime satisfies $p \leq 4N(n,\varepsilon) = O(n/\varepsilon)$ by Bertrand's postulate. The reduction makes $GT = O(n \cdot pn) = O(n^3/\varepsilon)$ oracle calls, all on lattices of dimension $n+1$, and each trial costs $O(n^2)$ arithmetic operations for Lemma 3.4 and the shift, and one rational linear system for the membership test, on numbers of bit length polynomial in the input length and $\log p$. None of these quantities depends on γ, which only enters through the promise of the oracle.

A.4 Proof of Theorem 6.4

The reduction differs from the one of Theorem 6.3 only in its trials. Let $N = N(n,\varepsilon)$, let q be the least prime at least $\max(2N(n-1)+2n, h(\varepsilon)+1)$, and let $U \subset \mathbb{F}_q^n$ be a fixed set of $N(q-1)+2$ distinct vectors. For each guess i, the reduction runs one trial of Section 6 with the modulus q and $M = M_i$ for each pair $(\mathbf{z}, \mathbf{u}) \in \mathcal{Z}_{n,q} \times U$, and it outputs the candidate nearest to $\mathbf{t}$.

For the guess i^*, Lemma 6.2 applies unchanged, since it does not involve the modulus, so $|\mathcal{Y}| \le N$ and every $\mathbf{y} \in \mathcal{Y}$ satisfies $0 < \|\mathbf{y}\| < h(\varepsilon)\lambda < q\lambda$, hence $\bar{\mathbf{y}} \neq 0$ modulo q. The set $B = \{0\} \cup \bigcup_{\mathbf{y}\in\mathcal{Y}} \{a\bar{\mathbf{y}} : a \in \mathbb{F}_q\}$ has at most $1 + N(q-1)$ elements, because each line through 0 contributes $q-1$ nonzero points. The map $\mathbf{u} \mapsto \bar{\mathbf{c}} + \mathbf{u}$ is injective and $|U| > |B|$, so there is $\mathbf{u} \in U$ with $\mathbf{x} = \bar{\mathbf{c}} + \mathbf{u} \notin B$. Then $\mathbf{x} \neq 0$, and every $\bar{\mathbf{y}}$ with $\mathbf{y} \in \mathcal{Y}$ is linearly independent from $\mathbf{x}$, since two nonzero vectors are dependent only if the first is a multiple of the second. Lemma 4.3 with $A = N$ and the prime $q \ge 2N(n-1) + 2n$ gives $\mathbf{z} \in \mathcal{Z}_{n,q}$ with $\langle \mathbf{z}, \mathbf{x} \rangle = 0$ and $\langle \mathbf{z}, \bar{\mathbf{y}} \rangle \neq 0$ for all $\mathbf{y} \in \mathcal{Y}$. This is the event E of the proof of Theorem 6.3 for the pair $(\mathbf{z}, \mathbf{u})$, so the corresponding trial returns a nearest vector $\mathbf{c}$, and the output is a nearest vector. The number of calls is at most $G|U||\mathcal{Z}_{n,q}| \le G(N(q-1)+2)q^3$, which is polynomial in n and $1/\varepsilon$ because $N = O(n/\varepsilon)$ and $q = O(n^2/\varepsilon)$, and it does not depend on γ.

A.5 Proof of Theorem 7.2

If $d \le \lambda/\gamma_1$, Lemma 7.1 shows that $\mathcal{E}(L, \mathbf{t}, k_\gamma d)$ satisfies the promise of uSVP_γ and that its only shortest nonzero vectors are $\pm(\mathbf{t} - \mathbf{c}, k_\gamma d)$ for a nearest vector $\mathbf{c}$. The oracle returns one of them, CAND returns $\mathbf{c}$, and since $\|\mathbf{t} - \mathbf{c}\| = d$ the algorithm stops after one call. Otherwise, or if the first call does not end the algorithm, the trials use $M = \sqrt{\varepsilon} d$, which satisfies $\sqrt{\varepsilon} d \le M \le 1.04\sqrt{\varepsilon} d$, so for $d \le (1-\varepsilon)\lambda/(\sqrt{2}\gamma)$ the argument of the proof of Theorem 6.3 for the guess i^* applies to each trial without change, which succeeds with probability at least $1/(2p)$ for p the least prime at least $2N(n, \varepsilon)$, independently of the other trials. The number of trials until the first success is therefore dominated by a geometric variable with mean at most $2p$, so the expected number of calls is at most $1 + 2p$, and the probability that $T = \lceil 2pn \rceil$ trials all fail is at most e^{-n}. Each oracle input is a basis of $\mathcal{E}(L, \mathbf{t}, k_\gamma d)$ or of $\mathcal{E}(L_{\mathbf{z}}, \mathbf{t} + \mathbf{w}, M)$ of dimension $n+1$. The entries of the basis of $L_{\mathbf{z}}$ are those of $\mathbf{b}_i - c_i \mathbf{b}_j$ or $p\mathbf{b}_j$ with $0 \le c_i < p$, so they are bounded by $p \max|B_{ij}|$ in absolute value, and the entries of $\mathbf{t} + \mathbf{w}$ are bounded by $\max|t_i| + n(p-1)\max|B_{ij}| \le (n+1)p2^\beta$, which gives the stated bit lengths.

A.6 Proof of Theorem 8.3

We prove the statement for every lattice L with a shortest vector $\mathbf{s}$ of norm λ and every target $\mathbf{t} = \mathbf{c} + \alpha\mathbf{s}$ with $\mathbf{c} \in L$, which contains the case $L = \lambda\mathbb{Z}^n$, $\mathbf{s} = \lambda\mathbf{e}_1$ and $\mathbf{c} = 0$ of Theorem 8.3; these are the targets that Section 9 calls hard. Let $\mathbf{e} = \mathbf{t} - \mathbf{c} = \alpha\mathbf{s}$ and $d = \alpha\lambda$. For every $\mathbf{x} \in L$ with $\mathbf{x} \neq \mathbf{c}$, $\|\mathbf{t} - \mathbf{x}\| \ge \|\mathbf{x} - \mathbf{c}\| - \|\mathbf{e}\| \ge \lambda - d > d$ because $\alpha < 1/2$, so $\mathbf{c}$ is the unique nearest lattice vector and $\operatorname{dist}(\mathbf{t}, L) = d$. Since $\mathcal{E}(L, \mathbf{t}, M)$ is generated by the columns of $\left(\begin{smallmatrix} \mathbf{B} & \mathbf{t} \\ 0 & M \end{smallmatrix}\right)$ and $\mathbf{c} \in L$, it also contains the vector $(\mathbf{e}, M) = (\mathbf{t}, M) - (\mathbf{c}, 0)$, and its shortest vectors could only be $\pm(\mathbf{e}, M)$ in the sense of the statement.

Let $k = M/d$, $\beta = 1/\alpha$ and $a = \lfloor \gamma \rfloor \ge 1$, so that $1/\gamma_1 < \alpha < 1/2$ means $2 < \beta < \gamma_1$. Assume that $\mathcal{E}(L, \mathbf{t}, M)$ is an instance of uSVP_γ whose shortest vectors are $\pm(\mathbf{e}, M)$. Then every vector $\mathbf{v}$ of the embedding that is not parallel to $(\mathbf{e}, M)$ satisfies $\|\mathbf{v}\|^2 \ge \gamma^2 \|(\mathbf{e}, M)\|^2 = \gamma^2 d^2 (1 + k^2)$. The embedding contains the three vectors

$$\mathbf{v}_1 = (\mathbf{s}, 0), \quad \mathbf{v}_2 = a(\mathbf{e}, M) - (\mathbf{s}, 0), \quad \mathbf{v}_3 = (a+1)(\mathbf{e}, M) - (\mathbf{s}, 0).$$

None of them is parallel to $(\mathbf{e},M)$. The last coordinate of $\mathbf{v}_1$ is 0 while its first part is $\mathbf{s}\neq 0$, and if $\mathbf{v}_2=\mu(\mathbf{e},M)$, then comparing last coordinates gives $\mu=a$ and comparing first parts gives $\mathbf{s}=0$, which is impossible; the same argument applies to $\mathbf{v}_3$. Since $\mathbf{e}=\alpha\mathbf{s}$, the first parts of the three vectors are $\mathbf{s}$, $(a\alpha-1)\mathbf{s}$ and $((a+1)\alpha-1)\mathbf{s}$, and dividing their squared norms by $d^2=\alpha^2\lambda^2$ gives the three conditions

$$\begin{aligned}\beta^2 &\geq \gamma^2(1+k^2),\\ (\beta-a)^2+a^2k^2 &\geq \gamma^2(1+k^2),\\ (\beta-a-1)^2+(a+1)^2k^2 &\geq \gamma^2(1+k^2).\end{aligned}$$

The first condition gives $\beta\geq\gamma\sqrt{1+k^2}>\gamma\geq a$, so $y=\beta-a$ is positive.

If $\gamma=a$ is an integer, the second condition reads $y^2\geq\gamma^2+(\gamma^2-a^2)k^2=\gamma^2$, so $\beta\geq a+\gamma=2\gamma$, and $\gamma_1=2\gamma$ for integer γ. This contradicts $\beta<\gamma_1$.

If $\gamma>a$, let $A_0=\gamma^2-a^2>0$ and $B_0=(a+1)^2-\gamma^2>0$, so that $A_0+B_0=2a+1$. The second condition is equivalent to $A_0k^2\leq y^2-\gamma^2$, and the third is equivalent to $B_0k^2\geq\gamma^2-(y-1)^2$. Multiplying the first inequality by B_0 and the second by A_0 gives $B_0(y^2-\gamma^2)\geq A_0B_0k^2\geq A_0(\gamma^2-(y-1)^2)$, which rearranges to $Q(y)\geq 0$ for the quadratic

$$Q(y)=(2a+1)y^2-2A_0y+A_0-(2a+1)\gamma^2.$$

The product of the roots of Q equals $(A_0-(2a+1)\gamma^2)/(2a+1)$, which is negative because $A_0<\gamma^2$, so Q has one negative root and one positive root y_+, and $Q(y)<0$ for $0\leq y<y_+$. We claim that $y_+=\gamma_1-a=(2\gamma^2+a)/(2a+1)$. Indeed, for $y_0=(2\gamma^2+a)/(2a+1)$,

$$\begin{aligned}(2a+1)Q(y_0)&=(2\gamma^2+a)^2-2A_0(2\gamma^2+a)+(2a+1)A_0-(2a+1)^2\gamma^2\\ &=a(2a+1)(2\gamma^2+a)-a(2a+1)(a+2\gamma^2)=0,\end{aligned}$$

where the first two terms equal $(2\gamma^2+a)(2\gamma^2+a-2\gamma^2+2a^2)=a(2a+1)(2\gamma^2+a)$ and the last two equal $(2a+1)(\gamma^2-a^2-(2a+1)\gamma^2)=-a(2a+1)(a+2\gamma^2)$. Hence $Q(y)\geq 0$ and $y>0$ imply $y\geq\gamma_1-a$, that is $\beta\geq\gamma_1$, which again contradicts $\beta<\gamma_1$. In both cases the assumption fails, which proves the theorem.

The proof also shows what happens at the limit $\beta=\gamma_1$. For non-integer γ, the two bounds on k^2 become $k^2\leq(y_0^2-\gamma^2)/A_0$ and $k^2\geq(\gamma^2-(y_0-1)^2)/B_0$, and $Q(y_0)=0$ means that they coincide. Their common value is $k_\gamma^2=(4\gamma^2-1)/(2a+1)^2$, which one checks by substituting y_0; for $\gamma=1.5$, for example, both bounds equal $8/9=k_{1.5}^2$. So the scaling k_γ of the balanced embedding is the only scaling that reaches the radius $1/\gamma_1$ on these targets. Theorem 8.3 completes the line of analyses of Kannan's embedding with one call, from the radius $1/(2\gamma)$ of [5] over the radius of [18] for $\gamma<1.9318$ to the balanced radius $1/\gamma_1$ in the appendix of [6], which by Theorem 8.3 cannot be improved by any choice of M.

A.7 Proof of Theorem 8.4

Let $\mathbf{e}=\mathbf{t}-\mathbf{c}=\alpha\mathbf{s}$ and $d=\alpha\lambda$. As in the proof of Theorem 8.3, $\mathbf{c}$ is the unique nearest lattice vector because $\alpha<1/2$. Fix $\mathbf{z}\neq 0$ and $\mathbf{w}\in L$, and let $\mathbf{t}'=\mathbf{t}+\mathbf{w}$ and $\mathbf{c}'=\mathbf{c}+\mathbf{w}$, so that $\mathbf{t}'-\mathbf{c}'=\mathbf{e}$. The vectors of $\mathcal{E}(L_\mathbf{z},\mathbf{t}',M)$ are $(\mathbf{b}+m\mathbf{t}',mM)$ with $\mathbf{b}\in L_\mathbf{z}$ and $m\in\mathbb{Z}$.

If $\mathbf{c}' \notin L_{\mathbf{z}}$, then $(\mathbf{e},M)$ is not a vector of the embedding, since $(\mathbf{b}+m\mathbf{t}',mM)=(\mathbf{e},M)$ forces $m=1$ and $\mathbf{b}=\mathbf{e}-\mathbf{t}'=-\mathbf{c}' \notin L_{\mathbf{z}}$, and the statement holds. So assume $\mathbf{c}' \in L_{\mathbf{z}}$, in which case $(\mathbf{e},M)$ is a vector of the embedding. Let m be the integer nearest to p/α, which is at least $2p \geq 4$. Since $p\mathbf{s} \in pL \subseteq L_{\mathbf{z}}$ and $\mathbf{c}' \in L_{\mathbf{z}}$, the vector $\mathbf{b}=-p\mathbf{s}-m\mathbf{c}'$ lies in $L_{\mathbf{z}}$, and the embedding contains

$$\mathbf{v}=(\mathbf{b}+m\mathbf{t}',mM)=(-p\mathbf{s}+m\mathbf{e},mM)=((m\alpha-p)\mathbf{s},mM).$$

Its first part has norm $\lambda|m\alpha - p| = d|m - p/\alpha| \leq d/2$, and $m \leq p/\alpha + 1/2$, so $\|\mathbf{v}\|^2 \leq d^2/4+(p/\alpha+1/2)^2k^2d^2$. The assumption $p < \alpha(\sqrt{\gamma^2+(\gamma^2-1/4)/k^2}-1/2)$ is equivalent to $(p/\alpha+1/2)^2k^2 < \gamma^2k^2+\gamma^2-1/4$, hence $\|\mathbf{v}\|^2 < \gamma^2d^2(1+k^2) = \gamma^2\|(\mathbf{e},M)\|^2$. The vector $\mathbf{v}$ is not parallel to $(\mathbf{e},M)$, since $\mathbf{v}=\mu(\mathbf{e},M)$ would force $\mu=m$ and then $(m\alpha-p)\mathbf{s}=m\alpha\mathbf{s}$, that is $p=0$. If $(\mathbf{e},M)$ is a shortest vector of the embedding, then $\lambda_2 \leq \|\mathbf{v}\| < \gamma\lambda_1$ and the promise of uSVP_γ fails; otherwise the shortest vectors are not $\pm(\mathbf{e},M)$. In both cases the statement holds.

For the scaling $k=1/(n-1)$ of [6], the threshold of Theorem 8.4 is at least $\alpha((n-1)\sqrt{\gamma^2-1/4}-1/2)$. When $\alpha \geq 0.9/(\sqrt{2}\gamma)$, which holds in the upper part of the range of [6], this is at least $(0.9/\sqrt{2})((n-1)\sqrt{1-1/(4\gamma^2)}-1/(2\gamma)) \geq 0.55(n-1)-0.32$, because $\sqrt{1-1/(4\gamma^2)} \geq \sqrt{3}/2$ and $1/(2\gamma) \leq 1/2$ for $\gamma \geq 1$. For the scaling $k=\sqrt{\varepsilon}$ of TSE and $\alpha \leq 1/(\sqrt{2}\gamma)$, the threshold is at most $\alpha\gamma\sqrt{1+1/\varepsilon} \leq \sqrt{1+1/\varepsilon}/\sqrt{2} \leq h(\varepsilon)$, since $\sqrt{1+1/\varepsilon} \leq 1+1/\sqrt{\varepsilon}$. Theorem 8.4 is therefore tight for TSE up to the term 1 in $h(\varepsilon)+1$, with below its threshold no prime works, and above $h(\varepsilon)$ every prime keeps all unwanted vectors outside pL, which is what the proof of Theorem 6.3 needs. The thresholds are compared with the precise failure sets of real instances in Section 10.1.

A.8 Proof of Theorem 10.3

If the first call returns a candidate at distance d, the algorithm stops with a nearest vector. Otherwise let $F \geq 1$ be the index of the first successful trial of the fixed stream and A that of the adaptive stream, with $A=\infty$ if no adaptive trial succeeds. The streams use independent coins, so F and A are independent. Since the i-th fixed trial precedes the i-th adaptive trial and the adaptive stream has at most $|\mathcal{S}|$ trials, the algorithm stops after at most $1+2\min(F,A)$ calls, and $E[\text{calls}] \leq 1+2E[\min(F,A)]$.

A trial of the fixed stream succeeds with probability at least $1/(2p^*)$ by the proof of Theorem 7.2, because $|\mathcal{Y}| \leq N_0 \leq p^*/2$ by Corollary 8.2 and $p^* > h(\varepsilon)$. Hence F is almost surely finite, the algorithm returns a nearest vector with probability 1, and $E[\min(F,A)] \leq E[F] \leq 2p^*$, which gives the first bound.

For the second bound, consider a trial of level $j \geq j^*$. Its prime exceeds $h(\varepsilon)$, so no vector of $\mathcal{Y}^* \subseteq \mathcal{Y}$ lies in p_jL by Lemma 6.2, and Corollary 10.2 gives $\pi_{p_j} > 1-\ell/p_j \geq 1/2$, so the trial succeeds with probability at least $1/(2p_j)$ by Proposition 10.1. All $T_j = \lceil 2\ln 4 \cdot p_j \rceil$ trials of the level fail with probability at most $(1-1/(2p_j))^{T_j} \leq e^{-\ln 4} = 1/4$, independently of the other levels. Let J be the number

of levels and $S_j = T_0 + \cdots + T_j$. Then

$$E[\min(F,A)] = \sum_{t \geq 0} \Pr[\min(F,A) > t]$$
$$\leq \sum_{j<J} T_j \Pr[A > S_{j-1}] + \Pr[A = \infty] E[F],$$

because for $t < S_{J-1}$ the event $\min(F,A) > t$ implies $A > t$, and for larger t it implies $A = \infty$ and $F > t$. By Bertrand's postulate, $p_0 2^j \leq p_j < 2p_0 2^j$, so $T_j \leq 5.55 p_0 2^j + 1$ and $p_0 2^{j^*} \leq p_{j^*}$. The levels $j \leq j^*$ contribute at most $\sum_{j \leq j^*} (5.55 p_0 2^j + 1) \leq 11.1 p_{j^*} + j^* + 1$. A level $j > j^*$ is reached with probability at most $4^{-(j-j^*)}$, so these levels contribute at most $\sum_{i \geq 1} (5.55 p_{j^*} 2^i + 1) 4^{-i} \leq 5.55 p_{j^*} + 1/3$. Finally $\Pr[A = \infty] \leq 4^{-(J-j^*)}$, $E[F] \leq 2p^*$ and $p^* \leq p_J < 2p_0 2^J \leq 2p_{j^*} 2^{J-j^*}$ give $\Pr[A = \infty] E[F] \leq 4 p_{j^*} 2^{-(J-j^*)} \leq 2p_{j^*}$, since $J > j^*$. The sum is at most $18.7 p_{j^*} + j^* + 2$. If $j^* = 0$, then $p_{j^*} = p_0 < 2p_0$. Otherwise $p_{j^*-1} < 2\ell$ and $p_{j^*-1} \geq p_0 2^{j^*-1}$, so $p_0 2^{j^*} < 4\ell$ and $p_{j^*} < 2p_0 2^{j^*} < 8\ell$.

A.9 Derivation of the complexity comparison

Here, we derive the entries of Table 2 and evaluate them for one parameter set, including the cost of not knowing the distance d. Let β bound the bit length of the entries of the input basis and target.

For the three embeddings with one call, the oracle input is the basis of $\mathcal{E}(L,\mathbf{t},M)$ with the last coordinate M rounded to the precision of the input, so its entries have $\beta + O(1)$ bits, the reduction performs $O(n^2)$ operations to write it down and one membership test to check the answer, and it uses no randomness. Their radii are $\Theta(1/(\gamma\sqrt{n}))$, $1/(2\gamma)$ and $1/\gamma_1$ by the argument of Lemma 7.1 for the scalings $d/\sqrt{n}$, d and $k_\gamma d$, and Theorem 8.3 shows that no scaling exceeds $1/\gamma_1$.

A trial of the reduction of [6] draws $\mathbf{z}$ and $\mathbf{u}$ uniformly, that is $2n\log_2 p_B$ random bits, computes the basis of $L_{\mathbf{z}}$ and a shift with $\|\mathbf{t}+\mathbf{w}\| \geq (n+1)\lambda/\sqrt{2}$, and succeeds with probability at least $1/p_B - N_B/p_B^2 - N_B/p_B^{n-1}$ by [6, Lemma 13], where $N_B \leq 2\gamma n^2 \leq p_B/2$. The expected number of trials is therefore at most about $2p_B$, and Corollary 10.2 shows that it is about $p_B + 1$ when the unwanted vectors lie on one line. The Hermite normal form used in [6, Lemma 9] costs $\Theta(n^3)$ operations with elimination, and the entries of the oracle input have at most $\beta + \log_2 p_B$ bits for the sublattice basis and $\beta + \log_2 p_B + \log_2(n^2\lambda) + O(1)$ bits for the shifted target, whose norm is at least $(n+1)\lambda/\sqrt{2}$. When d is unknown, the analysis of [6] needs d and λ within a factor $1 \pm 1/n$, which takes $\Theta(n^2)$ guesses for each of them over a range of ratio 2^n, so $\Theta(n^4)$ guesses in total.

TSE draws the same number of random bits for its prime $p = \Theta(n/\varepsilon)$, computes the basis of Lemma 3.4 in $O(n^2)$ operations, and its entries have at most $\beta + \lceil \log_2 p \rceil + \lceil \log_2(n+1) \rceil + 1$ bits by Theorem 7.2. When d is unknown, the proof of Theorem 6.3 uses $G = 1 + \lceil 35.01n \rceil$ guesses of M with ratio 1.02 and $T = \lceil 2pn \rceil$ trials per guess, which gives failure probability at most e^{-n} and $GT = O(n^2 p)$ calls. The adaptive variant has the same bit lengths, since its primes are at most p^*, and Theorem 10.3 bounds its expected number of calls. The pencil reduction of

Table 6: Proven numbers of oracle calls for $n = 20$, $\gamma = 2$ and $\varepsilon = 0.1$, with the distance d known and unknown, the random bits per call and the largest bit length of an oracle entry measured on the hard instances of Section 9. For [6], the count for unknown d is given up to constants.

Reduction	Known d	Unknown d	Coins	Bits
Embeddings with one call	1	$O(n^2)$	0	21
Sparsified [6], $p_B = 3203$	≈ 6406	$\Theta(n^5 p_B)$	466	32
TSE, $p = 809$	≤ 1619	$\leq 2.3 \cdot 10^7$	386	29
Adaptive TSE, one line	≤ 117	–	≤ 386	28
Pencils, $q = 15241$	$\leq 2.2 \cdot 10^{19}$	$\leq 1.5 \cdot 10^{22}$	0	30
Shifts, $q = 7603$	$\leq 5.8 \cdot 10^7$	$\leq 4.1 \cdot 10^{10}$	0	29

Theorem 6.4 and the shift reduction of Theorem A.4 use no randomness, their primes q are about $2Nn$ and Nn, their entries have $\beta + O(\log q)$ bits, and their numbers of calls for the right guess are $(N(q-1)+2)q^3$ and $(N_0(n-1)+1)q$.

Table 6 evaluates these expressions for $n = 20$, $\gamma = 2$ and $\varepsilon = 0.1$, the parameters of the sweep in Figure 8. With a known distance, TSE is guaranteed 1619 calls in expectation against about 6406 for [6], and the adaptive variant is guaranteed 117 calls when the unwanted vectors lie on one line, since then $\ell = 1$, $p_0 = 3$ and $j^* = 0$ in Theorem 10.3. Without the distance, TSE runs $G = 702$ guesses with $T = 32360$ trials each, $2.3 \cdot 10^7$ calls to fail with probability at most e^{-20}, while [6] needs $\Theta(n^4)$ guesses with about $2p_B n$ trials each. The deterministic reductions pay for determinism with their worst-case counts, although Table 10 shows that they need at most $2.9q$ calls on our instances. The random bits per call are $2n\log_2 p$, 386 for TSE and 466 for [6], and the adaptive variant uses fewer on its small primes.

A.10 Thin pencils and a lower bound for isolating families

The pencil family $\mathcal{Z}_{n,p}$ of Lemma 4.3 contains a vector on each line of each plane spanned by two points of the moment curve, although the proof of Lemma 4.3 only needs one good pair (α, α'). A pair is good as soon as $f_{\bar{\mathbf{x}}}(\alpha') \neq 0$ and α avoids the roots of the univariate polynomials $D_{\mathbf{y}}(\cdot, \alpha')$, and since $f_{\bar{\mathbf{x}}}$ has at most $n-1$ roots, any n fixed values of α' contain one with $f_{\bar{\mathbf{x}}}(\alpha') \neq 0$. For a set $S \subseteq \mathbb{F}_p$ of n elements, the thin pencil family is

$$\mathcal{Z}^S_{n,p} = \{\mu(\alpha) + \beta\mu(\alpha') : \alpha' \in S,\ \alpha \in \mathbb{F}_p \setminus \{\alpha'\},\ \beta \in \mathbb{F}_p\},$$

of size at most $n(p-1)p$, and we take $S = \{0,1,\ldots,n-1\}$ when $p \geq n$. The Lemma A.1 tackles isolation with thin pencils.

Lemma A.1. *Let $n \geq 2$, $A \geq 0$, let $p \geq \max(A(n-1)+2, n)$ be a prime and let $S \subseteq \mathbb{F}_p$ have n elements. For every nonzero $\mathbf{x} \in \mathbb{F}_p^n$ and every set $Y \subseteq \mathbb{F}_p^n$ of at most A vectors that are each linearly independent from $\mathbf{x}$, there is $\mathbf{z} \in \mathcal{Z}^S_{n,p}$ with $\langle \mathbf{z}, \mathbf{x} \rangle = 0$ and $\langle \mathbf{z}, \mathbf{y} \rangle \neq 0$ for all $\mathbf{y} \in Y$.*

Proof. The polynomial $f_{\mathbf{x}}$ is nonzero of degree at most $n-1$, so it has at most $n-1$ roots, and there is $\alpha' \in S$ with $f_{\mathbf{x}}(\alpha') \neq 0$. For $\mathbf{y} \in Y$ let $g_{\mathbf{y}}(X) = D_{\mathbf{y}}(X, \alpha') =$

$f_{\mathbf{x}}(\alpha')f_{\mathbf{y}}(X)-f_{\mathbf{y}}(\alpha')f_{\mathbf{x}}(X)$, a polynomial of degree at most $n-1$ in X. If $g_{\mathbf{y}}$ were zero, then $f_{\mathbf{y}}=(f_{\mathbf{y}}(\alpha')/f_{\mathbf{x}}(\alpha'))f_{\mathbf{x}}$, so $\mathbf{y}$ would be a multiple of $\mathbf{x}$ because $\mathbf{v}\mapsto f_{\mathbf{v}}$ is linear and injective, which contradicts the independence of $\mathbf{y}$ from $\mathbf{x}$. Hence each $g_{\mathbf{y}}$ has at most $n-1$ roots. The values $\alpha\in\mathbb{F}_p$ that equal α' or are a root of some $g_{\mathbf{y}}$ number at most $1+A(n-1)<p$, so there is $\alpha\neq\alpha'$ with $g_{\mathbf{y}}(\alpha)\neq 0$ for all $\mathbf{y}\in Y$. Let $\mathbf{z}=\mu(\alpha)+\beta\mu(\alpha')$ with $\beta=-f_{\mathbf{x}}(\alpha)/f_{\mathbf{x}}(\alpha')$, which lies in $\mathcal{Z}^S_{n,p}$. Then $\langle\mathbf{z},\mathbf{x}\rangle=f_{\mathbf{x}}(\alpha)+\beta f_{\mathbf{x}}(\alpha')=0$, and for $\mathbf{y}\in Y$, $\langle\mathbf{z},\mathbf{y}\rangle=f_{\mathbf{y}}(\alpha)+\beta f_{\mathbf{y}}(\alpha')=g_{\mathbf{y}}(\alpha)/f_{\mathbf{x}}(\alpha')\neq 0$. □

Lemma A.1 needs a prime about half as large as Lemma 4.3 and one factor p less in the size of the family. Replacing the pencil family in Theorem 5.1 gives the following bound. The Theorem A.2 tackles deterministic isolation with thin pencils.

Theorem A.2. *Let $\gamma\geq 1$ and $n\geq 2$. There is a deterministic algorithm that receives a basis of a lattice $L\subset\mathbb{R}^n$ and an integer $A\geq A_\gamma(L)$, has access to a uSVP_γ oracle, and outputs a shortest nonzero vector of L with at most $n(p-1)p$ oracle calls on sublattices of index p, where p is the least prime with $p\geq\max(A(n-1)+2,n,2\gamma+1)$.*

Proof. The algorithm is the one of the proof of Theorem 5.1 in Appendix A.2, run over $\mathcal{Z}^S_{n,p}$ with $S=\{0,\ldots,n-1\}$ in place of $\mathcal{Z}_{n,p}$. That proof uses the pencil family only once, to obtain $\mathbf{z}$ with $\langle\mathbf{z},\bar{\mathbf{x}}\rangle=0$ and $\langle\mathbf{z},\bar{\mathbf{y}}\rangle\neq 0$ for the set Y of fewer than A residues, each independent from $\bar{\mathbf{x}}$, and it uses the prime only through $p>2\gamma-1$ and $\bar{\mathbf{y}}\neq 0$. Lemma A.1 provides such a $\mathbf{z}$ in $\mathcal{Z}^S_{n,p}$ because $p\geq\max(A(n-1)+2,n)$, and the rest of the proof applies without change. The number of calls is $|\mathcal{Z}^S_{n,p}|\leq n(p-1)p$. □

For rotations of $\mathbb{Z}^n$ and $s=\lceil\gamma^2\rceil$, the bound $A\leq(2e^3n/s)^s/2$ of Corollary 5.2 gives $p\approx A(n-1)$ for Theorem A.2 against $p\approx 2A(n-1)$ for Theorem 5.1, so the number of calls drops from about $8A^3n^3$ to about A^2n^3, a factor of about $8A$, and Corollaries 5.2 and 5.3 hold with this smaller count. The next result shows that no family can be much smaller than Ap. The Theorem A.3 handles the size of isolating families.

Theorem A.3. *Let $n\geq 3$, $A\geq 1$, and let $\mathcal{Z}\subseteq\mathbb{F}_p^n\setminus\{0\}$ be a family such that for every nonzero $\mathbf{x}$ and every set Y of at most A vectors that are each linearly independent from $\mathbf{x}$, some $\mathbf{z}\in\mathcal{Z}$ satisfies $\langle\mathbf{z},\mathbf{x}\rangle=0$ and $\langle\mathbf{z},\mathbf{y}\rangle\neq 0$ for all $\mathbf{y}\in Y$. Then $|\mathcal{Z}|\geq(A+1)(p^n-1)/(p^{n-1}-1)>(A+1)p$.*

Proof. For a nonzero $\mathbf{x}$ let $\mathcal{Z}_{\mathbf{x}}=\{\mathbf{z}\in\mathcal{Z}:\langle\mathbf{z},\mathbf{x}\rangle=0\}$. Suppose that $|\mathcal{Z}_{\mathbf{x}}|\leq A$ for some $\mathbf{x}$. For each $\mathbf{z}\in\mathcal{Z}_{\mathbf{x}}$, the hyperplane $\mathbf{z}^\perp$ has dimension $n-1\geq 2$, so it contains a vector $\mathbf{y}_{\mathbf{z}}$ outside the line $\mathbb{F}_p\mathbf{x}$, and $\mathbf{y}_{\mathbf{z}}$ is linearly independent from $\mathbf{x}$. The set $Y=\{\mathbf{y}_{\mathbf{z}}:\mathbf{z}\in\mathcal{Z}_{\mathbf{x}}\}$ has at most A elements, and every $\mathbf{z}\in\mathcal{Z}$ with $\langle\mathbf{z},\mathbf{x}\rangle=0$ satisfies $\langle\mathbf{z},\mathbf{y}_{\mathbf{z}}\rangle=0$, which contradicts the property of $\mathcal{Z}$. Hence $|\mathcal{Z}_{\mathbf{x}}|\geq A+1$ for every nonzero $\mathbf{x}$. Each nonzero $\mathbf{z}$ is perpendicular to precisely $p^{n-1}-1$ nonzero vectors, so counting the pairs $(\mathbf{x},\mathbf{z})$ with $\mathbf{x}\neq 0$, $\mathbf{z}\in\mathcal{Z}$ and $\langle\mathbf{z},\mathbf{x}\rangle=0$ in two ways gives $|\mathcal{Z}|(p^{n-1}-1)=\sum_{\mathbf{x}\neq 0}|\mathcal{Z}_{\mathbf{x}}|\geq(A+1)(p^n-1)$, and $(p^n-1)/(p^{n-1}-1)>p$. □

Theorem A.3 bounds every deterministic reduction that isolates a shortest vector by querying the sublattices of a fixed family, which includes the reductions of Theorems 5.1 and A.2. For the prime $p\approx A(n-1)$ of Theorem A.2, the thin family has about np^2 members and the lower bound is about $Ap\approx p^2/n$, so the two differ by a

Table 7: The deterministic reduction from ZSVP of Corollary 5.2, over five random bases of $\mathbb{Z}^n$ per row, with the count A of primitive pairs of $\mathbb{Z}^n$ of norm at most γ, the prime p, the size of the pencil family, and the mean and largest number of calls until a unit vector, against the mean for uniform random $\mathbf{z}$ with the same prime.

n	γ^2	A	p	$\lvert\mathcal{Z}_{n,p}\rvert$	Mean	Largest	Random
4	2	16	107	$1.2{\cdot}10^6$	21.2	36	32.0
4	3	32	211	$9.3{\cdot}10^6$	53.6	90	68.6
4	4	40	251	$1.6{\cdot}10^7$	22.8	41	43.6
5	2	25	211	$9.3{\cdot}10^6$	62.6	198	29.0
5	3	65	541	$1.6{\cdot}10^8$	79.4	149	110.8
5	4	105	853	$6.2{\cdot}10^8$	139.6	293	102.8
6	2	36	373	$5.2{\cdot}10^7$	49.6	163	83.8
6	3	116	1181	$1.6{\cdot}10^9$	140.6	211	355.4
6	4	236	2377	$1.3{\cdot}10^{10}$	245.4	485	333.6
7	2	49	607	$2.2{\cdot}10^8$	61.0	151	52.4
7	3	189	2287	$1.2{\cdot}10^{10}$	303.2	465	222.0
7	4	469	5647	$1.8{\cdot}10^{11}$	737.2	1000	1021.0
8	2	64	919	$7.8{\cdot}10^8$	203.8	459	45.4
8	3	288	4049	$6.6{\cdot}10^{10}$	273.6	408	642.0
8	4	848	11897	$1.7{\cdot}10^{12}$	1585.2	4642	2209.4

factor of about n^2, while the gap for the pencil family is about $8An^2$. The polynomial method of Lemma A.1 needs $p > A(n-1)$ because each polynomial $g_{\mathbf{y}}$ may have $n-1$ roots, as in the lemma of Schwartz and Zippel [32, 33], and whether families of size $O(Ap)$ exist for primes $p = O(A)$, which Theorem A.3 does not exclude, is open.

Table 8 compares the two families on the rotations of Section 9, extended to $n \in \{9,\ldots,12\}$, with $\gamma^2 \le 3$ for $n \ge 11$. The thin family is smaller by a factor between 110 and 16296, near the predicted factor $8A$, and it is within a factor between 12 and 132 of the lower bound of Theorem A.3, which grows like n^2 as predicted. All three strategies find a unit vector on all 125 bases, and the thin family needs 400.4 calls on average against 583.6 for the pencil family and 873.8 for random vectors with the larger prime. On every basis, the thin family stops within its first $0.75p$ members, which have $\alpha' = 0$, so that $\mathbf{z} = \mu(\alpha) + \beta\mathbf{e}_1$; as for the pencil family, the bound on the number of calls is far from the number of calls that the reduction needs on these bases.

A.11 Comparison with the sampler of Bennett et al.

The randomized reduction of [11] from ZSVP to uSVP_γ repeats a sampler that picks a prime p between $50A'\log(100A')$ and $100A'\log(100A')$ for an upper bound $A' \ge A$, draws $\mathbf{a}$ uniformly from $\mathbb{Z}_p^n$ and queries the sublattice $\{\mathbf{v} \in L : \langle \mathbf{B}^{-1}\mathbf{v}, \mathbf{a}\rangle \equiv 0 \pmod p\}$ once [11, Theorem 5.2 and Theorem 5.3]. We implemented this sampler with $A' = A$, the least prime above $50A\ln(100A)$, and the sublattice basis of Lemma 3.4. We gave it the precise count A instead of the guesses of [11, Theorem 5.3], which only helps it, and stopped it at the first unit vector.

Table 9 shows that both reductions find a unit vector on all 125 bases. The sampler needs 23934.5 calls on average, against 400.4 for the deterministic thin family, and between 17.1 and 247.6 times more calls and between 18.4 and 263.6 times more time per row. The prime of the sampler is between $369A$ and $617A$ for our values of A, so a sample keeps one of the n pairs of unit vectors with probability

Table 8: Deterministic reductions from ZSVP on five random bases of $\mathbb{Z}^n$ per row. For each row, A is the number of pairs of primitive vectors of $\mathbb{Z}^n$ of norm at most γ, followed by the prime and size of the pencil family of Theorem 5.1, the prime and size of the thin family of Theorem A.2, the lower bound of Theorem A.3 for the thin prime, and the mean number of calls until a unit vector for the thin family, the pencil family and uniform random $\mathbf{z}$ with the pencil prime.

			Pencil family		Thin family			Mean calls		
n	γ^2	A	p	$\lvert\mathcal{Z}_{n,p}\rvert$	p	$\lvert\mathcal{Z}^S_{n,p}\rvert$	Lower bound	Thin	Pencil	Random
4	2	16	107	$1.2{\cdot}10^6$	53	$1.1{\cdot}10^4$	$9.0{\cdot}10^2$	13.8	21.2	32.0
4	3	32	211	$9.3{\cdot}10^6$	101	$4.0{\cdot}10^4$	$3.3{\cdot}10^3$	31.2	53.6	68.6
4	4	40	251	$1.6{\cdot}10^7$	127	$6.4{\cdot}10^4$	$5.2{\cdot}10^3$	27.0	22.8	43.6
5	2	25	211	$9.3{\cdot}10^6$	103	$5.3{\cdot}10^4$	$2.7{\cdot}10^3$	26.2	62.6	29.0
5	3	65	541	$1.6{\cdot}10^8$	263	$3.4{\cdot}10^5$	$1.7{\cdot}10^4$	68.8	79.4	110.8
5	4	105	853	$6.2{\cdot}10^8$	431	$9.3{\cdot}10^5$	$4.6{\cdot}10^4$	162.6	139.6	102.8
6	2	36	373	$5.2{\cdot}10^7$	191	$2.2{\cdot}10^5$	$7.1{\cdot}10^3$	25.6	49.6	83.8
6	3	116	1181	$1.6{\cdot}10^9$	587	$2.1{\cdot}10^6$	$6.9{\cdot}10^4$	103.0	140.6	355.4
6	4	236	2377	$1.3{\cdot}10^{10}$	1187	$8.4{\cdot}10^6$	$2.8{\cdot}10^5$	137.0	245.4	333.6
7	2	49	607	$2.2{\cdot}10^8$	307	$6.6{\cdot}10^5$	$1.5{\cdot}10^4$	41.6	61.0	52.4
7	3	189	2287	$1.2{\cdot}10^{10}$	1151	$9.3{\cdot}10^6$	$2.2{\cdot}10^5$	112.8	303.2	222.0
7	4	469	5647	$1.8{\cdot}10^{11}$	2819	$5.6{\cdot}10^7$	$1.3{\cdot}10^6$	328.0	737.2	1021.0
8	2	64	919	$7.8{\cdot}10^8$	457	$1.7{\cdot}10^6$	$3.0{\cdot}10^4$	71.2	203.8	45.4
8	3	288	4049	$6.6{\cdot}10^{10}$	2027	$3.3{\cdot}10^7$	$5.9{\cdot}10^5$	138.0	273.6	642.0
8	4	848	11897	$1.7{\cdot}10^{12}$	5939	$2.8{\cdot}10^8$	$5.0{\cdot}10^6$	348.4	1585.2	2209.4
9	2	81	1319	$2.3{\cdot}10^9$	653	$3.8{\cdot}10^6$	$5.4{\cdot}10^4$	71.0	49.6	151.6
9	3	417	6691	$3.0{\cdot}10^{11}$	3343	$1.0{\cdot}10^8$	$1.4{\cdot}10^6$	451.4	724.8	353.8
9	4	1425	22853	$1.2{\cdot}10^{13}$	11411	$1.2{\cdot}10^9$	$1.6{\cdot}10^7$	635.2	1854.0	5840.2
10	2	100	1823	$6.1{\cdot}10^9$	907	$8.2{\cdot}10^6$	$9.2{\cdot}10^4$	68.0	130.4	64.8
10	3	580	10463	$1.1{\cdot}10^{12}$	5227	$2.7{\cdot}10^8$	$3.0{\cdot}10^6$	378.6	682.0	839.4
10	4	2260	40709	$6.7{\cdot}10^{13}$	20347	$4.1{\cdot}10^9$	$4.6{\cdot}10^7$	4642.6	2454.4	3088.6
11	2	121	2447	$1.5{\cdot}10^{10}$	1213	$1.6{\cdot}10^7$	$1.5{\cdot}10^5$	85.6	226.4	99.6
11	3	781	15643	$3.8{\cdot}10^{12}$	7817	$6.7{\cdot}10^8$	$6.1{\cdot}10^6$	684.8	1827.6	1750.8
12	2	144	3203	$3.3{\cdot}10^{10}$	1597	$3.1{\cdot}10^7$	$2.3{\cdot}10^5$	26.4	123.0	121.0
12	3	1024	22567	$1.1{\cdot}10^{13}$	11273	$1.5{\cdot}10^9$	$1.2{\cdot}10^7$	1330.6	2538.0	4184.2

about n/p, and indeed the mean number of calls times n/p is 0.96 on average. The thin family needs a prime of only about $A(n-1)$ because its analysis does not use a union bound over random choices. Moreover, it finds a unit vector early in its enumeration, because on these bases some $\mathbf{z}=\mu(\alpha)+\beta\mathbf{e}_1$ with small α keeps a unit vector and removes the other short vectors. The comparison is not a statement about the asymptotic result of [11], whose constants were not optimized, but it shows that the deterministic reduction costs no efficiency on these instances.

A.12 A deterministic decoding reduction from shifts

The deterministic reduction of Theorem 6.4 searches over pairs $(\mathbf{z},\mathbf{u})$ with $\mathbf{z}$ in the pencil family, because Lemma 4.3 must keep a residue $\bar{\mathbf{c}}+\mathbf{u}$ that nobody knows. For decoding, however, the shift is free, and it can absorb the unknown residue completely. It suffices to choose $\mathbf{z}$ so that it is not perpendicular to any unwanted residue, which does not involve $\mathbf{c}$, and then to try all q values of $\langle\mathbf{z},\bar{\mathbf{c}}+\mathbf{u}\rangle$ through shifts along one basis vector. This gives a much simpler and much smaller deterministic family that does not need pencils.

Table 9: The sampler of [11] with its prime against the thin pencil family of Theorem A.2 on five random bases of $\mathbb{Z}^n$ per row, with the prime of the sampler, and the mean number of calls and time in seconds until a unit vector.

		Sampler of [11]			Thin family	
n	γ^2	p	Calls	Time	Calls	Time
4	2	5903	2034.8	0.32	13.8	0.001
4	3	12917	4125.8	0.63	31.2	0.003
4	4	16603	3809.0	0.63	27.0	0.002
5	2	9781	1597.6	0.41	26.2	0.003
5	3	28537	5795.6	1.46	68.8	0.009
5	4	48611	9586.0	2.52	162.6	0.040
6	2	14741	3215.2	1.22	25.6	0.015
6	3	54287	8330.0	3.02	103.0	0.043
6	4	118819	11911.2	5.12	137.0	0.051
7	2	20849	2328.6	1.15	41.6	0.025
7	3	93059	14844.8	8.28	112.8	0.052
7	4	252223	21299.4	10.01	328.0	0.159
8	2	28051	1215.4	0.84	71.2	0.046
8	3	147863	32155.4	23.34	138.0	0.113
8	4	481171	57994.2	38.77	348.4	0.254
9	2	36451	3977.6	3.84	71.0	0.068
9	3	221813	20121.8	20.00	451.4	0.433
9	4	845531	87468.8	78.30	635.2	0.617
10	2	46061	4370.6	5.81	68.0	0.087
10	3	318103	25634.2	32.84	378.6	0.507
10	4	1393097	187671.6	208.14	4642.6	6.292
11	2	56891	3323.2	4.68	85.6	0.124
11	3	439949	19459.8	31.13	684.8	1.025
12	2	68947	6536.6	11.37	26.4	0.044
12	3	590713	59554.8	123.58	1330.6	2.343

Algorithm 3 Deterministic trials from shifts, for one guess of M

function SHIFTTRIALS($\mathbf{B}$,$\mathbf{t}$,M)
  $N_0 \leftarrow \min(N(n,\varepsilon),N'(\varepsilon))$; $q \leftarrow$ least prime $\geq \max(N_0(n-1)+1,h(\varepsilon)+1)$
  for $\alpha=0,1,\ldots,N_0(n-1)$ **do**
    $\mathbf{B}_\alpha \leftarrow$ basis of $L_{\mu(\alpha)}$ from Lemma 3.4
    for $\tau=0,1,\ldots,q-1$ **do**
      $\mathbf{w} \leftarrow \tau\mathbf{b}_1$; $\hat{\mathbf{c}} \leftarrow$ CAND($\mathcal{O}(\mathcal{E}(L_{\mu(\alpha)},\mathbf{t}+\mathbf{w},M)),\mathbf{t},M$)
      if $\hat{\mathbf{c}} \neq \perp$ **and** $\hat{\mathbf{c}}$ is the nearest candidate so far **then** keep $\hat{\mathbf{c}}$
    end for
  end for
end function

Algorithm 3 lists the trials of one guess of the scaling M. It uses the points $\mu(\alpha)$ of the moment curve for the first $N_0(n-1)+1$ values of α, where N_0 is the count of Corollary 8.2, and for each of them all q shifts $\tau\mathbf{b}_1$ along the first basis vector, whose residue is $\tau\mathbf{e}_1$. The candidates are extracted by the helper CAND of Algorithm 1, which returns the failure symbol unless the answer has last coordinate $\pm M$, and the output is the candidate nearest to $\mathbf{t}$. When d is known, as in our experiments, the reduction stops at the first candidate at distance d; otherwise it runs all trials of all G guesses of Theorem 6.3. The Theorem A.4 goes through the deterministic reduction from shifts.

Theorem A.4. *For every $\gamma \geq 1$ and every $\varepsilon \in (0,1/2]$ with $1/\varepsilon \leq \mathrm{poly}(n)$, there is a deterministic polynomial-time reduction from $\mathsf{BDD}_{(1-\varepsilon)/(\sqrt{2}\gamma)}$ to uSVP_γ whose oracle*

inputs have dimension $n+1$ *and which makes at most* $G(N_0(n-1)+1)q$ *oracle calls, where* $N_0=\min(N(n,\varepsilon),N'(\varepsilon))$*,* q *is the least prime at least* $\max(N_0(n-1)+1,h(\varepsilon)+1)$ *and* $G=1+\lceil 35.01n\rceil$*. The number of calls does not depend on* γ*, and it is* $O(n^3/\varepsilon^4)$ *for every* ε *and* $O(n^3)$ *for constant* ε*.*

Proof. The reduction computes $\mathbf{c}_0$ and the guesses M_i as in the proof of Theorem 6.3 in Appendix A.3, runs Algorithm 3 for each guess, and outputs the nearest vector to $\mathbf{t}$ among $\mathbf{c}_0$ and all candidates, all of which are lattice vectors. Consider the guess i^* of that proof, for which $\sqrt{\varepsilon}d\le M_{i^*}\le 1.04\sqrt{\varepsilon}d$, and a fixed nearest vector $\mathbf{c}$ with offset $\mathbf{e}=\mathbf{t}-\mathbf{c}$. By Lemma 6.2 and Corollary 8.2, the unwanted set $\mathcal{Y}$ has at most N_0 elements, and each $\mathbf{y}\in\mathcal{Y}$ satisfies $0<\|\mathbf{y}\|<h(\varepsilon)\lambda<q\lambda$, so $\bar{\mathbf{y}}\neq 0$ modulo q and $f_{\bar{\mathbf{y}}}$ is a nonzero polynomial of degree at most $n-1$. The set of $\alpha\in\mathbb{F}_q$ that are a root of some $f_{\bar{\mathbf{y}}}$ has at most $N_0(n-1)$ elements, and the $N_0(n-1)+1$ values $0,1,\ldots,N_0(n-1)$ are distinct elements of $\mathbb{F}_q$ because $q>N_0(n-1)$. So some α in this range satisfies $\langle\mu(\alpha),\bar{\mathbf{y}}\rangle=f_{\bar{\mathbf{y}}}(\alpha)\neq 0$ for all $\mathbf{y}\in\mathcal{Y}$. For this α let $\tau\in\{0,\ldots,q-1\}$ with $\tau\equiv -f_{\bar{\mathbf{c}}}(\alpha)\pmod q$, and let $\mathbf{w}=\tau\mathbf{b}_1$, so that $\bar{\mathbf{w}}=\tau\mathbf{e}_1$ and $\langle\mu(\alpha),\bar{\mathbf{c}}+\bar{\mathbf{w}}\rangle=f_{\bar{\mathbf{c}}}(\alpha)+\tau=0$, since the first coordinate of $\mu(\alpha)$ is 1. This is the event E of the proof of Theorem 6.3 for $\mathbf{z}=\mu(\alpha)$ and the shift $\mathbf{w}$, so the trial returns $\mathbf{c}$ and the output is a nearest vector. The number of calls is $G(N_0(n-1)+1)q$. Since $q\le 2(N_0(n-1)+h(\varepsilon)+1)$ by Bertrand's postulate and $G=O(n)$, the count is $O(n\cdot(nN_0)^2)=O(n^3N_0^2)$, and $N_0\le N'(\varepsilon)=O(1/\varepsilon^2)$ makes it $O(n^3/\varepsilon^4)$, which is $O(n^3)$ for constant ε. Each trial costs $O(n^2)$ arithmetic operations and one membership test, so the running time is polynomial. □

Theorem A.4 improves Theorem 6.4 in every parameter. For $\varepsilon=1/n$, which Corollary 6.5 uses, Theorem 6.4 makes $G(N(q-1)+2)q^3$ calls with $N=\Theta(n^2)$ and $q=\Theta(n^3)$, that is $\Theta(n^{15})$, while Theorem A.4 makes $G(N(n-1)+1)q=\Theta(n^7)$ calls with $q=\Theta(n^3)$. Corollary 6.5 therefore holds with $O(n^7)$ calls per instance of the self-reduction of [5]. The pencil family remains necessary for Theorem 5.1, where no target and hence no shift is available, and the shift argument is the reason why decoding is easier to derandomize than isolating a shortest vector.

Table 10 runs both deterministic reductions on the hard instances with the known distance d. The shift reduction solves all 200 instances up to $n=28$ and never needs more than $2.9q$ calls, although its bound is $(N_0(n-1)+1)q$, between $7.0\cdot 10^6$ and $2.3\cdot 10^8$, and the pencil reduction solves all 120 instances up to $n=20$ with at most 1.9 times its prime, against bounds of 10^{17} to 10^{19}. On the common instances, the shift reduction needs between 1.6 and 5.3 times fewer calls and between 2.9 and 8.0 times less time than the pencil reduction, because its prime is half as large and because a successful α is found after one pass over the shifts. Its number of calls depends only on the lattice and on the nearest point. The point $\mu(0)=\mathbf{e}_1$ avoids the unwanted residues whenever the first coordinate of the residue of $\mathbf{s}$ is nonzero, which holds on 19 of the 25 lattices, and the successful shift is then $\tau\equiv -x_1 \pmod q$ for the first coefficient x_1 of the nearest point; since our nearest points have coefficients in $\{-4,\ldots,4\}$, it is found after at most 5 calls when $x_1\le 0$ and after about q calls otherwise, which explains the medians of 4 calls and of about q calls. On the other 6 lattices, $\alpha=1$ or $\alpha=2$ succeeds after one or two further passes over the shifts, which gives the largest values, up to $2.9q$ for $n=28$. Both deterministic reductions are thus practical on these instances, and they need between 1.3 and 58 times more calls than TSE, which is the price of avoiding all randomness.

Table 10: Deterministic decoding reductions with $\varepsilon = 0.1$ on the hard instances of Section 9, with 40 runs per dimension. For each reduction, the table gives the prime, the proven number of calls for the right guess of M, and the mean, median and largest number of calls, the mean time and the mean memory until the nearest vector, for the pencil reduction of Theorem 6.4 and the shift reduction of Theorem A.4. The pencil reduction was run up to $n = 20$.

n	N	q	Bound	Mean	Median	Max	Time (s)	KiB
Pencils, Theorem 6.4								
12	240	5309	$1.9 \cdot 10^{17}$	2814.8	3489	5311	9.23	31.7
16	320	9643	$2.8 \cdot 10^{18}$	7163.2	7031	18082	38.26	51.4
20	400	15241	$2.2 \cdot 10^{19}$	14310.8	15241	24386	128.06	76.8
Shifts, Theorem A.4								
12	240	2647	$7.0 \cdot 10^{6}$	532.2	4	2647	1.16	28.9
16	320	4801	$2.3 \cdot 10^{7}$	2880.4	4799	4801	9.80	47.7
20	400	7603	$5.8 \cdot 10^{7}$	9121.6	7609	15199	43.41	72.8
24	480	11047	$1.2 \cdot 10^{8}$	6628.2	11044	11047	46.09	97.4
28	560	15121	$2.3 \cdot 10^{8}$	14745.2	15123	43473	136.13	126.7

A.13 The dimension-free count in numbers

Corollary 8.2 replaces the count $N(n,\varepsilon)$ by $N_0 = \min(N(n,\varepsilon), N'(\varepsilon))$. The two counts cross at $n = 58$, 30, 16 and 7 for $\varepsilon = 0.05$, 0.1, 0.2 and 0.5, where $N'(\varepsilon)$ equals 2112, 595, 171 and 36. Beyond the crossing, the prime of TSE no longer grows with n. For $\varepsilon = 0.1$, for example, it is 1193 for all $n \geq 30$, against 2579 and 10243 with the count $N(n,\varepsilon)$ for $n = 64$ and $n = 256$, and for $\varepsilon = 0.5$ it is 73 for all $n \geq 7$. The prime of [6] for $\gamma = 1$ is 16411 for $n = 64$ and 262147 for $n = 256$.

Figure 13 shows the three bounds for $8 \leq n \leq 256$. The count $2\gamma n^2$ of [6] grows quadratically in n and linearly in γ, the count $N(n,\varepsilon)$ grows linearly in n and does not depend on γ, and $N'(\varepsilon)$ depends on neither. The actual unwanted sets of our hard instances have at most 4 elements, far below all three bounds, and the tube experiment of Figure 4 finds at most 6 lattice points in the tube for n up to 40. The gap between $N'(\varepsilon)$ and these sizes comes from the worst case over lattices. The factor $1/\varepsilon$ per slice in $N'(\varepsilon)$ is necessary up to a constant. For $\eta = 1/k$ with an integer $2 \leq k \leq n$, the lattice $L_{k,b}$ of Proposition A.7, placed in the first k coordinates and completed by $\sqrt{2}k\mathbb{Z}^{n-k}$ in the others, has minimum $\sqrt{2}k$ and $k = 1/\eta$ lattice points in one slice of a tube of radius $\sqrt{k(k-1)}$, while Lemma 8.1 allows at most $\kappa_\theta(\eta) \leq 1.5/\eta$ for $\theta = 1/\sqrt{3}$.

A.14 Consequences for the decision problem

Lyubashevsky and Micciancio reduce uSVP_γ to the decision problem GapSVP_γ for polynomially bounded γ, by a deterministic procedure that queries sublattices of the input and a binary search on the distance parameter [5, Theorem 6.1]. Their reduction applies to instances with $\lambda_2 > \gamma\lambda_1$. Combined with their deterministic reduction from $\mathsf{BDD}_{1/(2\gamma)}$ to uSVP_γ, it reduces $\mathsf{BDD}_{1/(2\gamma)}$ to GapSVP_γ deterministically, and the reduction of [6] improves the factor to $1/(\sqrt{2}\gamma)$ with randomness. The Corollary A.5 handles the decision problem.

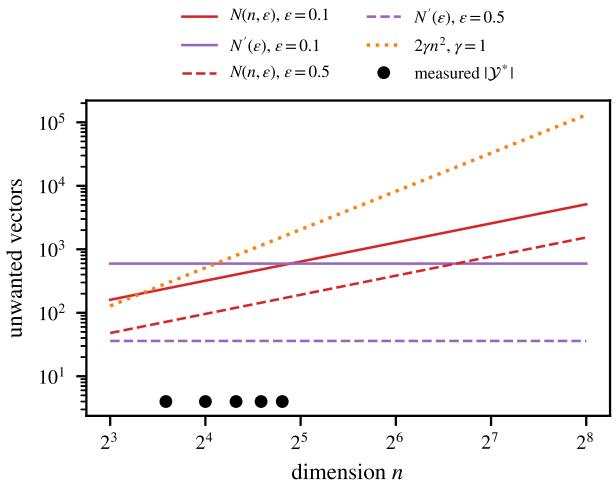


Fig. 13: Bounds on the number of unwanted vectors as a function of the dimension, with the tube count $N(n,\varepsilon)$ of Lemma 6.2, the dimension-free count $N'(\varepsilon)$ of Corollary 8.2, the count $2\gamma n^2$ of [6] for $\gamma=1$, and the largest size of $\mathcal{Y}^*$ on the hard instances of Section 9 for $\varepsilon=0.1$.

Corollary A.5. *For every polynomially bounded $\gamma \geq 1$, there is a deterministic polynomial-time reduction from $\mathsf{BDD}_{1/(\sqrt{2}\gamma)}$ to GapSVP_γ whose oracle inputs have dimension $n+1$ if $\gamma \geq \sqrt{2}$ and $n+2$ otherwise.*

Proof. Run the deterministic reduction of Corollary 6.5, and answer each of its uSVP_γ queries with the procedure of [5, Theorem 6.1], stopped after its polynomial number of steps, with the output discarded if the procedure does not return a nonzero vector of the queried lattice. By the proof of Theorem 6.3, every query on which the reduction relies satisfies the conclusion of Lemma 6.1, so every vector that is not parallel to the shortest vector is longer than γ times the minimum, $\lambda_2 > \gamma\lambda_1$, and the procedure returns a shortest vector. On the other queries its output is arbitrary, which the reduction tolerates, because it keeps only candidates that are lattice vectors and outputs the one nearest to the target. All queries of the procedure are on sublattices of the queried lattice, whose dimension is given by Corollary 6.5. □

Corollary A.5 improves the deterministic factor from $1/(2\gamma)$ to $1/(\sqrt{2}\gamma)$. For super-polynomial γ, the procedure of [5] is not polynomial, because it enumerates the $p+1$ sublattices of index p of a plane for a prime $p > \gamma$. There our reduction to uSVP_γ remains deterministic and polynomial, and the missing link is a reduction from uSVP_γ to GapSVP_γ for large γ.

A.15 Unwanted multiples and the distribution of calls

This section describes the precise unwanted set of the hard instances and compares the measured mean numbers of calls with Corollary 10.2.

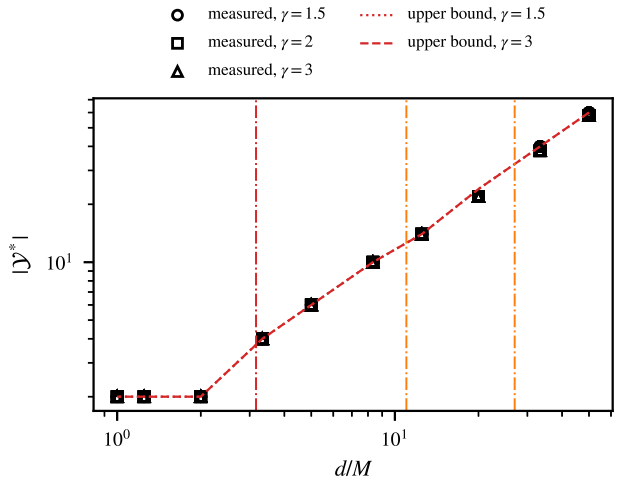


Fig. 14: Size of the precise unwanted set $\mathcal{Y}^*$ on hard instances with $\gamma\alpha \approx 0.6$ as a function of $d/M = 1/k$, for $n \in \{12,20,28\}$, three lattices and three gaps, against the upper count $2(\lceil \alpha\gamma(1+k^2)/k \rceil - 1)$ of Proposition A.6. The dash-dotted lines mark the scaling of TSE and the scalings of [6] for $n = 12$ and $n = 28$.

On the hard instances, the unwanted set can be described precisely. The Proposition A.6 addresses the unwanted multiples.

Proposition A.6. *Let* $\mathbf{s}$ *be a shortest vector of* L *with* $\lambda = \|\mathbf{s}\|$, $\mathbf{t} = \mathbf{c} + \alpha\mathbf{s}$ *with* $\mathbf{c} \in L$ *and* $0 < \alpha < 1/2$, $d = \alpha\lambda$ *and* $M = kd$. *For* $j \neq 0$, $j\mathbf{s} \in \mathcal{Y}^*$ *precisely when some integer* m *satisfies* $(j/\alpha + m)^2 + k^2m^2 < \gamma^2(1+k^2)$. *This holds for every* j *with* $0 < |j| < \alpha(1+k^2)\sqrt{\gamma^2 - 1/4}/k$ *and for no* j *with* $|j| \geq \alpha\gamma(1+k^2)/k$.

Proof. Since $\mathbf{e} = \alpha\mathbf{s}$,

$$\|j\mathbf{s} + m\mathbf{e}\|^2 + m^2M^2 = \alpha^2\lambda^2\big((j/\alpha + m)^2 + k^2m^2\big)$$

and $\gamma^2(d^2 + M^2) = \gamma^2\alpha^2\lambda^2(1+k^2)$, which gives the equivalence. The function $f(x) = (j/\alpha + x)^2 + k^2x^2$ is minimized over the reals at $x^* = -(j/\alpha)/(1+k^2)$, where $f(x^*) = (j/\alpha)^2k^2/(1+k^2)$, and $f(x^* + \delta) = f(x^*) + (1+k^2)\delta^2$. If $|j| \geq \alpha\gamma(1+k^2)/k$, then $f(x) \geq f(x^*) \geq \gamma^2(1+k^2)$ for all x. If $|j| < \alpha(1+k^2)\sqrt{\gamma^2 - 1/4}/k$, the integer m nearest to x^* satisfies $f(m) \leq f(x^*) + (1+k^2)/4 < (\gamma^2 - 1/4)(1+k^2) + (1+k^2)/4 = \gamma^2(1+k^2)$. □

Proposition A.6 shows that the multiples of $\mathbf{s}$ in $\mathcal{Y}^*$ number about $2\alpha\gamma/k$ for small k, so the length of the tube in units of λ is inversely proportional to the scaling. Theorem 8.4 is the special case $j = p$, since $p\mathbf{s} \in \mathcal{Y}^*$ lies in every sublattice $L_{\mathbf{z}}$.

Figure 14 computes $\mathcal{Y}^*$ for 270 combinations of dimension, gap, lattice and ten scalings k between 0.02 and 1. In all 270 cases, every vector of $\mathcal{Y}^*$ is a multiple of $\mathbf{s}$, so no other lattice vector enters the tube on these instances. The number of multiples lies between the two bounds of Proposition A.6 in all cases and equals the upper count in 207 of them. It does not depend on γ for a fixed normalized

Table 11: Precise prediction of Corollary 10.2 against measurements on the 40 hard runs per dimension of Section 9, with the prime, the sizes of $\mathcal{Y}^*$ and its number of lines, the predicted and the measured mean number of oracle calls. The last two rows repeat TSE 40 times with independent coins on each of five lattices.

	TSE				[6]			
n	p	$\lvert\mathcal{Y}^*\rvert$, lines	Pred.	Meas.	p_B	$\lvert\mathcal{Y}^*\rvert$, lines	Pred.	Meas.
12	487	2–4, 1	489.0	418.4	727–1733	10–12, 1	1123.5	1228.5
16	641	2–4, 1	643.0	328.4	1283–3079	14–18, 1	1990.5	1696.5
20	809	2–4, 1	811.0	246.0	2003–4801	18–22, 1	3105.5	2653.8
24	967	2–4, 1	969.0	1057.6	2887–6917	24–26, 1	4471.5	6328.3
28	1123	2–4, 1	1125.0	1397.4	3923–9413	26–32, 1	6084.5	5395.4
all			807.4	689.6			3355.1	3460.5
12, ×40	487	2–4, 1	489.0	542.4				
20, ×40	809	2–4, 1	811.0	823.7				

radius $\gamma\alpha$, grows like $1/k$, and ranges from 2 to 60. For the scaling $k=\sqrt{0.1}$ of TSE, the proposition gives $|j|\leq 2$ and at most 4 unwanted vectors, and for the scaling $k=1/(n-1)$ of [6] about $1.2(n-1)$, which matches the sizes of Table 11.

Table 11 computes $\mathcal{Y}^*$ precisely for each hard instance by enumerating, for every integer m with $m^2M^2<\gamma^2(d^2+M^2)$, the lattice vectors in the ball of radius $(\gamma^2(d^2+M^2)-m^2M^2)^{1/2}$ around $-m\mathbf{e}$ with fplll [39], and evaluates Proposition 10.1 for the prime of each reduction. For the scaling of TSE, $\mathcal{Y}^*$ consists of $\pm\mathbf{s}$ and possibly $\pm 2\mathbf{s}$, on a single line, so the prediction is $1+p/\pi_p\approx p+2$ calls, while the longer tube of [6] contains 10 to 32 vectors, still on the line of $\mathbf{s}$. The measured means over the 40 runs per dimension deviate from the prediction because the 40 runs of a dimension use only five lattices, nearest points and coin sequences, as explained in Section 9; averaged over all 200 hard runs, the predictions are 807.4 and 3355.1 calls against the measured 689.6 and 3460.5. With 40 independent coin sequences per lattice, the measured means of TSE are 542.4 and 823.7 calls for $n=12$ and $n=20$ against the predicted 489.0 and 811.0, which is within 1.6 standard errors of the mean.

A.16 Further measurements of the adaptive variant

This section measures the adaptive variant along the radius and in larger dimensions. Figure 15 shows the number of calls along the radius sweep of Figure 8, where all four reductions solve all 1400 instances. Both variants of TSE make one call up to the balanced radius and switch to sparsification beyond it, where TSE needs 836.3 and 819.1 calls and the adaptive variant 11.0 and 11.1 calls on average for $\gamma=1.5$ and $\gamma=2$. The reduction of [6] needs about its prime, 2411 and 3203, at each radius. The mean number of calls of adaptive [6] grows in steps from 8.4 and 7.0 at the smallest radius to 228.1 and 211.4 at the largest, with single runs between 2 and 397 calls, because the threshold of Theorem 8.4 grows linearly with α and each step moves its first successful level up. Over all radii, adaptive TSE uses 3.6 and 4.0 calls on average, against 52.5 and 65.8 for adaptive [6], 215.8 and 241.7 for TSE and 2457.5 and 3312.4 for [6].

Table 12 extends the comparison to dimensions 32, 36 and 40, where one oracle call takes between 10 ms and more than one second because the enumeration of our oracle grows exponentially with the dimension. All reductions that we ran solve

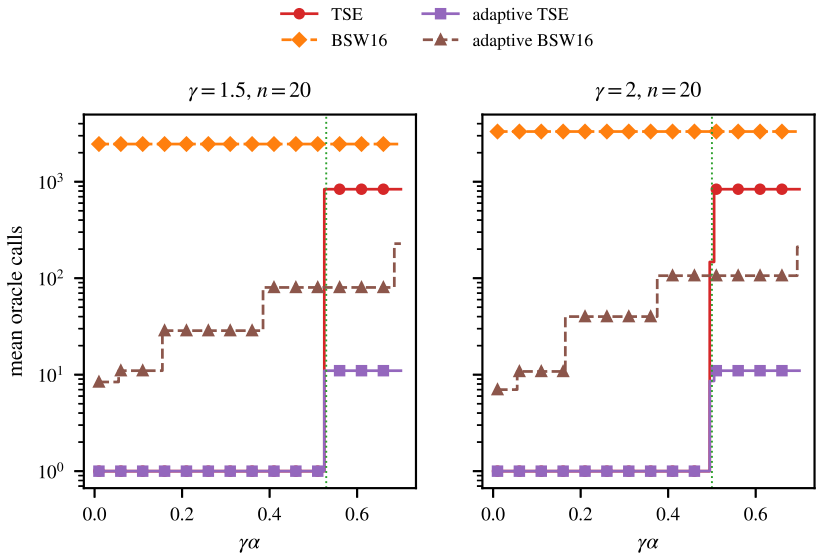


Fig. 15: Mean oracle calls along the radius sweep of Figure 8, for $n = 20$ and ten offsets along a shortest vector per radius, of TSE, of [6] and of their adaptive variants, with markers at every fifth radius. The dotted line is the balanced radius, below which the curve of TSE lies under that of its adaptive variant.

all instances, including TSE with the dimension-free prime 1193 of Corollary 8.2, which needs 1237.6 and 833.6 calls for $n=32$ and $n=36$, of the order of its prime, as Corollary 10.2 predicts. For $n=32$, adaptive TSE needs 14.4 calls and 0.26 s against 1433.2 calls and 47.3 s for TSE and 6936.4 calls and 256.1 s for [6], and for $n=40$ it needs 10.6 calls and 10.9 s, against 208.2 calls and 237.6 s for adaptive [6]. Since the cost of the oracle dominates in these dimensions, the number of calls translates directly into time, and only the adaptive variant of TSE keeps the reduction within a small constant number of oracle calls in all dimensions.

A.17 Choice of the slack

The slack ε trades the radius $(1-\varepsilon)/(\sqrt{2}\gamma)$ against the prime $\Theta(n/\varepsilon)$ of TSE, and Theorem 10.3 predicts that the adaptive variant depends on ε only through $p_0 \approx h(\varepsilon)$. We ran both variants with seven slacks from 0.02 to 0.5 on the hard lattices of dimension 20, for $\gamma \in \{1.5,2\}$ and the normalized radii $\gamma\alpha \in \{0.55,0.60\}$, with five instances per combination.

Table 13 lists the results, which coincide for the four combinations of gap and radius because they share the lattice, the nearest point and the coins, and only the unwanted set could differ. The number of calls of TSE follows its prime, which grows like $1/\varepsilon$, from 241 for $\varepsilon = 0.5$ to 3449 for $\varepsilon = 0.02$, and the adaptive variant needs between 7 and 18 calls for every slack. All runs succeed, also for the slack 0.2 with $\gamma\alpha = 0.60$ and for the slacks 0.3 and 0.5, whose proven radius lies below the radius of

Table 12: Hard instances of dimension 32 to 40, with ten runs per row over $\gamma \in \{1.5,2\}$ and the radius R3b, with fixed prime, mean and largest number of calls, mean time in seconds and largest bit length of an oracle entry. The balanced embedding fails on all of them, and dashes mark reductions that we did not run because of their time.

n	Reduction	Prime	Mean	Largest	Time (s)	Bits
32	Balanced (solves 0)	–	1.0	1	0.01	22
32	Adaptive TSE	1283	14.4	29	0.26	30
32	TSE	1283	1433.2	3309	47.32	30
32	TSE with N_0	1193	1237.6	2871	32.15	30
32	Adaptive [6]	6151	192.0	274	3.56	33
32	[6]	6151	6936.4	19188	256.10	34
36	Balanced (solves 0)	–	1.0	1	0.05	21
36	Adaptive TSE	1447	9.4	15	0.66	30
36	TSE	1447	643.4	1934	107.25	30
36	TSE with N_0	1193	833.6	1717	113.41	30
36	Adaptive [6]	7789	197.6	382	15.99	34
36	[6]	–	–	–	–	–
40	Balanced (solves 0)	–	1.0	1	0.98	21
40	Adaptive TSE	1601	10.6	19	10.89	30
40	TSE	–	–	–	–	–
40	TSE with N_0	–	–	–	–	–
40	Adaptive [6]	9601	208.2	302	237.59	34
40	[6]	–	–	–	–	–

Table 13: Oracle calls of TSE and adaptive TSE for seven slacks ε on the hard lattices with $n = 20$, over 20 runs per slack for $\gamma \in \{1.5,2\}$ and $\gamma\alpha \in \{0.55,0.60\}$, with the prime, the largest radius $\gamma\alpha = (1-\varepsilon)/\sqrt{2}$ with a guarantee, and the mean number of calls. All runs succeed.

ε	p	Proven $\gamma\alpha$	TSE	Adaptive TSE
0.02	3449	0.693	1791	18
0.05	1481	0.672	1331	10
0.10	809	0.636	246	9
0.15	563	0.601	316	14
0.20	443	0.566	462	7
0.30	331	0.495	228	9
0.50	241	0.354	305	7

the instance. There the guarantee of Theorem 7.2 does not apply, and for $\varepsilon = 0.5$ and $\gamma\alpha = 0.60$ even $r_0 = \gamma d\sqrt{1.5}$ exceeds $\lambda_1/\sqrt{2}$, but the unwanted set of these instances still lies on the line of $\mathbf{s}$. In practice, the slack can therefore be chosen as large as the radius of the instance permits, which makes the prime of TSE small, while the adaptive variant does not need this choice at all.

A.18 Tightness of the tube lemma and a lower bound for sparsification

Lemma 4.2 allows n lattice points in each slice, and Lemma 8.1 replaces n by $\kappa_\theta(\eta)$, which is smaller when η is larger than about $1/n$. This section shows that n points in one slice do occur when $\eta = 1/n$, and that such configurations force every sparsified trial with uniform sublattices to fail with a probability that grows with n. For an integer b, let $\mathbf{1} = (1,\ldots,1)$ and let $L_{n,b}$ be the lattice generated by

$$\mathbf{w}_i = (b-1)\mathbf{1} + n\mathbf{e}_i, \qquad i = 1,\ldots,n.$$

The Proposition A.7 addresses n points in one slice.

Proposition A.7. *Let $n \geq 2$ and $b \geq \sqrt{n+1}$. For $\mathbf{x} \in \mathbb{Z}^n$ with $k = \sum_i x_i$, $\|\sum_i x_i \mathbf{w}_i\|^2 = n(b^2-1)k^2 + n^2\|\mathbf{x}\|^2$, so $\lambda_1(L_{n,b}) = \sqrt{2}n$. For $\mathbf{u} = \mathbf{1}/\sqrt{n}$, the n vectors $\mathbf{w}_i$ satisfy $\langle \mathbf{w}_i, \mathbf{u} \rangle = b\sqrt{n}$ and lie at distance $r = \sqrt{n(n-1)} < \lambda_1/\sqrt{2}$ from the point $b\sqrt{n}\mathbf{u}$, and $\eta = 1 - 2r^2/\lambda_1^2 = 1/n$. Hence the tube of radius r around every segment in the direction $\mathbf{u}$ that contains $b\sqrt{n}\mathbf{u}$ has n lattice points in one slice, which attains the bound per slice of Lemmas 4.2 and 8.1.*

Proof. The vector $\sum_i x_i \mathbf{w}_i = (b-1)k\mathbf{1} + n\mathbf{x}$ has squared norm $n(b-1)^2k^2 + 2n(b-1)k^2 + n^2\|\mathbf{x}\|^2 = n(b^2-1)k^2 + n^2\|\mathbf{x}\|^2$. For $\mathbf{x} \neq 0$ with $k = 0$ we have $\|\mathbf{x}\|^2 \geq 2$, and for $k \neq 0$ we have $\|\mathbf{x}\|^2 \geq 1$ and $n(b^2-1)k^2 \geq n^2$, so the squared norm is at least $2n^2$ in both cases, with equality for $\mathbf{x} = \mathbf{e}_1 - \mathbf{e}_2$. Next, $\langle \mathbf{w}_i, \mathbf{1} \rangle = n(b-1) + n = nb$, so $\langle \mathbf{w}_i, \mathbf{u} \rangle = b\sqrt{n}$, and $\mathbf{w}_i - b\mathbf{1} = n\mathbf{e}_i - \mathbf{1}$ has squared norm $(n-1)^2 + (n-1) = n(n-1)$. Since $n(n-1) < n^2 = \lambda_1^2/2$, the distance r is below $\lambda_1/\sqrt{2}$, and $\eta = 1 - 2n(n-1)/(2n^2) = 1/n$. All n vectors have the same value of σ, so they lie in one slice of every partition used in the proofs of Lemmas 4.2 and 8.1. For $\eta = 1/n$ the bound of Lemma 8.1 per slice is $\min(n, \kappa_\theta(1/n)) = \min(n, \lceil (n-1)/(1-\theta^2) \rceil) = n$ for every $\theta \in (0,1)$, which is attained. □

We verified Proposition A.7 with the enumeration of Section 9, which finds precisely $2n$ nonzero lattice points in the tube of radius $\sqrt{n(n-1)}$ around the segment from $-b\sqrt{n}\mathbf{u}$ to $b\sqrt{n}\mathbf{u}$, which are $\pm\mathbf{w}_i$, with precisely n of them in each of the two slices, for every even n from 4 to 24 and $b = \lceil \sqrt{n+1} \rceil$. The next theorem places the n points of one slice in the unwanted set of a decoding instance. Since their residues are linearly independent, a uniform sublattice of index p removes all of them with probability only about $(1-1/p)^n$, whatever the prime.

The Theorem A.8 addresses a lower bound for uniform sparsification.

Theorem A.8. *Let $n \geq 3$, $\gamma \geq 1$ and $0 < \varepsilon \leq 1/(40n)$, let $\tau_0 = (1-\varepsilon)\sqrt{n}/\gamma$ and let τ be rational with $(1-\varepsilon)\tau_0 \leq \tau \leq \tau_0$. Let K be the least integer $K \geq 2$ with $K\tau \geq \sqrt{n+1} + 1/2$, let b be an integer nearest to $K\tau$, $L = L_{n,b}$ and $\mathbf{t} = \tau\mathbf{1}$. Then $\mathbf{t}$ is an instance of $\mathsf{BDD}_{(1-\varepsilon)/(\sqrt{2}\gamma)}$ whose unique nearest lattice vector is 0, and for every M with $\sqrt{\varepsilon}d \leq M \leq 1.04\sqrt{\varepsilon}d$, where $d = \tau\sqrt{n}$, the set $\mathcal{Y}^*$ contains $\mathbf{w}_1, \ldots, \mathbf{w}_n$. Consequently, with the failure-symbol oracle, a trial with any prime p, uniform nonzero $\mathbf{z}$ and uniform $\mathbf{u}$ succeeds with probability at most $(p-1)^n/(p(p^n-1)) \leq 4/(3en)$, and every sequence of such trials, with primes chosen in any way, needs at least $3en/4 > 2n$ trials in expectation.*

Proof. Since $b \geq K\tau - 1/2 \geq \sqrt{n+1}$, Proposition A.7 gives $\lambda = \lambda_1(L) = \sqrt{2}n$. Then $d = \tau\sqrt{n} \leq \tau_0\sqrt{n} = (1-\varepsilon)n/\gamma = (1-\varepsilon)\lambda/(\sqrt{2}\gamma)$.

We show that 0 is the unique nearest lattice vector. For $\mathbf{v} = \sum_i x_i \mathbf{w}_i$ with $k = \sum_i x_i$, write $\mathbf{v} = bk\mathbf{1} + (n\mathbf{x} - k\mathbf{1})$, where the second term is perpendicular to $\mathbf{1}$ and has squared norm $n^2\|\mathbf{x}\|^2 - nk^2$. Hence $\|\mathbf{t} - \mathbf{v}\|^2 = n(bk-\tau)^2 + n^2\|\mathbf{x}\|^2 - nk^2$, and $n^2\|\mathbf{x}\|^2 - nk^2 \geq 0$ by the Cauchy–Schwarz inequality. For $\mathbf{x} \neq 0$ with $k = 0$, $\|\mathbf{t} - \mathbf{v}\|^2 = n\tau^2 + n^2\|\mathbf{x}\|^2 > d^2$. For $k \leq -1$, $(bk-\tau)^2 \geq (b+\tau)^2 > \tau^2$. For $k \geq 2$, $bk - \tau \geq 2\sqrt{n+1} - \sqrt{n} > \sqrt{n} \geq \tau$. For $k = 1$, $\|\mathbf{x}\|^2 \geq 1$ gives $\|\mathbf{t} - \mathbf{v}\|^2 \geq n(b-\tau)^2 + n(n-1)$, and $b - \tau \geq (K-1)\tau - 1/2 \geq \tau - 1/2$. If $\tau \geq 1/2$, then $(b-\tau)^2 + n - 1 \geq \tau^2 - \tau + n - 3/4 > \tau^2$ because $\tau \leq \sqrt{n} < n - 3/4$ for $n \geq 3$; if $\tau < 1/2$, then $n - 1 > \tau^2$ already. In all cases $\|\mathbf{t} - \mathbf{v}\| > d$.

Table 14: The instances of Theorem A.8 with $\gamma = 2$ and $\varepsilon = 1/(40n)$, with the parameters K and b, the size and number of lines of $\mathcal{Y}^*$, and the mean, smallest and largest number of oracle calls of the sparsified phases of TSE over 5 runs and of adaptive TSE over 20 runs, against the lower bound $3en/4$.

				TSE		Adaptive TSE			
n	K	b	$\lvert\mathcal{Y}^*\rvert$, lines	p^*	Mean	Mean	Min	Max	$3en/4$
6	3	4	14, 7	4621	6683.8	52.7	2	246	12.2
8	3	4	18, 9	8117	4286.4	84.8	6	412	16.3
10	3	5	22, 11	12637	5259.8	83.1	4	398	20.4
12	3	5	26, 13	18077	11062.4	95.5	4	302	24.5

Next we show $\mathbf{w}_i \in \mathcal{Y}^*$ with $m = -K$. Since $\mathbf{w}_i - K\mathbf{t} = (b - K\tau)\mathbf{1} + (n\mathbf{e}_i - \mathbf{1})$ with perpendicular summands, $\|\mathbf{w}_i - K\mathbf{t}\|^2 = n(b - K\tau)^2 + n(n-1)$, and $M^2 \le 1.0816\,\varepsilon d^2 = 1.0816\,\varepsilon n\tau^2$. As $M^2 \ge \varepsilon d^2$, it suffices to show $(b-K\tau)^2 + (n-1) + 1.0816\varepsilon K^2\tau^2 < \gamma^2(1+\varepsilon)\tau^2$. We have $(b-K\tau)^2 \le 1/4$. If $K = 2$, then $K^2\tau^2 \le 4n$; if $K \ge 3$, the minimality of K gives $(K-1)\tau < \sqrt{n+1} + 1/2$, so $K\tau < 2\sqrt{n+1} + 1/2$ and $K^2\tau^2 < 4(n+1) + 2\sqrt{n+1} + 1/4 \le 5n + 6.25 \le 8.2n$. With $\varepsilon \le 1/(40n)$, the left side is therefore at most $n - 1 + 0.25 + 1.0816 \cdot 8.2/40 < n - 0.52$. On the right side, $\gamma^2\tau^2 \ge (1-\varepsilon)^2\gamma^2\tau_0^2 = (1-\varepsilon)^4 n \ge (1-4\varepsilon)n \ge n - 0.1$. Hence $\mathbf{w}_i \in \mathcal{Y}^*$ for all i.

For the last claim, the distribution of $L_{\mathbf{z}}$ for uniform nonzero $\mathbf{z}$ is uniform over the $(p^n-1)/(p-1)$ sublattices of index p that contain pL, whatever the basis, because a change of basis permutes the nonzero vectors $\mathbf{z}$. We may therefore take coordinates in the basis $\mathbf{w}_1, \ldots, \mathbf{w}_n$, in which $\bar{\mathbf{w}}_i = \mathbf{e}_i$, so $L_{\mathbf{z}} \cap \{\mathbf{w}_1, \ldots, \mathbf{w}_n\} = \emptyset$ precisely when all coordinates of $\mathbf{z}$ are nonzero, which has probability $(p-1)^n/(p^n-1)$. By Proposition 10.1, a trial succeeds only if this happens and $\langle \mathbf{z}, \bar{\mathbf{c}} + \mathbf{u} \rangle = 0$, which has probability $1/p$ independently. Since $p^n \ge 4$, $p^n/(p^n-1) \le 4/3$, and $(1/p)(1-1/p)^n \le e^{-n/p}/p \le 1/(en)$ because $x \mapsto e^{-n/x}/x$ is maximized at $x = n$. So every trial succeeds with probability at most $4/(3en)$, and the expected number of trials until the first success is at least $\sum_{t \ge 0}(1 - 4/(3en))^t = 3en/4$. □

Theorem A.8 shows that the linear dependence on n of the count $N(n,\varepsilon)$ cannot be avoided by any choice of primes when the slack is below $1/(40n)$, and that sparsification with uniform sublattices needs $\Omega(n)$ oracle calls on some instances. For such slacks, TSE uses $\Theta(n^2)$ calls and adaptive TSE uses $O(n)$ calls when the unwanted lines are the $n+1$ lines of this instance, so adaptive TSE is optimal up to a constant factor among reductions that use uniform sparsified sublattices on these instances. The instance is not hard for the balanced embedding, whose larger scaling $k_\gamma d$ makes $\mathbf{w}_i - K\mathbf{t}$ padded with $-Kk_\gamma d$ long, and the theorem therefore concerns the sparsified phases only; combining it with a target along a shortest vector, as in Theorem 8.3, is a natural next step.

Figure 16 measures the success probability of single trials on the instances of Theorem A.8 for five primes between $n/2$ and $8n$, with 2647 to 16476 trials per prime. The precise computation finds the $2n+2$ vectors $\pm\mathbf{w}_i$ and $\pm\sum_i \mathbf{w}_i$ in $\mathcal{Y}^*$, on $n+1$ lines, so the precise value lies slightly below the bound of the theorem, which only uses the first n lines. After multiplication by n, all curves nearly coincide as functions of p/n and peak near $p = n$ at about 0.33, below the maximum $4/(3e) \approx 0.49$ of the bound, and the measurements follow the precise values within the sampling error.

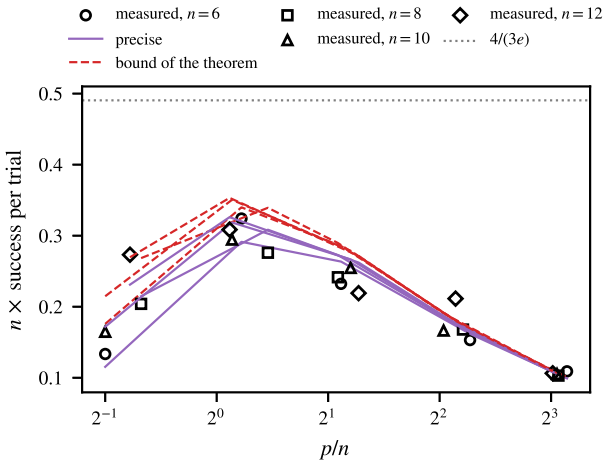


Fig. 16: Success probability of one sparsified trial on the instance of Theorem A.8 with $\gamma=2$ and $\varepsilon=1/(40n)$, multiplied by n, as a function of p/n, for $n\in\{6,8,10,12\}$, with measured values, the precise value of Proposition 10.1, the bound $(p-1)^n/(p(p^n-1))$ of Theorem A.8 and its maximum $4/(3e)$.

Table 14 runs the sparsified phases of both variants of TSE on these instances. The prime of TSE for $\varepsilon=1/(40n)$ is about $120n^2$, and its sparsified phase needs between 4286 and 11062 calls on average. Adaptive TSE needs between 52.7 and 95.5 calls, which is between 3.9 and 5.2 times the lower bound $3en/4$ of Theorem A.8 for all four dimensions, in agreement with the bound $O(\ell)$ of Theorem 10.3 for the $\ell=n+1$ unwanted lines of these instances.

A.19 Beyond the proven radius

All proven reductions stop at the normalized radius $\gamma\alpha=1/\sqrt{2}$, because the tube lemma needs $r_0<\lambda_1/\sqrt{2}$. Proposition A.6, however, describes the multiples of $\mathbf{s}$ in $\mathcal{Y}^*$ for every $\alpha<1/2$, and only the other lattice vectors near the segment depend on the radius $\lambda_1/\sqrt{2}$. We therefore ran the reductions on hard instances of dimension 20 with $\gamma=2$ and eight normalized radii between 0.60 and 0.95, the last one close to the largest radius $\gamma/2$ for which the nearest point is unique, with five lattices per radius. We also computed $\mathcal{Y}^*$ precisely for each of these instances.

In Table 15, all sparsified reductions have a guarantee at $\gamma\alpha=0.60$, at 0.65 only [6], whose radius is 0.672 for $n=20$, and TSE with a slack $\varepsilon\le0.08$, and at 0.70 only TSE with a slack $\varepsilon<0.01$. All three sparsified reductions solve all 40 instances, including those with $\gamma\alpha\ge0.75$, where no known reduction has a guarantee and r_0 exceeds $\lambda_1/\sqrt{2}$. The precise unwanted set consists of 4 multiples of $\mathbf{s}$ up to $\gamma\alpha=0.85$ and of 6 multiples for 0.90 and 0.95, as Proposition A.6 predicts, and it contains no other lattice vector at any radius. The success probability of a trial is therefore still about $1/p$, and TSE and [6] use the same numbers of calls at every radius, since their coins and nearest points do not depend on the radius. Adaptive TSE needs

Table 15: Hard instances with $n=20$, $\gamma=2$ and radii beyond the proven range, five lattices per row, with the reductions that have a guarantee at the radius, the size of $\mathcal{Y}^*$ and the number of its vectors outside the line of $\mathbf{s}$, the number of runs solved by the balanced embedding, and the mean number of calls of three sparsified reductions, which solve all runs.

$\gamma\alpha$	Guarantee	$\lvert\mathcal{Y}^*\rvert$	Off line	Bal.	A-TSE	TSE	[6]
0.60	all	4–4	0	0	9.4	246.0	2302.8
0.65	[6], small ε	4–4	0	0	9.4	246.0	2302.8
0.70	small ε	4–4	0	0	9.4	246.0	2302.8
0.75	none	4–4	0	0	9.4	246.0	2302.8
0.80	none	4–4	0	0	9.4	246.0	2302.8
0.85	none	4–4	0	0	9.4	246.0	2302.8
0.90	none	6–6	0	0	27.0	246.0	2302.8
0.95	none	6–6	0	0	27.0	246.0	2302.8

Table 16: Gap λ_2/λ_1 of the oracle inputs on the 50 hard instances per gap γ, with the number of instances on which $(\mathbf{e},M)$ is a shortest vector and the range of the gap for the embeddings with one call, and for one successful trial of TSE the number of instances whose gap exceeds $\gamma+1$ and the gaps of the others.

	$M=d/\sqrt{n}$		$M=d$ [5]		Balanced		TSE, one success	
γ	Short.	Gap	Short.	Gap	Short.	Gap	$>\gamma+1$	Others
1.25	0	–	50	1.03–1.14	50	1.03–1.17	48	2.14, 2.21
1.5	0	–	50	1.25–1.39	50	1.27–1.41	50	–
2	10	1.00–1.56	50	1.70–1.88	50	1.61–1.85	50	–
3	45	1.00–1.51	50	2.54–2.82	50	2.46–2.78	50	–

9.4 calls up to $\gamma\alpha=0.85$ and 27 calls from 0.90 on, where the multiple $3\mathbf{s}$ joins $\mathcal{Y}^*$ and the prime 3 of the first level no longer keeps all unwanted vectors outside $3L$. On random q-ary lattices of this dimension, the barrier $1/\sqrt{2}$ is thus not reached. It comes from lattices with many vectors near a single point, such as those of [35], and a reduction beyond $1/(\sqrt{2}\gamma)$ would have to handle them.

A.20 The role of the promise

The embeddings with one call fail on the hard instances although the wanted vector $(\mathbf{e},M)$ is often still a shortest vector of their input. The input then violates the promise $\lambda_2 \geq \gamma\lambda_1$, and the oracle may answer arbitrarily. This section separates the two effects. It measures the gap λ_2/λ_1 of the inputs, and it repeats the experiment with an oracle that ignores the promise.

Table 16 computes the gap of the inputs by enumeration. For each hard instance, the table uses the embeddings with one call and one successful sparsified trial of TSE, drawn uniformly among the pairs $(\mathbf{z},\mathbf{u})$ that satisfy the event of Proposition 10.1. For the embeddings of [5] and the balanced embedding, $(\mathbf{e},M)$ is a shortest vector on all 200 instances, but the gap lies between 0.80γ and 0.94γ, below the promise. For the scaling $d/\sqrt{n}$, $(\mathbf{e},M)$ is not even a shortest vector on most instances. The successful sparsified trial of TSE has gap above $\gamma+1$ on 198 instances, which is the largest gap that our enumeration resolves, and 1.72γ and 1.77γ on the remaining two. Sparsification there-

Table 17: Four lattice families with $\gamma \in \{1.5,2\}$, the radius R3b and offsets along a shortest vector, ten runs per row: the largest bit length of an entry of the input basis after scaling, and the mean number of oracle calls and mean time in seconds of the sparsified reductions. The balanced embedding fails on all 80 runs, and all sparsified reductions solve all of them.

			TSE		Adaptive TSE		[6]		Adaptive [6]	
Family	n	Bits	Calls	Time	Calls	Time	Calls	Time	Calls	Time
q-ary	16	20	678.4	2.87	14.2	0.05	1491.5	7.09	68.5	0.26
	24	21	646.2	6.59	11.0	0.09	3731.1	43.67	184.9	1.73
uniform	16	21	527.6	2.19	17.8	0.06	2345.8	11.20	106.5	0.41
	24	22	1214.4	12.20	5.8	0.05	6903.4	75.93	160.2	1.47
basis of $\mathbb{Z}^n$	16	89	260.8	2.16	9.4	0.07	1467.4	13.13	82.6	0.65
	24	93	739.4	27.61	9.8	0.34	5624.7	186.54	168.4	5.99
$L_{n,b}$	16	20	1011.0	3.23	7.4	0.02	1895.7	6.12	82.8	0.23
	24	20	806.2	6.03	11.0	0.06	5569.0	37.80	157.9	1.01

fore does not make the wanted vector shorter. It removes the unwanted vectors that sit between λ_1 and $\gamma\lambda_1$ and so raises the gap above γ, which is what the promise asks for.

The same effect shows in the success rates with an oracle that returns a shortest vector of every input, whether the promise holds or not. With this oracle, the embeddings of [5] and the balanced embedding solve all 700 instances of Section 9 with one call, and so does the first call of TSE, while the embedding with $M = d/\sqrt{n}$ solves 440 of them, 90 more than with the promise oracle. An oracle for SVP therefore solves these decoding instances directly, which is the classical reduction of [4]. A uSVP_γ oracle may fail, and a reduction to uSVP_γ must therefore create the gap. This distinction is also the one that matters in cryptanalysis, where lattice reduction finds the embedded vector when the gap is large enough for the block size [7, 8]. The failure-symbol oracle of Section 9 is the one that realizes the worst case the promise allows, and our comparison measures what each reduction guarantees, not what a stronger oracle happens to provide.

A.21 Other lattice families

The main comparison uses random q-ary lattices. To check that the conclusions do not depend on this family, we ran the sparsified reductions and the balanced embedding on four families in dimensions 16 and 24, with $\gamma \in \{1.5,2\}$, the radius R3b and an offset along a shortest vector, and five instances per combination. The families are random q-ary lattices as in Section 9 with fresh nearest points and coins, lattices generated by random integer matrices with entries of 6 bits after LLL reduction, random bases of $\mathbb{Z}^n$ generated as for Table 7, whose entries have about 90 bits after scaling and whose minimum is attained by $2n$ vectors, and the lattices $L_{n,b}$ of Proposition A.7, whose minimum is attained by $n(n-1)$ vectors.

Table 17 shows the same picture in all four families. The balanced embedding fails on all 80 runs, because the radius lies beyond the balanced radius, while all four sparsified reductions solve all runs. Adaptive TSE needs between 5.8 and 17.8 calls on average, TSE between 260.8 and 1214.4, adaptive [6] between 68.5 and 184.9, and [6] between 1467.4 and 6903.4, and the ordering of the four reductions is the same in every row. The bases of $\mathbb{Z}^n$ have entries of about 90 bits, which makes each oracle call of TSE

between two and four times slower than for the q-ary lattices of the same dimension, but the numbers of calls are of the same order, as Proposition 10.1 predicts, because they depend on the primes and on the unwanted lines and not on the size of the entries. The lattices $L_{n,b}$, whose minimum is attained by $n(n-1)$ vectors, do not make any of the reductions harder. Their offsets along a shortest vector $\mathbf{w}_i - \mathbf{w}_j$ keep the other minimal vectors at angles of at least 60 degrees from the segment, far outside the tube.

A.22 Further related work

Here, we place the extensions of the Appendix in the literature, in addition to Section 1.1.

The reductions between BDD, uSVP and GapSVP of [5] combine three tools. The embedding of [4] maps decoding to unique shortest vectors. The technique of [40], which queries the sublattices obtained by multiplying one basis vector by a prime, maps unique shortest vectors to decoding. Sublattices of index p in a plane map unique shortest vectors to the decision problem. The last two tools are deterministic because a unique shortest vector has only one line to keep, and Appendix A.10 extends the third tool to A lines with a family whose size is within a factor n^2 of optimal. The reduction from GapSVP to BDD for large gaps of [41], which [5] use to complete the circle of equivalences, is randomized, and Corollary A.5 only concerns the direction from BDD to GapSVP. Dimension-preserving reductions based on sparsification with discrete Gaussian sampling and with a random shift are analyzed in [14, 19, 20]. Their analyses bound the probability that a random sublattice of prime index keeps one vector and removes a given set, as Proposition 10.1 does, but for general sets of short vectors, while Proposition 10.1 computes this probability precisely for the unwanted set of the embedding.

Sparsification entered lattice problems through hardness proofs for SVP. These start from the randomized reduction of [42] and continue through [43, 12, 44], and their derandomization is the subject of [21, 22]. In these proofs the set to be removed is exponentially large and dense, which is why explicit constructions are hard. In our setting the unwanted set has at most $\min(N(n,\varepsilon),N'(\varepsilon))$ elements by Lemmas 6.2 and 8.1. For decoding, where the target provides a shift, Theorem A.4 shows that the moment curve alone suffices, and the pencil family is only needed when there is no target, as in Theorem 5.1.

In cryptanalysis, the embedding of [4] is the core of the primal attack on learning with errors. There, block reduction recovers the embedded vector once the gap between the embedded vector and the lattice is large enough for the block size, which [7] observed and [8, 45] quantified. The estimates of [46] extend this to instances with side information, which change the lattice and the embedded vector. Appendix A.20 measures the same quantity, the gap of the embedding, on worst-case offsets. It shows that the embeddings with one call lose the gap γ beyond their proven radius even when the embedded vector is still the shortest, and that sparsification restores a gap of more than $\gamma+1$. Sparsification multiplies the number of oracle calls by about the prime, which is why attacks do not use it, but Theorem 8.3 shows that no choice of the scaling alone keeps the gap on these offsets.

For rotations of $\mathbb{Z}^n$, ZSVP and the isomorphism problem were studied in [25–27], and [11, 15, 28, 29] relate them to SVP and to lattice reduction. The randomized

reductions of [11, 15] use sparsification and automorphisms respectively. Table 8 shows that the deterministic thin family needs fewer oracle calls on our bases than random sparsification with the same prime, and Appendix A.11 compares it with the sampler of [11] as specified there. The isomorphism problem underlies the signatures of [10] and the framework of [9], whose question on unusual lattices motivates Contribution 1.

A.23 Details of the promise oracle

The oracle receives a basis with integer entries, LLL-reduces it with fplll [39], and enumerates with a Gram–Schmidt decomposition in 160-bit floating point the $k = 2\lceil\gamma\rceil+3$ shortest nonzero vectors up to sign within the squared radius $\lfloor\gamma^2\|\mathbf{b}_1\|^2\rfloor+1$, where $\mathbf{b}_1$ is the first reduced basis vector. The first vector $\mathbf{v}_1$ of the list is a shortest vector, and the oracle declares the promise violated if the first vector of the list that is not parallel to $\mathbf{v}_1$ has squared norm smaller than $\gamma^2\|\mathbf{v}_1\|^2$, which it decides with precise rational arithmetic since γ^2 is rational in all experiments. This decision is correct as follows. If the lattice satisfies the promise, no vector outside the line of $\mathbf{v}_1$ is shorter than $\gamma\lambda_1$. If it does not, let $\mathbf{u}$ be a shortest vector outside this line, with $\|\mathbf{u}\| < \gamma\lambda_1 \le \gamma\|\mathbf{b}_1\|$. The vectors of norm at most $\|\mathbf{u}\|$ are, up to sign, the multiples $j\mathbf{v}_1$ with $j < \gamma$ and the vectors outside the line of norm at most $\|\mathbf{u}\|$, so among the $k > \lceil\gamma\rceil$ shortest vectors there is one outside the line of norm at most $\|\mathbf{u}\|$, and the violation is detected. On a valid input the oracle returns $\mathbf{v}_1$, and on an invalid input it returns a failure symbol, so a reduction that succeeds does so through a valid input. The oracle counts its calls and the valid ones, measures its time, and stores the largest bit length of an input entry.

A.24 Details of the instances

The lattices of Section 9 are generated by fpylll's q-ary generator with $n/2$ random rows and the parameter 10 for the bit length of the modulus, followed by LLL reduction, and the seed depends only on the dimension and the index of the instance. For each lattice, a shortest vector $\mathbf{s}$ is computed by enumeration, and the nearest point is $\mathbf{c} = \mathbf{B}\mathbf{x}$ with $\mathbf{x}$ uniform in $\{-4,\ldots,4\}^n$. The lattice and $\mathbf{c}$ are scaled by 2^{12}, and the offset is the rounding of $2^{12}\alpha\lambda_1$ times a unit vector, which is $\mathbf{s}/\lambda_1$ for a short offset and a normalized Gaussian vector for a random offset. The distance d given to all reductions is the norm of the rounded offset. The random choices of the reductions are seeded by the instance and the dimension, so the same lattice, nearest point and random choices appear in all regions and for all gaps, which makes the counts of Tables 3 and 19 comparable across regions and gaps. The same generator with $n \in \{32,36,40\}$ gives the lattices of Table 12, and the experiment of Figure 7 uses separate seeds with ten instances per gap. The bases of $\mathbb{Z}^n$ of Tables 7 and 8 start from the standard basis, apply $40n$ random elementary operations, each adding a multiple in $\{-3,\ldots,3\}$ of one basis vector to another, and permute the basis vectors at random. The instances of Theorem A.8 use $\gamma = 2$, $\varepsilon = 1/(40n)$ and $\tau = \lfloor 10^5\tau_0\rfloor/10^5$, and all coordinates are multiplied by 10^5 to make them integers.

Table 18: Parameters of the implemented reductions from BDD to uSVP_γ, with the scaling M of the last coordinate, the primes, the shift of the target, the first call, the order of the trials and the budget of oracle calls. Here p_B is the least prime above $4\gamma n^2$, p^* the least prime at least $2N(n,\varepsilon)$ with $\varepsilon = 0.1$, p_j the least prime at least $3 \cdot 2^j$, $N = N(n,\varepsilon)$, and $\tilde{\mathbf{u}}$ a uniform vector in $\{0,\ldots,p-1\}^n$.

Reduction	M	Primes	Shift $\mathbf{w}$	Trials	Budget
$M = d/\sqrt{n}$ [4, 8]	$d/\sqrt{n}$	none	none	one call	1
$M = d$ [5]	d	none	none	one call	1
Balanced [6]	$k_\gamma d$	none	none	one call	1
Sparsified [6]	$d/(n-1)$	p_B	$\mathbf{B}\tilde{\mathbf{u}} + hp_B\mathbf{b}_1$ with $\|\mathbf{t}+\mathbf{w}\| \geq (n+1)\lambda/\sqrt{2}$	uniform $\mathbf{z} \neq 0$, $\mathbf{u}$	$10p_B$
Adaptive [6]	$d/(n-1)$	p_B, $p_j < p_B$	as [6]	interleaved streams	$20p_B$
TSE, Alg. 1	$\sqrt{\varepsilon}d$	p^*	$\mathbf{B}\tilde{\mathbf{u}}$	balanced call, then uniform	$1+10p^*$
Adaptive TSE, Alg. 2	$\sqrt{\varepsilon}d$	p^*, $p_j < p^*$	$\mathbf{B}\tilde{\mathbf{u}}$	balanced call, then interleaved	$1+20p^*$
Pencils, Thm. 6.4	$\sqrt{\varepsilon}d$	$q \geq 2N(n-1)+2n$	$\mathbf{B}\tilde{\mathbf{u}}$, $\mathbf{u}$ in order	$\mathbf{u} \in U$, then $\mathcal{Z}_{n,q}$	$5q$
Shifts, Thm. A.4	$\sqrt{\varepsilon}d$	$q \geq N(n-1)+1$	$\tau\mathbf{b}_1$	α, then τ	$(N(n-1)+1)q$

A.25 Parameters of the implementations

Table 18 lists the parameters of all implementations. All sparsified reductions use the basis of Lemma 3.4, which is cheaper than the Hermite normal form of [6] and does not change the sublattice, and all of them receive the precise distance d, round M to an integer after scaling the lattice by 2^{12}, and stop at the first candidate at distance d. For the reduction of [6], the integer $h \geq 0$ is the least one for which $\|\mathbf{t}+\mathbf{w}\| \geq (n+1)\lambda/\sqrt{2}$, which follows the choice of the shift in [6]; the term $hp_B\mathbf{b}_1$ lies in p_BL and does not change which sublattices keep the nearest point. The adaptive variants use $p_0 = 3$, the least prime above $h(0.1) \approx 2.94$, and $\lceil 2\ln 4 \cdot p_j \rceil$ trials per level. The pencil reduction orders U by the integer encoding of its vectors and the pencil family as in the definition of $\mathcal{Z}_{n,p}$, and the shift reduction orders its trials as in Algorithm 3. No run reached its budget.

A.26 Implementation of the reductions

This section describes how the implementation realizes the steps of the algorithms, since several of them admit choices that affect the measurements.

All lattice vectors, bases and targets are lists of Python integers, and all comparisons that decide success, validity or membership in $\mathcal{Y}^*$ use precise integer or rational arithmetic. The library fplll [39] is used only inside the promise oracle, for LLL reduction, the Gram–Schmidt decomposition and enumeration, in the instance generator to find a shortest vector, and in the precise computations of $\mathcal{Y}^*$. The basis of $L_{\mathbf{z}}$ follows Lemma 3.4. The implementation takes the first index j with $z_j \neq 0$, computes $c_i = z_i z_j^{-1} \bmod p$ with a modular exponentiation, and forms the rows $\mathbf{b}_i - c_i\mathbf{b}_j$ for $i \neq j$ and $p\mathbf{b}_j$, in $O(n^2)$ operations. The shift of TSE is $\mathbf{w} = \sum_i \tilde{u}_i\mathbf{b}_i$ with $\tilde{u}_i$ uniform in $\{0,\ldots,p-1\}$, which adds at most $\log_2(np)$ bits to the entries of the target, and the reduction of [6] adds the multiple $hp_B\mathbf{b}_1$ described in Appendix A.25. The embedding consists of the rows of the sublattice basis padded with a zero and the row $(\mathbf{t}+\mathbf{w}, m)$, where the integer m is M rounded after scaling the lattice by 2^{12}.

The oracle returns an integer combination of the rows of the LLL-reduced input basis, so its answer always lies in the queried lattice, and the membership test that the theorems require is satisfied by construction. An answer with last coordinate $\pm m$ yields the candidate $\mathbf{t}+\mathbf{w}\mp\mathbf{v}'$ minus the shift, where $\mathbf{v}'$ is the answer without its last coordinate, and any other answer yields the failure symbol. Since the implementation knows d, a run stops at the first candidate $\hat{\mathbf{c}}$ with $\|\mathbf{t}-\hat{\mathbf{c}}\|^2=d^2$, which is a precise comparison of integers. Each candidate is a lattice vector, so this candidate is a nearest vector.

The random choices of a run come from one instance of Python's generator `random.Random`, seeded from the dimension, the index of the instance and a constant per experiment. This generator is not cryptographic, but the analyses only need independent uniform choices, and Proposition 10.1 and Figure 11 confirm that the measured distributions match uniform choices. The vector $\mathbf{z}$ is drawn uniformly from $\mathbb{F}_p^n$ and redrawn if it is zero, and $\mathbf{u}$ is drawn uniformly from $\mathbb{F}_p^n$. The adaptive stream of Algorithm 2 is a generator that produces the next prime on demand. The pencil reduction enumerates U as the vectors whose base-q digits encode 0,1,2,..., and the shift reduction enumerates α and τ in increasing order.

The precise computations of Section 10.1 and Appendix A.15 enumerate, for every integer $m \geq 1$ with $m^2M^2 < \gamma^2(d^2+M^2)$, the lattice vectors $\mathbf{y}$ with $\|\mathbf{y}+m\mathbf{e}\|^2 \leq \gamma^2(d^2+M^2)-m^2M^2$ with fplll's enumeration around $-m\mathbf{e}$. They keep those that satisfy the strict inequality in precise arithmetic, and add the negatives for $-m$. They compute the residues of these vectors from the precise rational inverse of the basis, reduce each residue to a normalized representative of its line, and evaluate Proposition 10.1 by inclusion and exclusion over the lines with ranks computed by Gaussian elimination modulo p. The time and memory of a run are measured with `time.perf_counter` and `tracemalloc` around the whole reduction, including the oracle, and the oracle measures its own time separately, which gives the split of Table 23. The runs of one experiment are distributed over worker processes that run one reduction at a time, and every finished run is appended to the result file, which makes the drivers of the long experiments resumable.

A.27 Additional measurements

Here, we list the measurements behind the figures of Section 9 per dimension and gap.

Table 19 lists the hard instances of Table 3 per dimension and gap. The prime of TSE depends only on n, while the prime of [6] grows linearly with γ and quadratically with n, and the mean number of calls of both reductions is of the order of their prime. TSE uses fewer calls in 19 of the 20 rows; in the row $n=28$, $\gamma=1.5$, the five instances of [6] happen to succeed after 1004.4 calls on average against 1397.4 for TSE, while the primes are 4721 and 1123, and TSE is still faster in time because each of its calls is cheaper. TSE is faster in time and has shorter oracle entries in all 20 rows.

Table 20 gives the numbers behind Figures 7 and 12. All reductions except the balanced embedding solve all 140 instances. The mean number of calls of [6] is between 0.5 and 2.1 times its prime in every row, as Corollary 10.2 predicts for one unwanted line, and its bit length grows by one bit each time γ doubles, from 29 to 36 bits, while TSE and its adaptive variant have constant calls and 28 bits. The time of TSE grows from 1.2 to 14.6 seconds with constant calls, because one call of our oracle

Table 19: Hard instances per dimension and gap, ten runs per row, with the primes, mean calls, mean time in seconds and largest bit length of an input entry of the reduction of [6] and of TSE.

		[6]				TSE			
n	γ	p_B	Calls	Time	Bits	p	Calls	Time	Bits
12	1.25	727	1135.4	2.43	29	487	418.4	0.79	28
12	1.5	877	1223.4	2.72	29	487	418.4	0.80	28
12	2	1153	700.4	1.56	30	487	418.4	0.80	28
12	3	1733	1854.8	4.58	30	487	418.4	0.87	28
16	1.25	1283	1459.8	5.37	30	641	328.4	1.05	28
16	1.5	1543	816.0	3.06	30	641	328.4	1.05	28
16	2	2053	1914.2	7.27	31	641	328.4	1.07	28
16	3	3079	2596.2	10.83	31	641	328.4	1.18	28
20	1.25	2003	1201.0	8.15	31	809	246.0	1.25	29
20	1.5	2411	2212.4	15.65	31	809	246.0	1.28	29
20	2	3203	2302.8	16.18	31	809	246.0	1.29	29
20	3	4801	4899.2	37.84	32	809	246.0	1.38	29
24	1.25	2887	1257.4	13.56	31	967	1057.6	8.36	29
24	1.5	3457	4329.8	46.34	31	967	1057.6	8.45	29
24	2	4621	5736.2	61.06	32	967	1057.6	8.65	29
24	3	6917	13989.8	185.63	32	967	1057.6	9.58	29
28	1.25	3923	1673.6	31.26	32	1123	1397.4	18.12	29
28	1.5	4721	1004.4	19.23	32	1123	1397.4	18.46	29
28	2	6277	7362.6	119.78	33	1123	1397.4	18.63	29
28	3	9413	11540.8	233.81	33	1123	1397.4	18.48	29

Table 20: Oracle calls as a function of the gap for $n \in \{12,16\}$, over ten instances per row with an offset along a shortest vector halfway between the balanced radius and $0.9/(\sqrt{2}\gamma)$, with the prime, mean calls, mean time in seconds and largest bit length of an oracle entry of [6] and of TSE, and the mean calls of the two adaptive variants. The prime of TSE is 487 for $n=12$ and 641 for $n=16$ in all rows.

		[6]				TSE			Adaptive, mean calls	
n	γ	p_B	Calls	Time	Bits	Calls	Time	Bits	TSE	[6]
12	1	577	670.8	1.28	29	639.8	1.22	28	7.6	35.6
12	2	1153	845.2	1.95	30	639.8	1.52	28	7.6	35.6
12	4	2309	1261.7	3.56	31	639.8	1.78	28	7.6	38.0
12	8	4621	4509.9	17.20	32	639.8	2.36	28	7.6	30.9
12	16	9221	10232.6	68.67	33	639.8	3.65	28	7.6	35.6
12	32	18433	12499.8	139.24	34	639.8	6.45	28	7.6	40.0
12	64	36871	52937.7	1333.57	35	639.8	12.11	28	7.6	39.6
16	1	1031	964.1	3.16	30	543.4	1.71	28	8.8	85.2
16	2	2053	1774.1	6.72	31	543.4	2.07	28	8.8	91.8
16	4	4099	8454.7	39.63	32	543.4	2.37	28	8.8	107.0
16	8	8209	6519.7	40.55	33	543.4	3.08	28	8.8	136.6
16	16	16411	9827.1	97.97	34	543.4	4.56	28	8.8	137.8
16	32	32771	30673.1	586.92	35	543.4	7.86	28	8.8	96.0
16	64	65537	74077.4	1996.82	36	543.4	14.58	28	8.8	96.0

enumerates $2\lceil\gamma\rceil+3$ vectors. Over the 200 hard runs of Section 9, only 0.94% of the calls of TSE and 0.87% of those of [6] are valid instances of uSVP_γ, about $1/p$ as Proposition 10.1 predicts, because the sublattice rarely keeps the shifted nearest point.

Table 21 extends Table 3 to the adaptive variants, marked A-, and adds the time. Adaptive TSE uses one call wherever the balanced embedding succeeds and 8.8 calls on the hard offsets, while adaptive [6], which has no first call with a single embedding,

Table 21: All seven reductions per region of the radius and direction of the offset on the 700 instances of Section 9, with instances solved out of 100, or 50 for R2, mean oracle calls of the sparsified reductions and mean time in milliseconds of [6] and of the two variants of TSE.

			Solved						Mean calls				Mean time (ms)		
Region	Offset	$d/\sqrt{n}$	[5]	Bal.	[6]	A-[6]	TSE	A-TSE	[6]	A-[6]	TSE	A-TSE	[6]	TSE	A-TSE
R1	short	0	100	100	100	100	100	100	3460.5	71.8	1.0	1.0	41094.3	3.1	3.2
	random	100	100	100	100	100	100	100	3460.5	7.0	1.0	1.0	42352.4	3.4	3.1
R2	short	0	25	50	50	50	50	50	1631.3	94.5	1.0	1.0	14733.4	3.0	3.0
	random	50	50	50	50	50	50	50	1631.3	7.4	1.0	1.0	14839.7	2.8	3.3
R3a	short	0	0	0	100	100	100	100	3460.5	102.4	689.6	8.8	41947.2	6106.3	44.2
	random	100	100	100	100	100	100	100	3460.5	7.0	1.0	1.0	41378.6	3.3	3.4
R3b	short	0	0	0	100	100	100	100	3460.5	122.0	689.6	8.8	40683.6	6049.2	43.6
	random	100	100	100	100	100	100	100	3460.5	7.0	1.0	1.0	39563.3	3.1	3.2

Table 22: Hard instances per dimension for the four sparsified reductions, with mean and largest number of oracle calls and mean time in seconds, over 40 runs per dimension up to $n=28$ and 10 runs for $n\geq 32$; dashes mark reductions that we did not run.

	Adaptive TSE			Adaptive [6]			TSE			[6]			
n	Mean	Max	Time	Mean	Max	Time	Mean	Max	Time	Mean	Max	Time	Runs
12	11.0	27	0.019	37.0	120	0.070	418.4	742	0.819	1228.5	4186	2.822	40
16	7.0	11	0.018	80.0	118	0.244	328.4	614	1.089	1696.5	6286	6.631	40
20	9.4	13	0.044	105.9	196	0.514	246.0	599	1.300	2653.8	13509	19.455	40
24	8.6	13	0.058	145.7	512	1.080	1057.6	3187	8.760	6328.3	45972	76.648	40
28	8.2	13	0.081	192.6	318	2.129	1397.4	3678	18.422	5395.4	23714	101.022	40
32	14.4	29	0.256	192.0	274	3.564	1433.2	3309	47.324	6936.4	19188	256.096	10
36	9.4	15	0.659	197.6	382	15.987	643.4	1934	107.255	–	–	–	10
40	10.6	19	10.888	208.2	302	237.590	–	–	–	–	–	–	10

needs about 7 calls on random offsets and between 72 and 122 calls on offsets along a shortest vector in every region, including R1. The reason is Theorem 8.4. On offsets along a shortest vector, the scaling $d/(n-1)$ excludes the small primes of the adaptive stream even for small radii, and on random offsets the multiples of the offset stay far from the lattice. The time of [6] is between 14.7 and 42.4 seconds per instance in every region, because it never uses a single call, while both variants of TSE need about 3 ms outside the hard offsets.

Table 22 gives the per-dimension numbers behind the upper panel of Figure 12, including the largest number of calls over all runs of a dimension. The largest number of calls of adaptive TSE is 29, reached for $n=32$, while the largest numbers of TSE and [6] are 3678 and 45972. The time per call grows quickly with the dimension because of the enumeration in our oracle, from about 2 ms for $n=12$ to about one second for $n=40$, so that the reduction in calls translates into a reduction in time of the same order; for $n=32$, adaptive TSE takes 0.26 seconds, TSE 47.3 seconds and [6] 256.1 seconds on average.

Table 23 separates the two parts of the time of one call. The promise oracle takes between 16% and 47% more time per call on the inputs of [6] than on those of TSE, because their entries are 2 to 4 bits longer and the shifted target is longer, which slows down both LLL reduction and enumeration. The work of the reduction itself, which builds the sublattice basis, the shift and the embedding and checks the answer, is between 24% and 38% of the oracle time for TSE and grows more slowly with n than the oracle time. The calls of the adaptive variants are cheaper still, because most of

Table 23: Time per oracle call on the hard instances, in milliseconds, split into the time of the promise oracle and the time of the reduction itself, for the four sparsified reductions.

	TSE		[6]		A-TSE		A-[6]	
n	Oracle	Own	Oracle	Own	Oracle	Own	Oracle	Own
12	1.42	0.54	1.65	0.65	1.26	0.45	1.34	0.55
16	2.40	0.91	2.82	1.09	1.90	0.68	2.15	0.90
20	3.95	1.33	5.48	1.85	3.54	1.09	3.55	1.31
24	6.39	1.90	9.37	2.74	5.25	1.48	5.56	1.86
28	10.65	2.54	15.18	3.54	7.89	2.02	8.58	2.47

their trials use small primes and hence shorter entries. The time ratios of Table 1 are therefore products of a ratio of calls and a ratio of times per call, and both favor TSE.

A.28 Reproducing the tables and figures

All numbers of the paper are produced by the code that accompanies it, which has one driver per experiment in the directory `experiments` and writes one line per run to a file in `results`. The figures are produced from the result files by `make_shows.py`, with the option `--llncs` for the page width of this paper, and the README of the code lists the driver of each experiment and its running time. The drivers derive all random choices from fixed seeds that depend on the dimension, the index of the instance and, where stated, the lattice family, so a second run reproduces the same instances, coins and numbers of calls, while the measured times depend on the machine.

AI Usage Policy. Grammarly's AI assisted in editing text, and Claude in implementing experiments.

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